CANONICAL SUBGROUPS AND PERFECTOID SHIMURA VARIETIES

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Abstract. We develop a group-theoretic analogue of the theory of canonical subgroups which works uniformly for integral canonical models of Shimura varieties defined over $\mathbf{Z}_p$, with bounds on its overconvergence radii in terms of group-theoretic Hasse invariants. This is achieved using prismatic Dieudonné theory. We apply this to prove that all minimally compactified Shimura varieties at infinite level are perfectoid spaces, including those of exceptional type for sufficiently large primes.

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1. Introduction

1.1. **Main results.** Since Scholze's breakthrough [Sch15] establishing that Shimura varieties of Hodge type at infinite level are perfectoid spaces, there has been a concerted effort to extend this result to all Shimura varieties, as conjectured in [Sch16, Conjecture 2.1]. Subsequent works have enlarged the scope of this theorem, encompassing abelian type [She17] and pre-abelian type Shimura varieties [HJ23]essentially by reducing it to the Hodge type case. More recently, He [He26] proved pointwise perfectoidness at infinite level for arbitrary Shimura varieties using Sen theory. However, the global perfectoidness of Shimura varieties of exceptional type has remained out of reach due to there being no reasonable way to relate these Shimura varieties with abelian varieties. The recent breakthrough of Bakker, Shankar, and Tsimerman [BST25a] establishing the existence of integral canonical models for general Shimura varieties for sufficiently large primes is what allows us to approach this problem, coupled with the prismatic Dieudonné theory formulated by Gardner and Madapusi [GM24]. The aim of this paper is to leverage these tools to establish the global perfectoidness of arbitrary Shimura varieties at infinite level, including those of exceptional type for sufficiently large primes`thm:perfectoid-shimura-varieties`

Theorem A (Theorem 5.6.6)**.** *Let (G, X) be a Shimura datum with reflex field E. For all sufficiently large primes p, the following holds. Fix a complete algebraically closed field $C/\mathbf{Q}_p$ and an embedding $E \hookrightarrow C$. Fix any open compact subgroup $K^p \subset G(\mathbf{A}_f^p)$. For any open compact subgroup $K_p \subset G(\mathbf{Q}_p)$, let $\mathcal{S}^*_{K^pK_p}$ denote the adic space over $\operatorname{Spa} C$ associated with the base change of the minimal compactification $\mathrm{Sh}^*_{K^pK_p}(G, X)$ along $E \hookrightarrow C$. Then there is a perfectoid space $\mathcal{S}^*_{K^p}$ such that*

$$\mathcal{S}^*_{K^p} \sim \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathcal{S}^*_{K^pK_p}$$

as diamonds over $\operatorname{Spd} C$. Moreover, there is a canonical $G(\mathbf{Q}_p)$-equivariant Hodge–Tate period map

$$\pi_{\mathrm{HT}} \colon \mathcal{S}^*_{K^p} \to \mathscr{F}\!\ell_{G,\mu}$$

of adic spaces over $\operatorname{Spa} C$ which is functorial in the tame level. Finally, the boundary of $\mathcal{S}^*_{K^p}$ is Zariski-closed.
`rem:extension-to-small-primes`

Remark 1.1.2. The restriction to sufficiently large primes p in Theorem A stems solely from our reliance on the existence of integral canonical models for Shimura varieties of exceptional type, which is currently established by Bakker, Shankar, and Tsimerman [BST25a] only when p is ineffectively sufficiently large. Consequently, once integral canonical models are known to exist for small primes p, the identical argument will apply without change to establish the perfectoidness at infinite level for all primes p.

Remark 1.1.3. It is also worth noting that we find a uniform group-theoretic proof of infinite-level perfectoidness of minimally compactified Shimura varieties provided (i) its reflex field is $\mathbf{Q}$, (ii) it admits an integral canonical model at p, and (iii) the codimension of the boundary of minimal compactification is at least 2.

Scholze's method relies on the theory of canonical subgroups of abelian varieties. To transpose this machinery to general Shimura varieties, we replace Barsotti–Tate groups with group-theoretic analogues. For Shimura data of Hodge type, this is usually done by considering 'p-divisible groups with G-structure'. Such an interpretation is not known for arbitrary Shimura data, even in the general abelian-type case. The theory of (n-truncated) $(\mathcal{G}, \mu)$-apertures developed by Gardner and Madapusi [GM24] serves as a group-theoretic replacement for (n-truncated) Barsotti–Tate groups functioning uniformly even when G is a group of exceptional type. Let $\mathcal{G}$ be a reductive group scheme over $\mathbf{Z}_p$ and $\mu\colon \mathbf{G}_{\mathrm{m}} \to \mathcal{G}$ a miniscule cocharacter. In brief, over a p-complete ring R, an n-truncated $(\mathcal{G}, \mu)$-aperture is the data of a $\mathcal{G}$-torsor over the mod-p^n syntomification $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ that is "bounded by μ".

When reduced modulo p, any such aperture naturally induces an F-zip with $\mathcal{G}$-structure, also called a $\mathcal{G}$-zip, over R/pR in the sense of Pink, Wedhorn, and Ziegler [PWZ15]. In the classical setting of de Rham cohomology of a smooth projective variety in characteristic p, an F-zip encodes the decreasing Hodge filtration and the increasing conjugate filtration together with the semilinear Cartier isomorphism of their associated graded. In the group-theoretic framework, a $\mathcal{G}$-zip consists of a $\mathcal{G}$-torsor equipped with two parabolic reductions: a reduction to the negative parabolic $\mathcal{P}_\mu^-$ (the de Rham realization, or Hodge filtration) and a reduction to the positive parabolic $\mathcal{P}_\mu^+$ (the conjugate realization, or conjugate filtration) together with a semilinear isomorphism of their associated Levi reductions (Cartier isomorphism). The work of Goldring and Koskivirta [GK19] explains how to attach Hasse invariants to any $\mathcal{G}$-zip, which they call group-theoretic Hasse invariants, measuring failure of ordinariness.

Our key result establishes a uniform *integral* theory of canonical subgroups for these group-theoretic analogues of Barsotti–Tate groups:

thm:intro-canonical-subgroup

Theorem D (Canonical subgroup for 1-truncated apertures, Theorem 4.1.3)**.** *Let $\mathcal{G}/\mathbf{Z}_p$ be a reductive group scheme and $\mu\colon \mathbf{G}_{\mathrm{m}} \to \mathcal{G}$ a minuscule cocharacter. Let $\mathcal{E}_1 \to R^{\mathrm{syn}} \otimes \mathbf{F}_p$ be a $\mathcal{G}$-torsor representing a 1-truncated $(\mathcal{G},\mu)$-aperture over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R. Let $\varepsilon \geqslant 0$ be the p-adic valuation of its group-theoretic Hasse invariant. If $\varepsilon < \frac{p}{p+1}$, then there exists a unique reduction $\mathcal{E}_1^+ \subset \mathcal{E}_1$ of the structure group to $\mathcal{P}_\mu^+$ that agrees with the reduction of structure induced by the conjugate filtration modulo $p^{1-\varepsilon}$.*

By iterating this construction, we obtain analogues of higher-level canonical subgroups as well (Theorem 4.2.8). Specializing to the case of $\mathcal{G} = \mathrm{GL}_h$, we deduce a parallel statement for Barsotti–Tate groups. Remarkably, even in this classical setting, we obtain stronger results:

cor:intro-bt

Corollary E (Canonical subgroup for 1-truncated Barsotti–Tate groups, Theorem 4.3.1)**.** *Let G be a 1-truncated Barsotti–Tate group over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R with $G_0 := G|_{R/pR}$ its reduction modulo p. If the Hasse invariant of G has p-adic valuation $\varepsilon < \frac{p}{p+1}$, then G admits a unique finite locally free subgroup scheme which agrees with the kernel of relative Frobenius $\mathrm{Ker}(G_0 \to G_0^{(p)})$ modulo $p^{1-\varepsilon}$.*

cor:intro-p-divisible

Corollary F (Level-n canonical subgroup for p-divisible groups, Theorem 4.3.3)**.** *Let $n \geqslant 1$ be an integer, and let G be a p-divisible group over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R with $G_0 := G|_{R/pR}$ its reduction modulo p. If the Hasse invariant of G has p-adic valuation $\varepsilon < \frac{1}{p^{n-2}(p+1)}$, then $G[p^n]$ admits a unique finite locally free subgroup scheme which agrees with the kernel of n-th iterated relative Frobenius isogeny $\mathrm{Ker}(G_0 \to G_0^{(p^n)})$ modulo $p^{1-p^{n-1}\varepsilon}$.*

The theory of overconvergent canonical subgroups has a rich historical development. The case of elliptic curves was completely resolved by Katz [Kat73] and Lubin [Lub79]. For abelian varieties of general dimension, Abbes and Mokrane [AM04] resolved the level-1 case (that is, for p-torsion points) by coupling the description given by Bloch and Kato of p-adic vanishing cycles on smooth projective varieties having good reduction with the ramification theory developed by Abbes and Saito. Andreatta and Gasbarri [AG06] subsequently recovered the result of Abbes–Mokrane via alternative global methods involving abelian schemes. Tian [Tia10] then extended the result of Abbes–Mokrane to truncated Barsotti–Tate groups of level 1 by utilizing resolutions of such groups by abelian schemes and Bloch–Kato results on the associated vanishing cycles. For higher levels, Conrad [Con06] established overconvergence in general for p^n-torsion points of abelian schemes for all n, though with inexplicit bounds. A comprehensive theory of canonical subgroups of arbitrary level $n \geqslant 1$ for Barsotti–Tate groups was established by Fargues [Far11] using Harder–Narasimhan and Abbes–Saito ramification filtrations, while Rabinoff [Rab12] investigated arbitrary level canonical subgroups via completely different methods. Hattori [Hat13] constructed level-n canonical subgroups for n-truncated Barsotti–Tate groups over complete discrete valuation rings using Breuil–Kisin modules.

Compared to this extensive literature, our results differ in both methodology and scope. A unifying characteristic of these classical approaches is their restriction to complete discrete valuation rings of mixed characteristic. In contrast, our linear-algebraic approach via prismatic theory operates uniformly over arbitrary flat p-adic formal schemes. Furthermore, as detailed in Section 4.3, our overconvergence valuation bounds ($\varepsilon < \frac{1}{p^{n-2}(p+1)}$ for level n) strictly improve upon the classical thresholds of Fargues [Far11] and Scholze [Sch15]. We obtain the same pointwise overconvergence bound as Hattori [Hat13] over complete discrete valuation rings for the existence of the Frobenius-kernel lift. Hattori imposes stronger Hasse-invariant bounds for additional assertions about the abstract group of geometric points of a general truncated Barsotti–Tate group and, still more restrictively, for comparison with ramification subgroups in rigid-analytic families. In the p-divisible setting considered below, the iterated level-1 construction recovers the expected geometric fibers in the full overconvergent range. Our prismatic approach avoids the latter gluing loss by constructing canonical subgroups directly over arbitrary flat p-adic formal schemes. These bounds are strong in the sense that they recover sharp valuation bounds of Katz *et al* when specialized to the case of elliptic curves. Rabinoff [Rab12] instead gives an explicit radius for a metric "closest to zero" subgroup on the rigid generic fiber; this is an important but different comparison, since his definition does not a priori involve lifting the kernel of the relative Frobenius. Similarly, Fargues characterizes canonical subgroups via Abbes–Saito ramification filtrations.

1.2. **Overview of proofs.** Let (G, X) be a Shimura datum with reflex field E, and let $\mathcal{G}$ be a reductive model of G over $\mathbf{Z}_p$. Fix an unramified place $v \mid p$ of E. The Shimura datum determines a minuscule cocharacter $\mu_v\colon \mathbf{G}_{\mathrm{m},\mathcal{O}_{E,v}} \to \mathcal{G}^c_{\mathcal{O}_{E,v}}$, where $\mathcal{G}^c$ is the *cuspidal quotient* of $\mathcal{G}$ (cf. Setup 3.1.1). For each integer $n \geqslant 1$, we can then attach the smooth p-adic formal Artin stack $\mathrm{BT}_n^{\mathcal{G}^c,-\mu_v}$ over $\operatorname{Spf}\mathcal{O}_{E,v}$ of n-truncated $(\mathcal{G}^c, -\mu_v)$-apertures constructed in [GM24]. Let $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v} := \varprojlim_n \mathrm{BT}_n^{\mathcal{G}^c,-\mu_v}$ denote their inverse limit. These stacks generalize the formal stacks of (truncated) polarized p-divisible groups of height $2g$ in the case of the Siegel Shimura datum, providing a group-theoretic framework for defining "p-divisible groups with $\mathcal{G}^c$-structure" without requiring one to actually work with p-divisible groups.

Following the construction of integral canonical models $\mathscr{S}_K$ over $\mathcal{O}_{E,(v)}$ for arbitrary Shimura varieties by Bakker, Shankar, and Tsimerman [BST25a], recent work of Madapusi and Youcis [MY26] provides an alternative characterization of these models via a Serre–Tate property: there is a canonical, formally étale *syntomic realization* map of p-adic formal stacks over $\operatorname{Spf}\mathcal{O}_{E,v}$,

$$\varpi\colon \widehat{\mathscr{S}_K} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v},$$

such that the induced $\mathcal{G}^c(\mathbf{Z}_p)$-local system on the generic fiber recovers the canonical étale local system $\mathbf{Et}_{K,p}$ (Theorem 3.1.5). In the Siegel case, this is precisely the map classifying the universal polarized p-divisible group over the p-adic completion of the integral canonical model of the Siegel modular variety. Once we have an analogue of a universal p-divisible group over integral canonical models of Shimura varieties of exceptional type, it is conceivable that a theory of canonical subgroups can be used to prove infinite-level perfectoidness, just as in Scholze's work [Sch15]. For technical simplicity, we prefer to develop a theory of canonical subgroups under the assumption that μ_v is defined over $\mathbf{Z}_p$, which corresponds to assuming that the reflex field E_v is $\mathbf{Q}_p$. This is sufficient to establish infinite-level perfectoidness for integral canonical models whose local reflex field is $\mathbf{Q}_p$; the general case is then deduced via an embedding argument using the fact that a closed adic subspace of a perfectoid space is perfectoid.

Let us now explain the proof of Theorem D. For simplicity, consider the case where $\mathfrak{X} = \operatorname{Spf} R$ for a p-torsion free perfectoid algebra R over $\mathbf{Z}_p^{\mathrm{cycl}}$. The general case is deduced by quasisyntomic descent. Let $R^\flat$ denote the tilt of R with pseudo-uniformizer $t = p^\flat$, and let φ be the absolute Frobenius on $R^\flat$. Then a 1-truncated $(\mathcal{G}, \mu)$-aperture over R is equivalent to the data of a $\mathcal{G}$-bundle $\mathcal{T}$ over $R^\flat$ equipped with a modification $m\colon \varphi^*\mathcal{T}[1/t] \xrightarrow{\sim} \mathcal{T}[1/t]$ that is "bounded by μ". When $\mathcal{G} = \mathrm{GL}_h$, this reduces to a 1-truncated Breuil–Kisin–Fargues module over R (cf. Example 2.2.17). In this setting, finite locally free subgroup schemes correspond to m-stable submodules. In our group-theoretic situation, this translates to an m-stable reduction of structure of $\mathcal{T}$ to a parabolic subgroup. For canonical subgroups, it suffices to find a reduction of structure to the positive parabolic $\mathcal{P}_\mu^+$ associated with μ such that it reduces modulo $p^{1-\varepsilon}$ to the reduction of structure induced by the conjugate filtration over $R^\flat/tR^\flat \simeq R/pR$. This latter condition encodes the fact that the subgroup lifts the Frobenius kernel. To conclude, we choose an arbitrary, not necessarily m-stable lift of the conjugate filtration and repeatedly apply m to obtain a unique "fixed-point" m-stable reduction via a t-adic contraction mapping principle argument applied to a flag variety.

Theorem D is sufficient to give a uniform proof of Theorem A for Shimura varieties with reflex field $\mathbf{Q}$ admitting a p-adic integral canonical model, essentially following Scholze's method [Sch15]. An additional ingredient is a relative perfectness result for the reduction modulo p of the syntomic realization map (Corollary 3.2.6). Interestingly, this yields a purely formal explanation for the existence of canonical Frobenius lifts over the μ-ordinary locus of Shimura varieties, which we record in Theorem 3.3.9.

For general Shimura varieties with reflex fields larger than $\mathbf{Q}$, we employ an embedding argument. A standard restriction-of-scalars construction handles the pre-abelian case and the exceptional E_7 case, whose reflex field is totally real. The remaining difficulty is outer type E_6, where the reflex field is CM and the standard construction does not provide the exceptional target needed here. To overcome it, Theorem 3.5.1 constructs a central lift of the E_6-datum and embeds that lift into an E_7-Shimura datum with reflex field $\mathbf{Q}$. Since the closed embedding spreads out to integral canonical models after excluding finitely many primes, the infinite-level perfectoidness can be pulled back from the ambient space.

1.3. **Structure of the paper.** In Section 2, we quickly review some basic facts about prismatic F-gauges, $(\mathcal{G},\mu)$-apertures, and the theory of group-theoretic Hasse invariants via $\mathcal{G}$-zips. In Section 3, we recall integral canonical models and p-Hecke correspondences, establish the relative perfectness of syntomic realization modulo p, briefly discuss bounded quasi-isogenies and minimal compactifications, and prove a result on embeddings of Shimura varieties of exceptional type. In Section 4, we establish our main local result Theorem D and construct Frobenius lift quasi-isogenies. Finally, in Section 5, we apply our result on canonical subgroups to prove Theorem A by mirroring Scholze's method [Sch15].

1.4. **Notations and conventions.**

- Throughout this article, p denotes a prime number.
- (Cyclotomic extensions) We denote by $\mathbf{Z}_p^{\mathrm{cycl}}$ the p-adic completion of $\mathbf{Z}_p[\mu_{p^\infty}]$, and by $\mathbf{Q}_p^{\mathrm{cycl}} := \mathbf{Z}_p^{\mathrm{cycl}}[1/p]$ the cyclotomic extension of $\mathbf{Q}_p$.
- (Completions) For any commutative ring or module M and an ideal $I \subset M$, $\widehat{M} := \varprojlim_n M/I^nM$ denotes its classical I-adic completion. Whenever we refer to a "p-complete" ring, module, formal scheme, or stack, we always mean *derived* p-completeness.
- (Witt vectors and $\mathrm{A}_{\mathrm{inf}}$) For any commutative ring R, we write $W(R)$ for the ring of p-typical Witt vectors with coefficients in R, and $W_n(R) := W(R)/V^nW(R)$ for the ring of n-truncated Witt vectors of length $n \geqslant 1$, equipped with the canonical Frobenius φ and Verschiebung V operators. For a perfectoid ring R, let $R^\flat := \varprojlim_\varphi R/p$ denote its tilt, which is a perfect $\mathbf{F}_p$-algebra of characteristic p. We denote by $\mathrm{A}_{\mathrm{inf}}(R) := W(R^\flat)$ the associated infinitesimal period ring of Fontaine, equipped with its canonical Frobenius φ and Fontaine's map $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$. The kernel of $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$ is a principal ideal generated by a non-zero-divisor ξ, and we let $\widetilde{\xi} := \varphi^{-1}(\xi)$ denote a generator of $\varphi^{-1}(\operatorname{Ker}\theta)$.
- (Reductive groups) For us, reductive group schemes have geometrically connected fibers. For a reductive group scheme $\mathcal{G}$ over a base ring S (such as $\mathbf{Z}_p$, $\mathbf{F}_p$, or an unramified ring of integers $\mathcal{O}$), we denote by $\mathcal{B} \subset \mathcal{G}$ a Borel subgroup scheme and by $\mathcal{T} \subset \mathcal{B}$ a maximal torus. Let $\mathbf{X}_\bullet(\mathcal{T}) := \operatorname{Hom}_S(\mathbf{G}_{\mathrm{m}}, \mathcal{T})$ and $\mathbf{X}^\bullet(\mathcal{T}) := \operatorname{Hom}_S(\mathcal{T}, \mathbf{G}_{\mathrm{m}})$ denote the groups of cocharacters and characters of $\mathcal{T}$, respectively. For a cocharacter $\mu\colon \mathbf{G}_{\mathrm{m}} \to \mathcal{G}$, we denote by $\mathcal{P}_\mu^+ \subset \mathcal{G}$ and $\mathcal{P}_\mu^- \subset \mathcal{G}$ the associated positive and negative parabolic subgroup schemes defined by the dynamic grading of μ and $-\mu$, respectively, with common Levi factor $\mathcal{M}_\mu := \mathcal{P}_\mu^+ \cap \mathcal{P}_\mu^-$. When writing bounds or cocharacters such as $-\nu$, $-\mu$, or $-\lambda$, it is implicitly understood that we take their dominant representatives in the chosen Weyl chamber, i.e., $-w_0\nu$, $-w_0\mu$, and $-w_0\lambda$, where w_0 is the longest element of the Weyl group.
- (Adic spaces) For a p-adic field $K/\mathbf{Q}_p$, we write $\operatorname{Spa} K$ for $\operatorname{Spa}(K, \mathcal{O}_K)$. We use script font $\widehat{\mathscr{S}}, \mathfrak{X}, \dots$ to denote p-adic formal schemes over $\operatorname{Spf} \mathcal{O}_K$, and calligraphic font $\mathcal{S}, \mathcal{X}, \dots$ to denote their generic fibers as analytic adic spaces over $\operatorname{Spa} K$. For an adic space $\mathcal{X}/K$, we denote by $\mathcal{X}^\diamondsuit$ its diamond over $\operatorname{Spd} K$. Finally, for an adic space or a diamond X, we denote by $|X|$ its underlying topological space.

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2. Prismatic Dieudonné theory

`sec:prismatic-dieudonne`

2.1. **Prismatic F-gauges.** We briefly recall some basic facts about cohomological stacks of Bhatt–Lurie and Drinfeld.

Definition 2.1.1 (Quasisyntomic morphisms)**.** A map $R \to S$ of p-complete rings is called **p-quasisyntomic** (or simply **quasisyntomic**) if it is p-completely flat (meaning $S/^{\mathbf{L}}p$ is flat over $R/^{\mathbf{L}}p$), and if the cotangent complex $\mathbf{L}_{S/R}$ has p-complete Tor amplitude in $[-1,0]$ (meaning $\mathbf{L}_{S/R} \otimes^{\mathbf{L}} \mathbf{F}_p$ has Tor amplitude in $[-1,0]$ over $S/^{\mathbf{L}}p$). Furthermore, $R \to S$ is a **quasisyntomic cover** if it is quasisyntomic and $S/^{\mathbf{L}}p$ is faithfully flat over $R/^{\mathbf{L}}p$. These properties are stable under base change along arbitrary morphisms of p-complete rings.

Example 2.1.2. The prototypical example of a quasisyntomic cover is $\mathbf{Z}_p\langle T\rangle \to \mathbf{Z}_p\langle T^{1/p^\infty}\rangle$.

Definition 2.1.3 (Quasisyntomic formal schemes)**.** A p-complete bounded p^∞-torsion ring R is **quasisyntomic** if the structure map $\mathbf{Z}_p \to R$ is quasisyntomic. A p-adic formal scheme X is **quasisyntomic** if it admits an open cover by affine formal schemes $\operatorname{Spf} R$ where each R is a quasisyntomic ring.

Remark 2.1.4 (Relation to classical notions)**.** In the p-complete context, this definition naturally extends the classical algebraic geometry concepts of locally complete intersection (lci) rings and syntomic morphisms—which are traditionally characterized as flat and lci—to the non-Noetherian and non-finite-type realm.

Definition 2.1.5 (Quasiregular semiperfectoid rings)**.** A ring R is called **quasiregular semiperfectoid (qrsp)** if it is a quasisyntomic ring and there exists a perfectoid ring S equipped with a surjective ring homomorphism $S \twoheadrightarrow R$.

Example 2.1.6. Any p-complete bounded p^∞-torsion quotient of a perfectoid ring by a finite regular sequence is quasiregular semiperfectoid.

Proposition 2.1.7. *Let R be a quasisyntomic ring. There exists a quasisyntomic cover $R \to R'$ such that R' is quasiregular semiperfectoid. Moreover, every term in the associated Čech nerve $R'^\bullet$ is a quasiregular semiperfectoid ring.*

Proof. See [BMS19, Lemmas 4.27 and 4.29]. □

`example:perfectoid-covers-of-smooth-rings`

Example 2.1.8 (Perfectoid covers of smooth rings)**.** Let R be a p-completely smooth $\mathbf{Z}_p$-algebra. While R itself is typically not perfectoid, it always admits a quasisyntomic cover by a perfectoid ring. To see this, we can first localize R in the p-completely étale topology to assume that there exists a p-completely étale morphism $\mathbf{Z}_p\langle T_1, \ldots, T_n\rangle \to R$. Letting $\mathbf{Z}_p^{\mathrm{cycl}}$ denote the p-adic completion of $\mathbf{Z}_p[\mu_{p^\infty}]$, we have the standard perfectoid cover $P = \mathbf{Z}_p^{\mathrm{cycl}}\langle T_1^{1/p^\infty}, \ldots, T_n^{1/p^\infty}\rangle_p^\wedge$ of $\mathbf{Z}_p\langle T_1, \ldots, T_n\rangle$ given by adjoining p^∞-th roots of the variables. Base changing along our étale map yields the p-complete tensor product $S = R\widehat{\otimes}_{\mathbf{Z}_p\langle T_1,\ldots,T_n\rangle} P$. Because p-completely étale extensions of perfectoid rings are again perfectoid, S is a perfectoid ring. The induced map $R \to S$ is a quasisyntomic cover.

To any (derived) p-adic formal scheme $\mathfrak{X}$, one can canonically attach three p-adic formal stacks: the **prismatization** $\mathfrak{X}^{\mathbb{\Delta}}$, the **(Nygaard) filtered prismatization** $\mathfrak{X}^{\mathcal{N}}$, and the **syntomification** $\mathfrak{X}^{\mathrm{syn}}$. When $\mathfrak{X} = \operatorname{Spf} R$, we often denote these by $R^{\mathbb{\Delta}}, R^{\mathcal{N}}$, and R^{syn}. These stacks are not classical in general: they are derived stacks that are covered flat locally by the spectra of animated rings.

`rem:prisms_and_prismatization`

Recollection 2.1.9 (Absolute prismatization)**.** The prismatization $\mathfrak{X}^{\mathbb{\Delta}}$ classifies Cartier–Witt divisors on $\mathfrak{X}$. Specifically, given an object $(A, I, \operatorname{Spf} A/I \to \mathfrak{X})$ in the absolute prismatic site $\mathfrak{X}_{\mathbb{\Delta}}$, there is a natural morphism $\iota_{(A,I)} \colon \operatorname{Spf} A \to X^{\mathbb{\Delta}}$. As explained in [BL22, Construction 3.10], this map sends an A-algebra B (with B being p-nilpotent) to the Cartier–Witt divisor $I \otimes_A W(B) \to W(B)$, where $A \to W(B)$ is the unique δ-ring homomorphism lifting the structure map $A \to B$.

`rem:nygaard_filtered_cohomology`

Recollection 2.1.10 (Nygaard filtration)**.** As demonstrated by Bhatt and Lurie, evaluating these stacks on quasiregular semiperfectoid (qrsp) rings yields objects governed by Nygaard-filtered absolute prismatic cohomology. For a qrsp ring R, its **absolute prismatic cohomology** is given by the initial prism $(\mathbb{\Delta}_R, I_R)$. The **Nygaard filtration** $\mathrm{Fil}^\bullet_{\mathcal{N}}\mathbb{\Delta}_R$ on $\mathbb{\Delta}_R$ is given by

$$\mathrm{Fil}^i_{\mathcal{N}}\mathbb{\Delta}_R = \{x \in \mathbb{\Delta}_R \;:\; \varphi(x) \in \mathrm{Fil}^i_{I_R}\mathbb{\Delta}_R\},$$

with $\mathrm{Fil}^\bullet_{I_R}\mathbb{\Delta}_R$ denoting the I_R-adic filtration. By construction, the Nygaard filtration satisfies quasisyntomic descent and preserves sifted colimits.

`rem:filtered-prismatization-syntomification`

Recollection 2.1.11 (Filtered prismatization and syntomification)**.** On a test qrsp ring R, the value of the filtered prismatization $\mathfrak{X}^{\mathcal{N}}$ is canonically identified with the formal Rees stack of the **Nygaard filtered absolute prismatic cohomology** $\mathrm{Fil}^\bullet_{\mathcal{N}}\mathbb{\Delta}_R$; see [GM24, Theorem 6.11.7]. Under this correspondence, j_{dR} corresponds to stripping away the filtration, and j_{HT} is induced by the divided Frobenius map $\mathrm{Fil}^\bullet_{\mathcal{N}}\mathbb{\Delta}_R \to \mathrm{Fil}^\bullet_{I_R}\mathbb{\Delta}_R$. The stack $\mathfrak{X}^{\mathcal{N}}$ is a *filtered stack*, meaning it sits over the formal Artin stack $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}}$ which classifies pairs of a line bundle $\mathcal{L}$ and a cosection $\mathcal{L} \to C$ on p-complete test rings C. The fiber over the open substack $\operatorname{Spf} \mathbf{Z}_p \times B\mathbf{G}_{\mathrm{m}} \subset \operatorname{Spf} \mathbf{Z}_p \times \mathbf{A}^1/\mathbf{G}_{\mathrm{m}}$

is canonically identified with $\mathfrak{X}^{\mathbbl{\Delta}}$. This open substack is referred to as the **de Rham locus** of $X^{\mathcal{N}}$, and we denote its inclusion by $j_{\mathrm{dR}}\colon X^{\mathbbl{\Delta}} \to X^{\mathcal{N}}$. Additionally, there is a second open immersion $j_{\mathrm{HT}}\colon X^{\mathbbl{\Delta}} \to X^{\mathcal{N}}$ whose image is disjoint from the de Rham locus, known as the **Hodge–Tate locus**. The **syntomification** X^{syn} is constructed as the coequalizer of j_{dR} and j_{HT}. Consequently, it comes with a natural open immersion $j_{\mathbbl{\Delta}}\colon X^{\mathbbl{\Delta}} \to X^{\mathrm{syn}}$.

rem:de_rham_point

Recollection 2.1.12 (The de Rham point and its filtered version)**.** There exist canonical morphisms

$$x_{\mathrm{dR}}^{\mathcal{N}}\colon \mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \times \mathfrak{X} \to \mathfrak{X}^{\mathcal{N}} \quad \text{and} \quad x_{\mathrm{dR}}\colon \mathfrak{X} \to \mathfrak{X}^{\mathbbl{\Delta}},$$

where the latter is simply the restriction of the former to $B\mathbf{G}_{\mathrm{m}}$. When evaluated on a test qrsp ring R, $x_{\mathrm{dR}}^{\mathcal{N}}$ translates to the morphism of formal Rees stacks induced by the canonical filtered map from $\mathrm{Fil}_{\mathcal{N}}^{\bullet}\mathbbl{\Delta}_R$ to R, endowing R with the trivial descending filtration. Using the canonical étale map $\mathfrak{X}^{\mathcal{N}} \to \mathfrak{X}^{\mathrm{syn}}$, we get maps $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \times \mathfrak{X} \to \mathfrak{X}^{\mathrm{syn}}$ and $\mathfrak{X} \to \mathfrak{X}^{\mathrm{syn}}$ which we also denote by $x_{\mathrm{dR}}^{\mathcal{N}}$ and x_{dR} respectively. These are called **(filtered) de Rham points**.

Example 2.1.13 (The case of perfectoid rings)**.** For a perfectoid ring R, its absolute prismatic cohomology is given by $\mathrm{A}_{\mathrm{inf}}(R)$ equipped with the Cartier divisor induced by Fontaine's map $\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R$. Thus, the prismatization $R^{\mathbbl{\Delta}}$ is canonically identified with the affine formal scheme $\operatorname{Spf} \mathrm{A}_{\mathrm{inf}}(R)$, and the de Rham point $x_{\mathrm{dR}}\colon \operatorname{Spf} R \to R^{\mathbbl{\Delta}}$ corresponds exactly to the closed immersion defined by θ. The Nygaard filtration is given by $\mathrm{Fil}_{\mathcal{N}}^{i}\mathrm{A}_{\mathrm{inf}}(R) = \widetilde{\xi}^{i}\mathrm{A}_{\mathrm{inf}}(R)$, where $\widetilde{\xi}$ is a generator of $\varphi^{-1}(\operatorname{Ker}\theta)$. The filtered prismatization $R^{\mathcal{N}}$ is the formal Rees stack associated to this filtration, which admits the explicit quotient presentation $R^{\mathcal{N}} \simeq [\operatorname{Spf} \mathrm{A}_{\mathrm{inf}}(R)\{u, v\}/(uv - \widetilde{\xi})/\mathbf{G}_{\mathrm{m}}]$, where the action of $\mathbf{G}_{\mathrm{m}}$ assigns weights 1 and -1 to u and v respectively. The syntomification R^{syn} is a stack whose category of vector bundles $\mathrm{Vect}(R^{\mathrm{syn}})$ is naturally equivalent to the category of Breuil–Kisin–Fargues modules over $\mathrm{A}_{\mathrm{inf}}(R)$, i.e., finite projective $\mathrm{A}_{\mathrm{inf}}(R)$-modules M equipped with an isomorphism $\varphi^*M[1/\widetilde{\xi}] \xrightarrow{\sim} M[1/\widetilde{\xi}]$.

prop:gm-etale-descent

Proposition 2.1.14. *For any p-completely étale map $R \to S$ of p-complete rings, the resulting stack morphisms*

$$S^{\mathcal{N}} \to R^{\mathcal{N}}, \qquad S^{\mathbbl{\Delta}} \to R^{\mathbbl{\Delta}}$$

are (p, I)-completely étale.

Proof. See, for instance, [GM24, Proposition 6.12.1]. □

A key feature of quasisyntomic covers is that taking prismatizations or syntomifications converts them into flat covers.

thm:pentland-quasisyntomic-descent

Theorem 2.1.15 (Quasisyntomic descent)**.** *Let $X \to Y$ be a quasisyntomic cover of bounded quasisyntomic p-adic formal schemes. Then the induced morphism of formal stacks*

$$X^{\mathbbl{\Delta}} \to Y^{\mathbbl{\Delta}}, \quad X^{\mathcal{N}} \to Y^{\mathcal{N}}, \quad \textit{and} \quad X^{\mathrm{syn}} \to Y^{\mathrm{syn}}$$

are surjective in the flat topology.

Proof. See [Pen26, Proposition 2.19, Corollary 2.21]. □

Definition 2.1.16 (Vector bundles in prismatic F-gauges)**.** A **vector bundle in prismatic F-gauges** (or **vector bundle F-gauge**) on $\mathfrak{X}$ is defined to be a vector bundle on the syntomification $\mathfrak{X}^{\mathrm{syn}}$. The category of such objects is denoted by $\mathrm{Vect}\,\mathfrak{X}^{\mathrm{syn}}$.

rem:prismatic_f-crystals

Recollection 2.1.17 (Prismatic F-crystals)**.** Let us recall the definitions of *prismatic crystals* and *prismatic F-crystals* on $\mathfrak{X}$. Consider the structure sheaf $\mathcal{O}_{\mathbbl{\Delta}}$ and the generalized Cartier divisor $\mathcal{I}_{\mathbbl{\Delta}} \to \mathcal{O}_{\mathbbl{\Delta}}$, which on an object $(A, I, \operatorname{Spf} A/I \to \mathfrak{X})$ evaluate to A and $I \to A$, respectively. A **prismatic crystal** of vector bundles on $\mathfrak{X}$ is a locally free $\mathcal{O}_{\mathbbl{\Delta}}$-module $\mathcal{E}$ of finite rank satisfying the crystal condition. A **prismatic F-crystal** consists of a prismatic crystal $\mathcal{E}$ together with an isomorphism

$$\varphi_{\mathcal{E}}\colon \varphi^*\mathcal{E}[1/\mathcal{I}_{\mathbbl{\Delta}}] \xrightarrow{\sim} \mathcal{E}[1/\mathcal{I}_{\mathbbl{\Delta}}].$$

The category of prismatic F-crystals is denoted by $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\mathbbl{\Delta}}, \mathcal{O}_{\mathbbl{\Delta}})$. More concretely, a prismatic crystal assigns a finite locally free A-module $\mathcal{E}(A, I)$ to each prism (A, I) over X, compatible with base change. A prismatic F-crystal structure then corresponds to a family of isomorphisms $\mathcal{E}(A, I)[1/I] \otimes_{A,\varphi} A \xrightarrow{\sim} \mathcal{E}(A, I)[1/I]$ that respect the crystal structure. Similarly, a **mod p^n prismatic F-crystal** is a vector bundle $\mathcal{E}$ on the ringed site $(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ together with an isomorphism $\varphi_{\mathcal{E}}\colon \varphi^*\mathcal{E}[1/\mathcal{I}_{\Delta}] \xrightarrow{\sim} \mathcal{E}[1/\mathcal{I}_{\Delta}]$. We denote the category of mod p^n prismatic F-crystals by $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$.

rem:f-gauges-prismatic

Recollection 2.1.18 (Prismatic F-gauges to prismatic F-crystals)**.** If a vector bundle on $\mathfrak{X}^{\Delta}$ is the pullback of a vector bundle F-gauge along $j_{\Delta}\colon \mathfrak{X}^{\Delta} \to \mathfrak{X}^{\mathrm{syn}}$, the gluing data defining $\mathfrak{X}^{\mathrm{syn}}$ automatically equips it with the structure of a prismatic F-crystal. This establishes a canonical functor

$$\mathrm{Vect}(\mathfrak{X}^{\mathrm{syn}}) \longrightarrow \mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}).$$

The same construction after reduction modulo p^n gives a functor

$$\mathrm{Vect}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \longrightarrow \mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n).$$

rem:tannakian-formalism

Recollection 2.1.19 (Tannakian formalism)**.** Let $\mathcal{G}$ be a smooth affine group scheme over $\mathbf{Z}_p$. Following [IKY24], a **prismatic $\mathcal{G}$-torsor with F-structure** on $\mathfrak{X}$ is a pair $(\mathcal{P}, \varphi_{\mathcal{P}})$, where $\mathcal{P}$ is a $\mathcal{G}$-torsor on the absolute prismatic site $\mathfrak{X}_{\Delta}$ and $\varphi_{\mathcal{P}}\colon (\varphi^*\mathcal{P})[1/\mathcal{I}_{\Delta}] \xrightarrow{\sim} \mathcal{P}[1/\mathcal{I}_{\Delta}]$ is an isomorphism of $\mathcal{G}$-torsors. The category of such objects is denoted by $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta})$.

A **$\mathcal{G}$-bundle in prismatic F-crystals** on $\mathfrak{X}$ is an exact $\mathbf{Z}_p$-linear $\otimes$-functor from the category $\mathrm{Rep}_{\mathbf{Z}_p}(\mathcal{G})$ of representations of $\mathcal{G}$ on finite free $\mathbf{Z}_p$-modules to the category $\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta})$ of prismatic F-crystals on $\mathfrak{X}$. We denote the category of such $\mathcal{G}$-objects by $\mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta})$.

By [IKY24, Proposition 1.28], there is a canonical equivalence

$$\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}) \xrightarrow{\sim} \mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}).$$

The mod p^n versions of these concepts are defined analogously, giving categories $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ and $\mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$. We will also use the finite-level variant

$$\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n) \xrightarrow{\sim} \mathcal{G}\text{-}\mathrm{Vect}^{\varphi}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n).$$

Although this variant is not part of the statement of the cited proposition, it follows from the same proof. Indeed, evaluation on a prism (A, I) turns an exact tensor functor into an exact tensor functor $\mathrm{Rep}_{\mathbf{Z}_p}(\mathcal{G}) \to \mathrm{Vect}(A/p^n)$. Applying [IKY24, Theorem A.19] over $\mathrm{Spec}(A/p^n)$ produces a $\mathcal{G}_{A/p^n}$-torsor. These torsors are compatible with flat descent and with change of prism, and the construction is inverse to taking associated vector bundles. Applying the same argument after inverting I, and then retaining the Frobenius isomorphisms, gives the displayed finite-level equivalence.

def:local-systems-group-schemes

Definition 2.1.20 (Local systems with $\mathcal{G}$-structure)**.** Let $\mathcal{G}$ be a smooth affine group scheme defined over $\mathbf{Z}_p$, and let X be a locally Noetherian scheme or adic space over $\mathbf{Z}_p$. A **$\mathcal{G}(\mathbf{Z}_p)$-local system** over X is defined to be a $\mathcal{G}(\mathbf{Z}_p)$-torsor on the proétale site of X. We denote by $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(X)$ the groupoid of such local systems on X. When X is affine (or affinoid) with ring of global sections R, we will frequently write $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(R)$ in place of $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(X)$. We also employ completely analogous notation for the finite-level variants, replacing the topological group $\mathcal{G}(\mathbf{Z}_p)$ with its finite quotients $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$ to obtain the groupoids $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(X)$ of proétale $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$-local systems.

rem:tannakian-local-systems

Remark 2.1.21 (Tannakian perspective on local systems)**.** By the general principles of Tannakian duality, providing a $\mathcal{G}(\mathbf{Z}_p)$-local system $\mathcal{P}$ over X is equivalent to specifying an exact $\mathbf{Z}_p$-linear symmetric monoidal functor $V \mapsto (V)_{\mathcal{P}}$ from the category $\mathrm{Rep}_{\mathbf{Z}_p}(\mathcal{G})$ of algebraic representations $\mathcal{G} \to \mathrm{GL}(V)$ on finite free $\mathbf{Z}_p$-modules to the category $\mathrm{Loc}_{\mathbf{Z}_p}(X)$ of finite locally free $\mathbf{Z}_p$-local systems on X; see [IKY24, Propositions 2.3 and 2.8]. Furthermore, the cited results demonstrate that this equivalence restricts cleanly to finite levels: specifying a $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$-local system $\mathcal{P}_n$ over X is precisely equivalent to providing an exact $\mathbf{Z}_p$-linear symmetric monoidal functor $\mathrm{Rep}_{\mathbf{Z}_p}(\mathcal{G}) \to \mathrm{Loc}_{\mathbf{Z}/p^n\mathbf{Z}}(X)$.

rem:etale-realization

Recollection 2.1.22 (Étale realization)**.** Let $\mathfrak{X}$ be a flat quasisyntomic p-adic formal scheme. Let $\mathcal{G}$ be a smooth affine group scheme over $\mathbf{Z}_p$. Write $\mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n)$ for the category of mod p^n prismatic $\mathcal{G}$-torsors with F-structure on $\mathfrak{X}$. We claim that, by Recollection 2.1.18 and the Tannakian formalism (Recollection 2.1.19), there is a natural functor

$$G\colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \to \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n).$$

Concretely, G pulls a torsor back to $\mathfrak{X}^{\Delta} \otimes \mathbf{Z}/p^n\mathbf{Z}$; the gluing defining $\mathfrak{X}^{\mathrm{syn}}$ supplies its Frobenius isomorphism after inverting $\mathcal{I}_{\Delta}$.

The finite-coefficient Laurent Riemann–Hilbert correspondence, obtained from [BS23, Corollary 3.7] by reduction modulo p^n, then gives the composition

$$\mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \xrightarrow{G} \mathbf{Tors}^{\varphi}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}/p^n) \xrightarrow{(-)[1/\mathcal{I}_{\Delta}]} \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}_{\Delta}, \mathcal{O}_{\Delta}[1/\mathcal{I}_{\Delta}]/p^n)^{\varphi^*=1} \simeq \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(\mathfrak{X}^{\mathrm{ad}}),$$

where the final equivalence uses the finite-level Tannakian argument of Recollection 2.1.19. This agrees with the direct construction of [MY26, Construction 3.8.1]. It is the **étale realization** $T_{\text{ét}} \colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}) \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})}(\mathfrak{X}^{\mathrm{ad}})$. Taking the inverse limit over n, we obtain the étale realization functor for $\mathcal{G}$-torsors on $\mathfrak{X}^{\mathrm{syn}}$:

$$T_{\text{ét}} \colon \mathbf{Tors}_{\mathcal{G}}(\mathfrak{X}^{\mathrm{syn}}) \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(\mathfrak{X}^{\mathrm{ad}}).$$

rem:modification-of-syntomic-torsors

Remark 2.1.23 (Modification of syntomic torsors)**.** Defining a modification of $\mathcal{G}$-torsors over $\mathfrak{X}^{\mathrm{syn}}$ along '$p = 0$' requires some care.

(1) **Evaluating on prisms**: By Recollection 2.1.19, a $\mathcal{G}$-torsor over $\mathfrak{X}^{\mathrm{syn}}$ gives a $\mathcal{G}$-bundle in prismatic F-crystals. For any p-torsion free prism $(A, I) \in (\mathfrak{X})_{\Delta}$, A is a genuine commutative ring where p is a non-zero divisor, making $A[1/p]$ well-defined. A *modification* of $\mathcal{G}$-torsors $\mathcal{P}_1 \dashrightarrow \mathcal{P}_2$ over $\mathfrak{X}^{\mathrm{syn}}$ can be defined as a compatible system of $\mathcal{G}$-torsor isomorphisms $\mathcal{P}_1(A, I)[1/p] \xrightarrow{\sim} \mathcal{P}_2(A, I)[1/p]$ across all p-torsion free prisms $(A, I) \in (\mathfrak{X})_{\Delta}$.
(2) **Pushforwards**: Alternatively, one can also avoid inverting p entirely. Given a cocharacter λ of $\mathcal{G}$ defined over $\mathbf{Z}_p$ and a $\mathcal{G}$-torsor $\mathcal{P}_1$ over $\mathfrak{X}^{\mathrm{syn}}$, a modification of constant type λ can be defined if $\mathcal{P}_1$ admits a reduction of structure to the group scheme $\mathcal{H}_\lambda$ defined in Remark 2.2.10 (using $d = p$). Specifically, the modified $\mathcal{G}$-torsor is defined as the pushforward $\mathcal{P}_2 := \mathcal{P}_1^- \times^{\mathcal{H}_\lambda, c_\lambda} \mathcal{G}$ (cf. Remark 2.2.10). By construction, this modification is of constant type λ.
(3) **Vinberg monoids**: The Vinberg monoid gives an algebraic repackaging of the previous approach and may be viewed as a 'local model' for the space of modifications. A brief explanation appears in Section 2.6.

These perspectives are compatible and give us a good notion of modification of syntomic torsors at least when $\mathfrak{X}$ is flat and quasisyntomic. Furthermore, a modification of $\mathcal{G}$-torsors over $\mathfrak{X}^{\mathrm{syn}}$ naturally induces an isomorphism between their associated $\mathcal{G}(\mathbf{Q}_p)$-local systems over $\mathfrak{X}^{\mathrm{ad}}$. This follows essentially by construction of the étale realization functor.

2.2. $(\mathcal{G}, \mu)$-apertures.

We recall the main results of [GM24]. For more details, see *loc. cit.* or [MY26, §3].

set:main

Setup 2.2.1. Let $\mathcal{G}$ be a smooth connected affine group scheme over $\mathbf{Z}_p$, $\mathcal{O}$ the ring of integers in a finite unramified extension of $\mathbf{Q}_p$, and $\mu \colon \mathbf{G}_{\mathrm{m},\mathcal{O}} \to \mathcal{G}_{\mathcal{O}}$ a 1-bounded cocharacter. This means that the weights on $\operatorname{Lie}\mathcal{G}_{\mathcal{O}}$ via the adjoint action are bounded above by 1. Let $\mathcal{P}_\mu^- \subset \mathcal{G}$ (resp. $\mathcal{P}_\mu^+ \subset \mathcal{G}$) be the smooth subgroup scheme whose Lie algebra is identified with the sum of the weight-i spaces in $\operatorname{Lie}\mathcal{G}_{\mathcal{O}}$ for $i \leqslant 0$ (resp. $i \geqslant 0$). In practice, $\mathcal{G}$ will be reductive in which case this is same as saying μ is miniscule and $\mathcal{P}_\mu^{\pm}$ is a pair of opposite parabolic subgroups associated with μ.

Definition 2.2.2 ($(\mathcal{G}, \mu)$-apertures)**.** For any p-complete (animated) $\mathcal{O}$-algebra R, an **n-truncated $(\mathcal{G}, \mu)$-aperture** over R is a $\mathcal{G}$-torsor $\mathcal{E}$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ satisfying the following condition: for every geometric point $R \to \kappa$, the restriction of $(x^{\mathcal{N}}_{\mathrm{dR}})^*\mathcal{E}$ over $B\mathbf{G}_{\mathrm{m}} \times \operatorname{Spec}\kappa$ is isomorphic to that of $\mathcal{Q}_\mu$, where $\mathcal{Q}_\mu$ is the $\mathcal{G}$-torsor over $B\mathbf{G}_{\mathrm{m},\mathcal{O}}$ classified by $B\mu \colon B\mathbf{G}_{\mathrm{m},\mathcal{O}} \to B\mathcal{G}_{\mathcal{O}}$. The ∞-groupoid of n-truncated apertures over R is denoted $\mathrm{BT}_n^{\mathcal{G},\mu}(R)$, and the assignment $R \mapsto \mathrm{BT}_n^{\mathcal{G},\mu}(R)$ is a (derived) p-adic formal prestack. Also define

$$\mathrm{BT}_\infty^{\mathcal{G},\mu} := \varprojlim_n \mathrm{BT}_n^{\mathcal{G},\mu}.$$

Objects in $\mathrm{BT}_\infty^{\mathcal{G},\mu}(R)$ are called **(∞-truncated) $(\mathcal{G}, \mu)$-apertures** over R.

thm:gm:main

Theorem 2.2.3. *The p-adic formal prestack $\mathrm{BT}_n^{\mathcal{G},\mu}$ is represented by a 0-dimensional quasi-compact smooth p-adic formal Artin stack over $\mathcal{O}$ with affine diagonal. Furthermore, the natural map $\mathrm{BT}_{n+1}^{\mathcal{G},\mu} \to \mathrm{BT}_n^{\mathcal{G},\mu}$ is smooth and surjective.*

Proof. See [GM24, Theorem D]. □

example:BT-for-GLn

Example 2.2.4 (The case of GL_n)**.** Consider the case when $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$ where $d \leqslant h$ and μ_d is the miniscule cocharacter defined over $\mathbf{Z}_p$ given by the diagonal matrix

$$\mu_d(z) = \mathrm{diag}(\underbrace{z, z, \ldots, z}_{h}, \underbrace{1, \ldots, 1}_{h-d}).$$

It is shown in [GM24, Theorem 11.2.7] that $\mathrm{BT}_n^{\mathrm{GL}_h,\mu_d}$ is canonically isomorphic to the moduli stack of n-truncated Barsotti-Tate groups of height h and dimension d. Hence, Theorem 2.2.3 recovers a classical result of Grothendieck.

Definition 2.2.5 (Breuil–Kisin frames)**.** Let R be a p-complete ring. A **Breuil–Kisin frame** for R is a prism $\underline{A} = (A, I)$ such that A is p-complete and p-torsion free, $I \subset A$ is an ideal and $R = A/I$.

Assumption 2.2.6. From now on, the underlying ideal I for Breuil–Kisin frames $\underline{A} = (A, I)$ we consider is going to be principal. This is a harmless assumption because any prism admits a faithfully flat cover by an orientable prism.

Example 2.2.7 (Frames for perfectoid rings)**.** If R is a perfectoid ring, then $\underline{\mathbb{\Delta}}_R := (\mathbb{\Delta}_R, \mathrm{Fil}^1_{\mathcal{N}}\mathbb{\Delta}_R)$ is a Breuil–Kisin frame for R. We call this the $\mathrm{A}_{\mathrm{inf}}$ **frame**.

example:classical-breuil-kisin-frames

Example 2.2.8 (Classical Breuil–Kisin frames)**.** Let K be a finite totally ramified extension of an unramified field $K_0/\mathbf{Q}_p$. Let π be a uniformizer of $\mathcal{O}_K$ with minimal polynomial $E(u) \in \mathcal{O}_{K_0}[u]$ over K_0. The classical Breuil–Kisin frame for $\mathcal{O}_K$ is the prism $\underline{\mathfrak{S}} = (\mathfrak{S}, (E(u)))$, where $\mathfrak{S} := \mathcal{O}_{K_0}[\![u]\!]$ is equipped with the Frobenius φ acting as the natural Frobenius on $\mathcal{O}_{K_0}$ and satisfying $\varphi(u) = u^p$. The structure map $\mathfrak{S} \to \mathcal{O}_K$ evaluates u at π.

rem:dilatations

Remark 2.2.9 (Dilatations)**.** Let R be a ring a $d \in R$ a non-zerodivisor. Let $\mathcal{G}$ be a flat affine group scheme over R, and let $H_N \hookrightarrow \mathcal{G}_{R/d^N R}$ be a closed subgroup scheme, $N \geqslant 1$. The dilatation of $\mathcal{G}$ along H_N is the flat affine R-group scheme $\mathcal{G}^{[H_N]}$ characterized by the following universal property: for every flat R-algebra A,

$$\mathcal{G}^{[H_N]}(A) = \{\, g \in \mathcal{G}(A) \mid g \bmod d^N \in H_N(A/d^N A) \,\}.$$

The structural group morphism $\mathcal{G}^{[H_N]} \longrightarrow \mathcal{G}$ is an isomorphism over $\operatorname{Spec} R[1/d]$. This is the Néron blow-up, or dilatation, of $\mathcal{G}$ along H_N; see [BLR90, §3.2].

rem:modification

Remark 2.2.10 (Modifications via pushforwards)**.** Let $\mathcal{G}$ be a reductive group scheme over R and $d \in R$ a non-zerodivisor. A **modification** of $\mathcal{G}$-torsors is a triple $(\mathcal{T}_1, \mathcal{T}_2, m)$, where $\mathcal{T}_1$ and $\mathcal{T}_2$ are $\mathcal{G}$-torsors on $\operatorname{Spec} R$ and $m : \mathcal{T}_1|_{d\neq 0} \xrightarrow{\sim} \mathcal{T}_2|_{d\neq 0}$ is an isomorphism of $\mathcal{G}$-torsors. After fixing $\mathcal{T}_1$, isomorphism classes of such pairs $(\mathcal{T}_2, m)$ are equivalently sections of the associated affine Grassmannian fibration $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G}}$, $\mathrm{Gr}_{\mathcal{G}} = L\mathcal{G}/L^+\mathcal{G}$, where $L\mathcal{G}(A) = \mathcal{G}(A[1/d])$ and $L^+\mathcal{G}(A) = \mathcal{G}(A)$. If λ is a dominant cocharacter for which the Schubert bound $\mathrm{Gr}_{\mathcal{G},\leqslant\lambda} \subset \mathrm{Gr}_{\mathcal{G}}$ is defined, then the modification is bounded by λ precisely when the associated section factors through $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G},\leqslant\lambda}$. The relative position may vary in the Schubert stratification of $\mathrm{Gr}_{\mathcal{G},\leqslant\lambda}$, taking values $\lambda' \leqslant \lambda$.

There is, however, a canonical group scheme construction for the open stratum. Suppose that $\lambda : \mathbf{G}_{m,R} \to \mathcal{G}$ is defined over R. The image $\lambda(d)$ lies in $\mathcal{G}(R[1/d])$. Define a group scheme $\mathcal{H}_\lambda$ characterised by: for every flat R-algebra A,

$$\mathcal{H}_\lambda(A) = \{\, h \in \mathcal{G}(A) \mid \lambda(d) h \lambda(d)^{-1} \in \mathcal{G}(A) \,\}.$$

Then the inclusion $i : \mathcal{H}_\lambda \hookrightarrow \mathcal{G}$ and the conjugation morphism

$$c_\lambda = \operatorname{Int} \lambda(d) : \mathcal{H}_\lambda \longrightarrow \mathcal{G}$$

are homomorphisms of R-group schemes. If $\mathcal{P}$ is a right $\mathcal{H}_\lambda$-torsor, then

$$\mathcal{T}_1 = \mathcal{P} \times^{\mathcal{H}_\lambda, i} \mathcal{G}, \qquad \mathcal{T}_2 = \mathcal{P} \times^{\mathcal{H}_\lambda, c_\lambda} \mathcal{G}$$

are $\mathcal{G}$-torsors, and over $\operatorname{Spec} R[1/d]$ there is a canonical isomorphism

$$m_\lambda : \mathcal{T}_1|_{d\neq 0} \xrightarrow{\sim} \mathcal{T}_2|_{d\neq 0}, \qquad [p, g] \longmapsto [p, \lambda(d) g].$$

This modification is, étale locally on $\operatorname{Spec} R$, the standard modification of relative position λ.

Conversely, a modification whose associated section of $\mathcal{T}_1 \times^{\mathcal{G}} \mathrm{Gr}_{\mathcal{G},\leqslant\lambda}$ factors through the open Schubert stratum $\mathrm{Gr}_{\mathcal{G},\lambda}$ is recovered from the $\mathcal{H}_\lambda$-torsor of isomorphisms with the standard modification. To conclude, the data of a modification $(\mathcal{T}_1, \mathcal{T}_2, m)$ of constant type λ is equivalent to the data of a reduction of structure of $\mathcal{T}_1$ to $\mathcal{H}_\lambda$.

rem:parabolic-dilatations

Remark 2.2.11 (Relation with parabolic dilatations)**.** Assume $\mathcal{G}$ is a reductive group scheme over R and $\lambda\colon \mathbf{G}_{m,R} \to \mathcal{G}$ is a cocharacter defined over R. Let $\mathcal{P}_\lambda^+ \subset \mathcal{G}$ (resp. $\mathcal{U}_\lambda^-$) be the parabolic subgroup (resp. opposite unipotent radical) corresponding to the non-negative (resp. negative) weights of $\mathrm{Ad}(\lambda)$.

The group scheme $\mathcal{H}_\lambda$ of Remark 2.2.10 can be explicitly described via its effect on the Lie algebra. The conjugation action of $\lambda(d)$ scales the weight i subspace $(i < 0)$ of $\mathrm{Lie}(\mathcal{U}_\lambda^-)$ by d^i. Consequently, $\mathcal{H}_\lambda$ is the unique flat group model over R that leaves $\mathcal{P}_\lambda^+$ unmodified while replacing each weight i root subgroup $U_i \subset \mathcal{U}_\lambda^-$ with its dilatation along the identity of depth $-i$.

Let $-N$ be the minimum weight of $\mathrm{Ad}(\lambda)$ on $\mathrm{Lie}(\mathcal{G})$. The depth-$N$ parabolic dilatation $\mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^NR}]}$ imposes d^N-divisibility on the entirety of $\mathcal{U}_\lambda^-$. Therefore, as subfunctors of $\mathcal{G}$, we have

$$\mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^NR}]} \subset \mathcal{H}_\lambda \subset \mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/dR}]},$$

with equality on the left if and only if all negative weights of λ are $-N$. In particular, if $\lambda = N\mu$ for a minuscule cocharacter μ, then $\mathcal{H}_\lambda = \mathcal{G}^{[(\mathcal{P}_\lambda^+)_{R/d^NR}]}$. Thus, when λ is miniscule, a modification bounded by λ is precisely the data of a reduction of structure of $\mathcal{T}_1|_{d=0}$ to $\mathcal{P}_\lambda^+$. In general, an $\mathcal{H}_\lambda$-torsor canonically induces a $\mathcal{P}_\lambda^+$-reduction only modulo d. Modulo d^N, the induced reduction is to a subgroup scheme intermediate between $(\mathcal{P}_\lambda^+)_{R/d^NR}$ and $\mathcal{G}_{R/d^NR}$.

def:windows-over-frames

Definition 2.2.12 ($(\mathcal{G},\mu)$-windows over frames)**.** Let $\underline{A} = (A, I)$ be a Breuil-Kisin frame for R equipped with a distinguished generator $E \in I$. An n**-truncated** $(\mathcal{G},\mu)$**-window** over $\underline{A}$ is a triple $(\mathcal{Q}, Q^-, \alpha)$ where $\mathcal{Q}$ is a $\mathcal{G}$-torsor over $A/^{\mathbf{L}}p^nA$, $Q^- \subset Q := \mathcal{Q}|_{R/^{\mathbf{L}}p^nR}$ is a reduction of structure group of Q to a $\mathcal{P}_\mu^-$-torsor, and $\alpha\colon \varphi^*\mathcal{Q}' \xrightarrow{\sim} \mathcal{Q}$ is an isomorphism of $\mathcal{G}$-torsors, where $\mathcal{Q}'$ is the modification of $\mathcal{Q}$ along Q^-. A(n ∞**-truncated)** $(\mathcal{G},\mu)$**-window** over $\underline{A}$ is the data of a compatible system of n-truncated $(\mathcal{G},\mu)$-windows over $\underline{A}$ for all $n \geqslant 1$.

rem:alternate-description-via-modifications

Remark 2.2.13 (Description in terms of modifications)**.** In the notations of Definition 2.2.12, assume A and R are p-torsion free. Then $A/^{\mathbf{L}}p^nA$ and $R/^{\mathbf{L}}p^nR$ can be replaced with A/p^nA and R/p^nR in the above definition respectively, and an n-truncated $(\mathcal{G},\mu)$-window over $\underline{A}$ can be equivalently described as a pair $(\mathcal{E}, m)$ consisting of a $\mathcal{G}$-torsor $\mathcal{E}$ over A/p^nA and a modification

$$m\colon \varphi^*\mathcal{E}[1/I] \xrightarrow{\sim} \mathcal{E}[1/I]$$

of type μ. A(n ∞-truncated) $(\mathcal{G},\mu)$-window over $\underline{A}$ can be described analogously. In practice, we prefer to use this description of windows.

Terminology 2.2.14. When R is a perfectoid ring, by a(n n-truncated) $(\mathcal{G},\mu)$-window over R, we always mean a(n n-truncated) $(\mathcal{G},\mu)$-window over the corresponding $\mathrm{A}_{\mathrm{inf}}$ frame.

prop:apertures-over-perfectoid-rings

Theorem 2.2.15 (Apertures over perfectoid rings are windows)**.** *Let R be a perfectoid ring and $n \in \mathbf{Z}_{\geqslant 1} \cup \{\infty\}$. There is canonical equivalence between the groupoid $\mathrm{BT}_n^{\mathcal{G},\mu}(R)$ and the groupoid of n-truncated $(\mathcal{G},\mu)$-windows over R.*

Proof. See [GM24, Lemma 9.2.3]. □

Remark 2.2.16 (Quotient description over perfectoid rings)**.** Let R be a p-torsion free perfectoid ring and ξ a generator of $\mathrm{Ker}(\theta\colon \mathrm{A}_{\mathrm{inf}}(R) \to R)$. Denote by $\overline{\xi}$ the image of ξ in $W_n(R^\flat)$. By Theorem 2.2.15, the stack $\mathrm{BT}_n^{\mathcal{G},\mu}$ restricted to perfectoid R-algebras is the (p-completely étale) stackification of the prestack given by sending a perfectoid R-algebra S to the action groupoid of $\mathcal{G}(W_n(S^\flat))$ acting on the double coset $\mathcal{G}(W_n(S^\flat))\mu(\overline{\xi})\mathcal{G}(W_n(S^\flat))$ via φ-conjugation: $g \in \mathcal{G}(W_n(S^\flat))$ acts on an element C by $g \cdot C = g^{-1}C\varphi(g)$. One gets an analogous description for $\mathrm{BT}_\infty^{\mathcal{G},\mu}$ restricted to perfectoid R-algebras by taking the limit $n \to \infty$.

example:bkf-modules

Example 2.2.17 (Breuil–Kisin–Fargues modules)**.** Let R be a p-torsion free perfectoid ring. When $(\mathcal{G},\mu) = (\mathrm{GL}_h, \mu_d)$, Theorem 2.2.15 combined with Example 2.2.4 and Remark 2.2.13 gives the familiar classification of p-divisible groups of height h and dimension d over R in terms of locally free modules M of rank h over $\mathrm{A}_{\mathrm{inf}}(R)$

equipped with an injective $\mathrm{A}_{\mathrm{inf}}(R)$-linear map $\Phi\colon \varphi^* M \to M$ whose cokernel is supported on R and locally free of rank d. See [Lau18] for details.

2.3. **$\mathcal{G}$-zips.** We review the theory of $\mathcal{G}$-zips as in [PWZ15] and their relationship with $(\mathcal{G}, \mu)$-apertures. set:gzips

Setup 2.3.1. Let k be the residue field of $\mathcal{O}$ and φ the Frobenius lift on $\mathcal{O}/\mathbf{Z}_p$. Let $\mathcal{P}^+_{\varphi(\mu)} \subset \mathcal{G}$ be the smooth subgroup Because $\mathcal{G}$ is defined over $\mathbf{Z}_p$, the Frobenius φ acts naturally on the cocharacter lattice of $\mathcal{G}_{\mathcal{O}}$. We denote by $\varphi(\mu)$ the Galois twist of the cocharacter μ by φ. Let $\mathcal{P}^+_{\varphi(\mu)} \subset \mathcal{G}$ be the smooth subgroup scheme whose Lie algebra is identified with the sum of the weight-i spaces in $\operatorname{Lie} \mathcal{G}_{\mathcal{O}}$ for $\varphi(\mu)$ with $i \geqslant 0$. Let $\mathcal{U}^-_\mu$ (resp. $\mathcal{U}^+_{\varphi(\mu)}$) be the smooth subgroup scheme of $\mathcal{P}^-_\mu$ (resp. $\mathcal{P}^+_{\varphi(\mu)}$) whose Lie algebra is the sum of the weight-i spaces for μ (resp. for $\varphi(\mu)$) with $i < 0$ (resp. $i > 0$). Set $\mathcal{M}_\mu = \mathcal{P}^-_\mu/\mathcal{U}^-_\mu$ and $\mathcal{M}_{\varphi(\mu)} = \mathcal{P}^+_{\varphi(\mu)}/\mathcal{U}^+_{\varphi(\mu)}$. We have a canonical isomorphism of group schemes $\varphi^*\mathcal{M}_\mu \xrightarrow{\sim} \mathcal{M}_{\varphi(\mu)}$.

Definition 2.3.2 ($\mathcal{G}$-zips)**.** Let R be a k-algebra. An **F-zip with $\mathcal{G}$-structure** or **$\mathcal{G}$-zip** of type μ over R (see [PWZ15, Definition 1.4]) is a tuple

$$(\mathcal{Q}, \mathcal{Q}^-, \mathcal{Q}^+, \alpha)$$

where $\mathcal{Q}$ is a $\mathcal{G}$-torsor over R, $\mathcal{Q}^- \subset \mathcal{Q}$ is a reduction of structure to a $\mathcal{P}^-_\mu$-torsor, $\mathcal{Q}^+ \subset \mathcal{Q}$ is a reduction of structure to a $\mathcal{P}^+_{\varphi(\mu)}$-torsor and

$$\alpha\colon \varphi^*(\mathcal{Q}^-/\mathcal{U}^-_\mu) \xrightarrow{\sim} \mathcal{Q}^+/\mathcal{U}^+_{\varphi(\mu)}$$

is an isomorphism of $\mathcal{M}_{\varphi(\mu)}$-torsors. We denote the groupoid of such objects by $\mathcal{G}\text{-zip}_\mu(R)$.

Example 2.3.3. The ith de Rham cohomology group of any smooth projective scheme X over a k-algebra R with degenerating Hodge-to-de Rham spectral sequence, when equipped with its decreasing Hodge filtration, its increasing conjugate filtration, and Cartier isomorphism, yields an example of a GL_n-zip over R where $n = \operatorname{rank}_R \mathrm{H}^i_{\mathrm{dR}}(X/R)$.

thm:gzip-stack

Theorem 2.3.4 (Stack of $\mathcal{G}$-zips)**.** *The assignment $R \mapsto \mathcal{G}\text{-zip}_\mu(R)$ for k-algebras R is a smooth 0-dimensional Artin stack over k, which can be presented as $[E\backslash\mathcal{G}_k]$ where*

$$E := \{(a, b) \in \mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\varphi(\mu),k} \colon \varphi(a \bmod \mathcal{U}^-_{\mu,k}) = b \bmod \mathcal{U}^+_{\varphi(\mu),k}\}$$

and the action is given by $(a, b) \cdot g = agb^{-1}$.

Proof. See [PWZ15, Theorem 1.5]. □

The following theorem follows from the proof of [GM24, Theorem 9.3.2]: thm:gzip-realization

Theorem 2.3.5 ($\mathcal{G}$-zip realization)**.** *There is a canonical smooth surjective map*

$$\mathrm{BT}_1^{\mathcal{G},\mu} \otimes_{\mathcal{O}} k \to \mathcal{G}\text{-zip}_\mu$$

that is a gerbe for a connected finite locally free commutative group scheme killed by p.

Remark 2.3.6. Note that $\mathcal{G}\text{-zip}_\mu$ only depends on the geometric conjugacy class of μ_k.

set:dominant

Setup 2.3.7. In what follows, $\mathcal{G}$ is *reductive*. Choose a maximal torus $\mathcal{T}$ contained in a Borel subgroup $\mathcal{B} \subset \mathcal{G}$ such that μ factors through $\mathcal{T}_{\mathcal{O}}$ and is dominant. This is always possible by conjugating μ by an element of $\mathcal{G}(\mathcal{O})$. rem:gzip-stratification

Remark 2.3.8 (Ekedahl–Oort stratification)**.** The underlying topological space of the stack $\mathcal{G}\text{-zip}_\mu$, denoted $|\mathcal{G}\text{-zip}_\mu|$, is a finite poset consisting of the orbits of the E-action on $\mathcal{G}_k$. These orbits define the **Ekedahl–Oort stratification** on $\mathcal{G}\text{-zip}_\mu$ by smooth locally closed substacks. Let W be the Weyl group of $\mathcal{G}_k$ and let $W_{\mathcal{M}_\mu}$ be the Weyl group of $\mathcal{M}_\mu$. The strata are parameterized by the set ${}^{\mathcal{M}_\mu}W$ of minimal-length representatives for the quotient $W_{\mathcal{M}_\mu}\backslash W$. The closure relations in the topology of $|\mathcal{G}\text{-zip}_\mu|$ corresponds to a partial order on ${}^{\mathcal{M}_\mu}W$ which in general is strictly finer than the Bruhat order– see [PWZ11, §6.3]. See [PWZ15, Theorem 3.20, Remark 3.21] for more details.

Definition 2.3.9 (The μ-ordinary stratum)**.** In the context of Remark 2.3.8, the unique open dense stratum of $\mathcal{G}\text{-zip}_\mu$ is called the **μ-ordinary stratum** or the **open zip stratum**, and we denote it by $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$.

Remark 2.3.10. The μ-ordinary stratum can be described as the zip stratum corresponding to the longest element of ${}^{\mathcal{M}_\mu}W$ or the image of E-orbit of the identity element. The latter statement is true because $\mathcal{B} \subseteq \mathcal{P}^+_{\varphi(\mu)}$ and $\mathcal{P}^-_\mu$ contains a Borel subgroup opposite to $\mathcal{B}$.

rem:p-rank-stratification

Remark 2.3.11 (p-rank stratification)**.** Assume μ is defined over $\mathbf{Z}_p$ so that $\mu = \varphi(\mu)$. By the presentation in Theorem 2.3.4, there is a natural map

$$\pi\colon \mathcal{G}\text{-zip}_\mu \longrightarrow [\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k} \backslash \mathcal{G}_k],$$

where the stack quotient in the target is formed via the action $(a, b) \cdot g = agb^{-1}$. The open dense big cell

$$\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subseteq \mathcal{G}_k$$

defines an open dense substack of the target, and the induced stratification of $\mathcal{G}_k$ is much coarser than the Ekedahl–Oort stratification: its strata are indexed by the double cosets

$$W_{\mathcal{M}_\mu} \backslash W / W_{\mathcal{M}_\mu},$$

and, in the context of Siegel Shimura varieties, this recovers the classical p-rank stratification of the special fibre. The preimage of the open dense stratum under π is exactly the μ-ordinary stratum $\mathcal{G}\text{-zip}^{\mathrm{ord}}_\mu$. Indeed, by Lang's theorem, the E-orbit of $1 \in \mathcal{G}_k$ is exactly $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k}$. On the complement of the big cell, however, many distinct Ekedahl–Oort strata may lie in a single double-coset stratum, and π is far from being a topological bijection.

rem:computing-gzip-realization

Remark 2.3.12 (Computing $\mathcal{G}$-zip realization over perfectoid rings)**.** Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G}, \mu)$-window over a p-torsion free perfectoid ring R. We explain how to access the $\mathcal{G}$-zip realization of the associated 1-truncated aperture obtained by base changing along $R \to R/pR$. Because m_1 is a modification of $\mathcal{E}_1$ of type μ, it induces a canonical reduction of structure $\mathcal{F}^- \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ to $\mathcal{P}^-_\mu$, called the **de Rham realization** and denoted $T_{\mathrm{dR}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}^-_\mu$. Similarly, the modification $(\sigma^{-1})^* m^{-1}$ gives a reduction of structure $\mathcal{F}^+ \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ to $\mathcal{P}^+_{\varphi(\mu)}$, called the **conjugate realization** and denoted $T_{\mathrm{conj}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}^+_{\varphi(\mu)}$. The isomorphism α of $\mathcal{M}_{\varphi(\mu)}$ is then induced by m_1. The tuple $(\mathcal{E}_1|_{R^\flat/tR^\flat}, \mathcal{F}^-, \mathcal{F}^+, \alpha)$ is a $\mathcal{G}$-zip of type μ over $R^\flat/tR^\flat \simeq R/pR$ and induces a morphism $\operatorname{Spec} R/pR \to \mathcal{G}\text{-zip}_\mu$.

rem:computing-gzip-realization-via-quotient-presentation

Remark 2.3.13 (Computing $\mathcal{G}$-zip realization via quotient presentation)**.** Assume that $\mathcal{E}_1$ is trivialized, and the modification m_1 is represented by $\mu(t)M \in \mathcal{G}(R^\flat[1/t])$ for some matrix $M \in \mathcal{G}(R^\flat)$. Under the stack quotient presentation $\mathcal{G}\text{-zip}_\mu \simeq [\mathcal{G}_k/E]$ defined in Theorem 2.3.4, the $\mathcal{G}$-zip associated to $(\mathcal{E}_1, m_1)$ corresponds exactly to the reduction $\overline{M} \in \mathcal{G}(R^\flat/tR^\flat)$ of M modulo t, defined upto E-action. In particular, if $(\mathcal{E}_1, m_1)$ is μ-ordinary, $\overline{M}$ lands in the open dense big cell $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subset \mathcal{G}_k$ associated to the μ-ordinary stratum.

rem:filtered-de-rham-realization

Remark 2.3.14 (Filtered de Rham realization)**.** Suppose $\mathcal{A} \in \mathrm{BT}^{\mathcal{G},\mu}_\infty(R)$. Then pulling back by the Hodge-filtered de Rham point $x^{\mathcal{N}}_{\mathrm{dR}}$ from Recollection 2.1.12, we get a $\mathcal{G}$-torsor over $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \times \operatorname{Spf} R$. The local triviality condition in the definition of aperture says that giving such a $\mathcal{G}$-torsor over $\mathbf{A}^1/\mathbf{G}_{\mathrm{m}} \times \operatorname{Spf} R$ is equivalent to giving a $\mathcal{P}^-_\mu$-torsor over R. Thus, we get a canonical classifying map

$$\mathrm{BT}^{\mathcal{G},\mu}_\infty \to B\mathcal{P}^-_\mu$$

which we call the **filtered de Rham realization**.

2.4. **Hasse invariants.** Group-theoretic Hasse invariants were constructed by Goldring and Koskivirta in [GK19]. These are sections of line bundles on $\mathcal{G}\text{-zip}_\mu$ whose nonvanishing loci cut out specific Ekedahl–Oort strata (Remark 2.3.8). Because this construction is intrinsic to $\mathcal{G}\text{-zip}_\mu$, it uniformly defines Hasse invariants for special fibers of arbitrary Shimura varieties via Theorems 2.3.5 and 3.1.5.

Setup 2.4.1. Maintain Setups 2.3.1 and 2.3.7 and the notations of Theorem 2.3.4.

rem:line-bundles-on-gzip

Remark 2.4.2 (Line bundles)**.** Any character $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ inflates to a character of $\mathcal{P}^-_{\mu,k}$, and by restriction via the first projection, defines a character of the zip group E. This induces a line bundle on $\mathcal{G}\text{-zip}_\mu$, which we denote by $\mathcal{V}(\chi)$.

Definition 2.4.3 (Hasse invariants)**.** Let $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ be a character. A (μ-ordinary) **Hasse invariant** of weight χ is a global section
$$h \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$$
whose non-vanishing locus is exactly the open dense μ-ordinary stratum $\mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}$.

Remark 2.4.4 (Hasse invariants are unique upto scaling)**.** Since the restriction to the open dense stratum
$$\mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi)) \to \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}, \mathcal{V}(\chi))$$
is injective, it follows that $\dim_k \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi)) \leqslant 1$.

thm:existence-of-hasse-invariants

Theorem 2.4.5 (Existence of Hasse invariants)**.** *Let $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ be an $\mathcal{M}_{\mu,k}$-ample character (see [GK19, Definition N.5.1]). Then there exists an integer $N \geqslant 1$ such that the line bundle $\mathcal{V}(N\chi)$ admits a Hasse invariant*
$$h \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(N\chi))$$
cutting out the μ-ordinary locus.

Proof. See [GK19, §3.2]. □

rem:hasse-invariants-via-p-rank-stratification

Remark 2.4.6 (Hasse invariants via p-rank stratification)**.** Assume μ is defined over $\mathbf{Z}_p$ so that $\mu = \varphi(\mu)$. Following Remark 2.3.11, there is a natural projection
$$\pi\colon \mathcal{G}\text{-zip}_\mu \longrightarrow [\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k} \backslash \mathcal{G}_k]$$
where the target stack quotient is formed via the action $(a,b)\cdot g = agb^{-1}$. Any character $\eta \in \mathbf{X}^\bullet(\mathcal{M}_{\mu,k})$ induces a character $(a,b) \mapsto \eta(a)^{-1}\eta(b)$ of $\mathcal{P}^-_{\mu,k} \times_k \mathcal{P}^+_{\mu,k}$, which defines a line bundle $\mathcal{L}(\eta)$ on the target stack. For an ample character η, a suitable representation of $\mathcal{G}_k$ provides a semi-invariant regular function $f\colon \mathcal{G}_k \to \mathbf{A}^1_k$ satisfying $f(agb^{-1}) = \eta(a)\eta(b)^{-1}f(g)$, whose non-vanishing locus is exactly the open dense big cell $\mathcal{P}^-_{\mu,k}\mathcal{P}^+_{\mu,k} \subseteq \mathcal{G}_k$. This function f defines a global section of $\mathcal{L}(\eta)$ cutting out the big cell. By definition of the zip group E (Theorem 2.3.4), the pullback $\pi^*\mathcal{L}(\eta)$ to $\mathcal{G}\text{-zip}_\mu$ is exactly the line bundle $\mathcal{V}(\chi)$, where $\chi = \eta \circ \varphi - \eta$. Since the preimage of the big cell under π is exactly the μ-ordinary stratum $\mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}$ (Remark 2.3.11), the pullback $\pi^* f \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$ is exactly a Hasse invariant of weight χ.

rem:hasse-invariant-of-windows-over-frames

Remark 2.4.7 (Hasse invariant of windows over frames)**.** Let $\underline{A} = (A, I)$ be a Breuil–Kisin frame for R such that A is p-torsion-free. Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G}, \mu)$-window over $\underline{A}$. Analogously to Remark 2.3.12, the modification $m_1\colon \varphi^*\mathcal{E}_1[1/I] \xrightarrow{\sim} \mathcal{E}_1[1/I]$ induces a canonical $\mathcal{G}$-zip of type μ over $A/(I,p) \simeq R/pR$, yielding a classifying morphism $z\colon \operatorname{Spec} R/pR \to \mathcal{G}\text{-zip}_\mu$. Given a Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$, we define the **Hasse invariant** $\mathrm{Ha}(\mathcal{E}_1, m_1)$ of the window as the pullback $z^*\mathrm{Ha} \in \mathrm{H}^0(\operatorname{Spec} R/pR, z^*\mathcal{V}(\chi))$. It therefore makes sense to speak of μ-ordinariness of $(\mathcal{G}, \mu)$-windows over arbitrary frames.

Remark 2.4.8 (Well-definedness of p-adic valuation of Hasse invariant)**.** For a general Breuil–Kisin frame $\underline{A} = (A, I)$, identifying the Hasse invariant $\mathrm{Ha}(\mathcal{E}_1, m_1)$ with an element of $A/(I,p) \cong R/pR$ requires a choice of local trivialization for the line bundle $z^*\mathcal{V}(\chi)$. However, any two such choices differ by multiplication by a unit in $(R/pR)^\times$. Consequently, the principal ideal $(\mathrm{Ha}(\mathcal{E}_1, m_1))$ generated by the Hasse invariant in R/pR is canonically well-defined and independent of the chosen trivialization. Furthermore, if the base ring R is equipped with a rank-1 valuation ν_p extending the p-adic valuation on $\mathbf{Z}_p$, we define the truncated p-adic valuation of the Hasse invariant as $\min(1, \nu_p(\widetilde{h})) \in [0,1]$, where $\widetilde{h} \in R$ is any lift of $\mathrm{Ha}(\mathcal{E}_1, m_1)$. Because any two lifts differ by a multiple of p (which has valuation 1), and any two trivializations differ by a unit (which has valuation zero), this truncated valuation is canonically well-defined. In the special case of an $\mathrm{A}_{\mathrm{inf}}$ frame for a perfectoid ring R, $A/(I,p)$ identifies with $R^\flat/tR^\flat$ (where $t = p^\flat$), and this recovers the well-definedness of the similarly truncated t-adic valuation $\min(1, \nu_t(\widetilde{h})) \in [0,1]$.

Remark 2.4.9 (Hasse invariant under base change and Frobenius twist)**.** The formation of the Hasse invariant intrinsically commutes with base change. Specifically, if $R \to S$ is a ring map and $\mathcal{A}$ is a 1-truncated $(\mathcal{G}, \mu)$-aperture over R with base change $\mathcal{A}_S$ over S, the classifying map to $\mathcal{G}\text{-zip}_\mu$ simply composes with $\operatorname{Spec} S/pS \to \operatorname{Spec} R/pR$. Consequently, $\mathrm{Ha}(\mathcal{A}_S)$ is precisely the image of $\mathrm{Ha}(\mathcal{A})$ under the map $R/pR \to S/pS$. As an obvious consequence, if R is a k-algebra and we base change along the absolute Frobenius $\varphi\colon R \to R$, the Frobenius pullback $\varphi^*\mathcal{A}$ is canonically a 1-truncated $(\mathcal{G}, \varphi(\mu))$-aperture and the Hasse invariant satisfies $\mathrm{Ha}(\varphi^*\mathcal{A}) = \mathrm{Ha}(\mathcal{A})^{\otimes p}$. In particular, the p-adic valuation gets multiplied by p.

2.5. **Cartier duality.** In order to establish a notion of Cartier duality for $(\mathcal{G},\mu)$-apertures, we require a group-theoretic operation on $\mathcal{G}$ that generalizes the map $g \mapsto (g^t)^{-1}$ on GL_n. This is provided by the Chevalley involution.

Assumption 2.5.1. $\mathcal{G}$ is *reductive.*

Definition 2.5.2 (Chevalley involution)**.** A **Chevalley involution** of $\mathcal{G}$ is an automorphism $C\colon \mathcal{G} \to \mathcal{G}$ of order 2 which restricts to the inversion map $t \mapsto t^{-1}$ on a maximal torus $\mathcal{T} \subset \mathcal{G}$, and maps a Borel subgroup $\mathcal{B}$ containing $\mathcal{T}$ to the opposite Borel subgroup $\mathcal{B}^-$ containing $\mathcal{T}$.

`thm:existence-of-chevalley-involution`

Theorem 2.5.3 (Existence of Chevalley involution)**.** *If $\mathcal{G}$ is a* split *reductive group, fixing a pinning (*épinglage*) $(\mathcal{T},\mathcal{B},\{X_\alpha\}_{\alpha\in\Delta})$ yields a unique automorphism C that satisfies:*

(1) $C(t) = t^{-1}$ for all $t \in \mathcal{T}$,
(2) $dC(X_\alpha) = -X_{-\alpha}$ for all $\alpha \in \Delta$, where $X_{-\alpha}$ is the unique basis of $\mathfrak{g}_{-\alpha}$ such that $[X_\alpha, X_{-\alpha}]$ is the corresponding simple coroot H_α.

This C is a Chevalley involution ($C^2 = \mathrm{id}_{\mathcal{G}}$). Because any two pinnings are conjugate by an element of the adjoint group $\mathcal{G}^{\mathrm{ad}}$, C is uniquely determined up to an inner automorphism.

Proof. See [DG70, Exp. XXIII, Théorème 4.1]. □

Remark 2.5.4 (Chevalley involution for non-split groups)**.** For a non-split reductive group $\mathcal{G}$ over $\mathbf{Z}_p$, its special fiber $\mathcal{G}_{\mathbf{F}_p}$ is quasi-split by Lang's theorem. By Hensel's lemma applied to the smooth and proper scheme of Borel subgroups, $\mathcal{G}$ itself is quasi-split over $\mathbf{Z}_p$. Thus, $\mathcal{G}$ is an outer form of a split group $\mathcal{G}_0$, obtained by Galois descent via pinned automorphisms. Because the canonical Chevalley involution on $\mathcal{G}_0$ commutes with all pinned automorphisms, it canonically descends to a Chevalley involution on $\mathcal{G}$ defined over $\mathbf{Z}_p$.

We thus obtain:

`thm:cartier-duality-apertures`

Theorem 2.5.5 (Cartier duality for apertures)**.** *Let $C\colon \mathcal{G}\to\mathcal{G}$ be a Chevalley involution. For any p-complete $\mathcal{O}$-algebra R and any $n \in \mathbf{Z}_{\geqslant 1}\cup\{\infty\}$, C induces an equivalence of groupoids*

$$\mathrm{BT}_n^{\mathcal{G},\mu}(R) \xrightarrow{\sim} \mathrm{BT}_n^{\mathcal{G},-w_0\mu}(R)$$

where w_0 is the longest element of the Weyl group of $\mathcal{G}$.

Example 2.5.6 (The case of GL_n)**.** When $(\mathcal{G},\mu) = (\mathrm{GL}_h, \mu_d)$, the standard pinning (diagonal torus, upper triangular Borel) yields exactly the inverse-transpose $C(g) = (g^\top)^{-1}$. This involution translates, on the level of representations, to taking the dual representation, which mirrors the classical Cartier duality of Barsotti-Tate groups via Dieudonné theory.

`subsec:quasi-isogenies`

2.6. **Quasi-isogenies.**

Setup 2.6.1. Let $\mathfrak{X}$ be a p-adic formal scheme, let λ be a dominant cocharacter of $\mathcal{G}$ defined over $\mathbf{Z}_p$, and let $\mathcal{A}_1$ and $\mathcal{A}_2$ be $(\mathcal{G},\mu)$-apertures over $\mathfrak{X}$, with underlying $\mathcal{G}$-torsors $\mathcal{P}_1$ and $\mathcal{P}_2$ over $\mathfrak{X}^{\mathrm{syn}}$.

Remark 2.6.2 (The Vinberg monoid)**.** The Vinberg monoid gives a useful algebraic model for relative-position bounds. Suppose first that $\mathcal{G}$ is split. Choose a Borel subgroup and a split maximal torus through which the dominant representative λ factors, let $\mathcal{T}$ be the resulting abstract Cartan, put $\mathcal{T}_{\mathrm{ad}} = \mathcal{T}/Z_{\mathcal{G}}$, and set

$$\mathcal{T}_{\mathrm{ad}}^+ := \operatorname{Spec} \mathbf{Z}_p[\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}],$$

where $\mathbf{X}^\bullet(\mathcal{T}_{\mathrm{ad}})_{\mathrm{pos}}$ is the monoid generated by the simple roots. The Vinberg monoid $V_{\mathcal{G}}$ is an affine monoid scheme equipped with a $(\mathcal{G}\times\mathcal{G})$-action and an abelianization map

$$\mathfrak{a}\colon V_{\mathcal{G}} \longrightarrow \mathcal{T}_{\mathrm{ad}}^+.$$

The image of $\lambda(p)$ in $\mathcal{T}_{\mathrm{ad}}(\mathbf{Q}_p)$ extends to a point of $\mathcal{T}_{\mathrm{ad}}^+(\mathbf{Z}_p)$, again denoted $\lambda(p)$, and we write

$$V_{\mathcal{G},\lambda(p)} := V_{\mathcal{G}} \times_{\mathcal{T}_{\mathrm{ad}}^+,\lambda(p)} \operatorname{Spec} \mathbf{Z}_p.$$

After inverting p, the point $\lambda(p)$ lies in the unit torus $\mathcal{T}_{\mathrm{ad}}$, so the generic fiber of $V_{\mathcal{G},\lambda(p)}$ is biequivariantly isomorphic to $\mathcal{G}_{\mathbf{Q}_p}$. See [XZ19, §3.2] and [Nat26, §2] for constructions of the Vinberg monoid.

Construction 2.6.3 (A $V_{\mathcal{G},\lambda(p)}$-bundle)**.** The product $\mathcal{P}_1 \times_{\mathfrak{X}^{\mathrm{syn}}} \mathcal{P}_2$ naturally forms a $(\mathcal{G} \times \mathcal{G})$-torsor. We form the associated $V_{\mathcal{G},\lambda(p)}$-fiber bundle over $\mathfrak{X}^{\mathrm{syn}}$:

$$V_{\lambda(p)}(\mathcal{P}_1, \mathcal{P}_2) := (\mathcal{P}_1 \times_{\mathfrak{X}^{\mathrm{syn}}} \mathcal{P}_2) \times^{\mathcal{G}\times\mathcal{G}} V_{\mathcal{G},\lambda(p)}.$$

def:quasi-isogeny

Definition 2.6.4 (Quasi-isogeny bounded by λ)**.** A **quasi-isogeny bounded by** λ (or λ**-modification**) $\mathcal{P}_1 \dashrightarrow \mathcal{P}_2$ over $\mathfrak{X}^{\mathrm{syn}}$ is a global section $s\colon \mathfrak{X}^{\mathrm{syn}} \to V_{\lambda(p)}(\mathcal{P}_1, \mathcal{P}_2)$.

rem:quasi-isogenies-preserve-mu-ordinariness

Remark 2.6.5 (Quasi-isogenies preserve μ-ordinariness)**.** Since μ-ordinariness is an open condition, it can be checked on geometric points. Let κ be an algebraically closed field of characteristic p. Using the Witt frame, a $(\mathcal{G},\mu)$-aperture over κ is just the data of a $\mathcal{G}$-torsor over $W(\kappa)$ equipped with a semilinear modification over $W(\kappa)[1/p]$ bounded by μ. This data gives an isocrystal with $\mathcal{G}$-structure over κ in the sense of Kottwitz (see, for instance, [MY26, §3.3]). A quasi-isogeny of $(\mathcal{G},\mu)$-apertures over κ then induces an isomorphism of respective isocrystals. Therefore, it suffices to observe that μ-ordinariness of a $(\mathcal{G},\mu)$-aperture over κ is witnessed by its isocrystal. This is implied by [Wor13, Proposition 7.2] (see also [MY26, Lemma 3.6.14]).

thm:relative-frobenius-quasi-isogeny

Theorem 2.6.6 (Relative Frobenius quasi-isogeny)**.** *Let R be an $\mathbf{F}_p$-algebra. For any $(\mathcal{G},\mu)$-aperture $\mathcal{A} \in \mathrm{BT}_\infty^{\mathcal{G},\mu}(R)$, its pullback $\mathcal{A}^{(p)}$ along the absolute Frobenius on R is naturally a $(\mathcal{G},\varphi(\mu))$-aperture. There is a canonical quasi-isogeny bounded by μ, called the* ***relative Frobenius quasi-isogeny****:*

$$\mathrm{Frob}_{\mathcal{A}}\colon \mathcal{A} \dashrightarrow \mathcal{A}^{(p)}.$$

Furthermore, if $\mathcal{A}$ is represented by a $\mathcal{G}$-torsor $\mathcal{E}$ over R^{syn} with its 1-truncation $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ equipped with its conjugate realization $T_{\mathrm{conj}}(\mathcal{E}_1)$, which is a reduction of structure to $\mathcal{P}^+_{\varphi(\mu)}$, then the quasi-isogeny $\mathrm{Frob}_{\mathcal{A}}$ *is precisely the parahoric pushforward of this reduction $T_{\mathrm{conj}}(\mathcal{E}_1)$.*

Proof. We can always perform the parahoric pushforward routine using the conjugate realization to produce a functorial quasi-isogeny $\mathcal{A} \dashrightarrow \mathcal{A}'$ bounded by μ. What needs to be shown is that there is a functorial isomorphism $\mathcal{A}' \simeq \mathcal{A}^{(p)}$. Since $\mathrm{BT}_\infty^{\mathcal{G},\mu} \otimes \mathbf{F}_p$ is smooth, we may assume R is a semiperfect ring by quasisyntomic descent. There is a universal map $R^\flat \to R$ from a perfect $\mathbf{F}_p$-algebra to R. For $R^\flat$, everything can be checked by hand using the the Witt frame $(W(R^\flat), (p))$. By base changing, we get the result over R too. $\square$

rem:mixed-apertures

Remark 2.6.7 (Mixed cocharacters)**.** Definition 2.6.4 naturally extends to the case where $\mathcal{A}_1$ and $\mathcal{A}_2$ are $(\mathcal{G},\mu_1)$ and $(\mathcal{G},\mu_2)$-apertures, respectively, for potentially distinct cocharacters μ_1 and μ_2. In this case, it still makes sense to talk about quasi-isogenies bounded by λ between them. We warn the reader that this gives a nontrivial notion only in characteristic p.

rem:cartier-duality-quasi-isogenies

Remark 2.6.8 (Cartier duality and quasi-isogenies)**.** Let $C\colon \mathcal{G} \to \mathcal{G}$ be a Chevalley involution. Recall from Theorem 2.5.5 that C induces an equivalence $\mathrm{BT}_\infty^{\mathcal{G},\mu} \xrightarrow{\sim} \mathrm{BT}_\infty^{\mathcal{G},-w_0\mu}$. Applying C and then inverting the resulting modification transforms a quasi-isogeny $\mathcal{A}_1 \dashrightarrow \mathcal{A}_2$ of $(\mathcal{G},\mu)$-apertures bounded by λ into a quasi-isogeny $\mathcal{A}_2^C \dashrightarrow \mathcal{A}_1^C$ of $(\mathcal{G},-w_0\mu)$-apertures bounded by λ in the opposite direction.

Remark 2.6.9. Construction of stacks of quasi-isogenies between apertures (and more) is the subject of forthcoming work of Lee and Madapusi [LM]. We do not use this theory here.

3. Shimura varieties

sec:shimuraeevarhimra

3.1. Shimura varieties.

set:shimura

Setup 3.1.1. As in [MY26], an **unramified Shimura datum** is a triple $(G,\mathcal{G},X)$ where (G,X) is a Shimura datum and $\mathcal{G}$ is a reductive model of G over $\mathbf{Z}_p$. We set $K_p = \mathcal{G}(\mathbf{Z}_p)$, and only consider level subgroups $K \subset G(\mathbf{A}_f)$ of the form K_pK^p for a neat compact open subgroup $K^p \subset G(\mathbf{A}_f^p)$. We call the tuple $(G,\mathcal{G},X,K)$ an **unramified Shimura tuple**. We also fix a place $v|p$ of the reflex field E of (G,X). Over the associated Shimura variety Sh_K, we have a canonical étale $\mathcal{G}^c(\mathbf{Z}_p)$-local system $\mathbf{Et}_{K,p}$, where $\mathcal{G}^c$ is a reductive model over $\mathbf{Z}_p$ of the *cuspidal quotient* G^c– this is a technical construction for which we refer the reader to [MY26, §6.1]. When G is adjoint, $G = G^c$. The Shimura datum gives an E_v-rational conjugacy class of cocharacters $\{\mu_v\}$. We in fact have a representative $\mu_v\colon \mathbf{G}_{\mathrm{m},\mathcal{O}_{E,v}} \to \mathcal{G}_{\mathcal{O}_{E,v}}$ for $\{\mu_v\}$. We may view each μ_v as a cocharacter of $\mathcal{G}^c_{\mathcal{O}_{E,v}}$. We can then form the formal stack $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$ over $\mathcal{O}_{E,v}$. For brevity, set $\mathcal{O} := \mathcal{O}_{E,v}$ and k be the residue field of $\mathcal{O}$.

not:shimura-varieties

Notation 3.1.2 (Shimura varieties)**.** For the remainder of this paper, we fix a neat tame level $K^p \subset \mathcal{G}(\mathbf{A}_f^p)$. Because this tame level remains unchanged throughout, we will suppress K^p from the notation of the Shimura varieties. Specifically, for any open compact subgroup $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, we denote the corresponding Shimura variety by Sh_{K_p} and its integral model(s) by $\mathscr{S}_{K_p}$. As usual, we denote formal completions by $\widehat{\mathscr{S}}_{K_p}$. Adic generic fibers are denoted by calligraphic letters $\mathcal{S}_{K_p}$. We use an asterisk in the superscript to denote the minimal compactification. For example, $\mathcal{S}^*_{K_p}$ is the analytification of the minimal compactification $\mathrm{Sh}^*_{K_p}$.

not:level-structures-at-p

Notation 3.1.3 (Level structures at p)**.** Recall from Setup 3.1.1 that our Shimura datum provides a cocharacter μ_v defined over $\mathcal{O}_{E,v}$. Although $-\mu_v$ might not be defined over $\mathbf{Z}_p$, we can sum all the Galois conjugates as in Setup 3.3.1 to get a cocharacter $-\nu$ defined over $\mathbf{Z}_p$. We define the level subgroups at p involved in our tower:

- $\Gamma_0(p^n)$**-level**: This is the preimage of the parabolic subgroup $\mathcal{P}^+_{-\nu}$ under the reduction modulo p^n map $\mathcal{G}(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$. Using the element $\delta_n = (-\nu)(p^n) \in \mathcal{G}(\mathbf{Q}_p)$, this subgroup can be equivalently expressed as the intersection $\mathcal{G}(\mathbf{Z}_p) \cap \delta_n \mathcal{G}(\mathbf{Z}_p)\delta_n^{-1}$.
- $\Gamma_1(p^n)$**-level**: This is the preimage of the unipotent radical of $\mathcal{P}^+_{-\nu}$ under the reduction modulo p^n map.
- $\Gamma(p^n)$**-level**: This is the kernel of the full reduction map $\mathcal{G}(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$.

These are the parahoric, pro-p Iwahori, and principal congruence levels, respectively.

rem:p-hecke-correspondences

Remark 3.1.4 (p-Hecke correspondences)**.** Fix an element $h \in G(\mathbf{Q}_p)$. Associated with this element, we can set $\mathcal{G}(\mathbf{Z}_p)_h := \mathcal{G}(\mathbf{Z}_p) \cap h\mathcal{G}(\mathbf{Z}_p)h^{-1}$. This subgroup naturally gives rise to a Hecke correspondence:

$$\begin{tikzcd} & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)_h} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\ \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} & & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} \end{tikzcd}$$

The two structure morphisms of this correspondence can be described as follows:

- The map t_h is the finite étale projection induced by the natural subgroup inclusion $\mathcal{G}(\mathbf{Z}_p)_h \subset \mathcal{G}(\mathbf{Z}_p)$.
- The map s_h is induced by the conjugation homomorphism $\mathcal{G}(\mathbf{Z}_p)_h \to \mathcal{G}(\mathbf{Z}_p)$ mapping $g \mapsto h^{-1}gh$, which geometrically corresponds to right multiplication by h^{-1}.

Let $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)} = [*/\underline{\mathcal{G}(\mathbf{Z}_p)}]$ be the classifying stack of étale $\mathcal{G}(\mathbf{Z}_p)$-local systems. The double coset $\mathcal{G}(\mathbf{Z}_p)h\mathcal{G}(\mathbf{Z}_p)$ carries a natural right action of $\mathcal{G}(\mathbf{Z}_p)$, allowing us to form the finite étale stack quotient over $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$:

$$\alpha_h \colon [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}] \to [*/\underline{\mathcal{G}(\mathbf{Z}_p)}] = \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}\,.$$

By Setup 3.1.1, the Shimura variety $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$ carries a canonical étale local system that induces a *p-adic étale realization map* $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$. The pullback of the cover α_h over $\mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$ along this realization map is canonically isomorphic to the cover $t_h \colon \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)_h} \to \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}$.

We recall the existence of universal $(\mathcal{G}^c, -\mu_v)$-apertures over the v-adic completion of the integral canonical models of Bakker, Shankar, and Tsimerman [BST25a].

thm:syntomic-realization

Theorem 3.1.5 (Syntomic realization for integral canonical models)**.** *For an unramified Shimura tuple $(G, \mathcal{G}, X, K)$, and all p sufficiently large, an integral model $\mathscr{S}_K$ of Sh_K over $\mathcal{O}_{E,(v)}$ exists with v-adic completion $\widehat{\mathscr{S}}_K$, for any $v \mid p$, such that*

(1) (Serre–Tate property) There exists a formally étale map, called the ***syntomic realization****, of p-adic formal stacks over $\mathrm{Spf}\,\mathcal{O}_{E,v}$*

$$\varpi \colon \widehat{\mathscr{S}}_K \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v},$$

such that the induced $\mathcal{G}^c(\mathbf{Z}_p)$-local system on the generic fiber is isomorphic to $\mathbf{Et}_{K,p}$.

(2) (Pointwise criterion) For any mixed characteristic $(0,p)$ complete discrete valuation field F over $\mathcal{O}_{E,v}$ with perfect residue field and $x \in \mathrm{Sh}_K(F)$, the following are equivalent:

(a) $x \in \mathscr{S}_K(\mathcal{O}_F)$;

(b) $\mathbf{Et}_{K,p}|_x$ is a crystalline $\mathcal{G}^c(\mathbf{Z}_p)$-local system;

(c) $\mathbf{Et}_{K,p}|_x$ *is a potentially crystalline* $\mathcal{G}^c(\mathbf{Z}_p)$*-local system.*

Proof. See [MY26, Theorem D]. □

Remark 3.1.6 (Shimura varieties of exceptional type)**.** One of the key points of the above result is that it also applies to Shimura varieties of *exceptional type*, but only for an ineffectively large bound on p because the integral canonical models in [BST25a] are obtained by spreading out from the generic fiber. By a Shimura variety of exceptional type, we refer to a Shimura variety associated to a reductive group G over $\mathbf{Q}$ such that its adjoint group G^{ad} has $\mathbf{R}$-simple factors that are either compact or isomorphic to an exceptional Lie group admitting a Hermitian symmetric domain. By Cartan's classification, there are exactly two such non-compact real exceptional simple Lie groups:

- $E_{6(-14)}$: The real form of E_6 with maximal compact subgroup $\mathrm{Spin}(10) \times \mathrm{U}(1)$. It has $\mathbf{R}$-rank 2.
- $E_{7(-25)}$: The real form of E_7 with maximal compact subgroup $E_6 \times \mathrm{U}(1)$. It has $\mathbf{R}$-rank 3.

3.2. **Relative perfectness of mod-p syntomic realization.** It is well known that a relatively perfect homomorphism of $\mathbf{F}_p$-algebras is formally étale but the converse is not necessarily true. It is shown in [DO25] that relative perfectness of a map of $\mathbf{F}_p$-algebras is equivalent to the map being what's called b-nil formally étale. The following is a straightforward generalization of this to stacks:

Definition 3.2.1 (B-nil formal smoothness, unramifiedness, and étaleness)**.** A morphism $f\colon \mathcal{X} \to \mathcal{Y}$ of (possibly nonalgebraic) stacks over $\mathbf{F}_p$ is called **b-nil formally smooth** (resp. **b-nil formally unramified**, **b-nil formally étale**) if for every affine scheme S over $\mathbf{F}_p$ and every closed subscheme $S_0 \subset S$ defined by an ideal sheaf $\mathcal{I} \subset \mathcal{O}_S$ such that $\mathcal{I}^{[p]} = 0$ ($\mathcal{I}^{[p]}$ is the ideal sheaf generated by pth powers of local sections of $\mathcal{I}$), the canonical restriction functor

$$\Phi\colon \mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$$

is essentially surjective (resp. fully faithful, an equivalence of groupoids).

Remark 3.2.2. Smooth (resp. unramified, étale) morphisms are b-nil formally smooth (resp. unramified, étale). Futhermore, these notions are stable under base change.

prop:bnil-etale-iff-relatively-perfect

Proposition 3.2.3 (B-nil formally étale $\Leftrightarrow$ relatively perfect)**.** *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a morphism of stacks over $\mathbf{F}_p$ that is representable by Deligne–Mumford stacks. Then f is b-nil formally étale if and only if f is relatively perfect (i.e., the relative Frobenius $F_{\mathcal{X}/\mathcal{Y}}\colon \mathcal{X} \to \mathcal{X}^{(p)} := \mathcal{X} \times_{\mathcal{Y}, F_{\mathcal{Y}}} \mathcal{Y}$ is an isomorphism).*

Proof. First, assume f is relatively perfect. Let $S_0 \hookrightarrow S$ be a closed embedding of affine schemes defined by an ideal $\mathcal{I}$ with $\mathcal{I}^{[p]} = 0$. The absolute Frobenius $F_S\colon S \to S$ factors canonically as $S \to S_0 \hookrightarrow S$. Applying the purely categorical argument of [DO25, Lemma 3.17] to the groupoids of points $\mathcal{X}(S)$ and $\mathcal{Y}(S)$ directly establishes that the canonical functor $\mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$ is an equivalence.

Conversely, assume f is b-nil formally étale. Let $V \to \mathcal{Y}$ be a map from an affine scheme. The projection $W := \mathcal{X} \times_{\mathcal{Y}} V \to V$ is b-nil formally étale. The assumption implies that W is a Deligne–Mumford stack. Choose an étale atlas $U \to W$ where U is an affine scheme. The composition $U \to W \to V$ is a composition of an étale map and a b-nil formally étale map, so it is a b-nil formally étale morphism of affine schemes. By [DO25, Theorem 3.21], its relative Frobenius $F_{U/V}\colon U \to U^{(p)}$ is an isomorphism. Because the map $U \to W$ is étale, the relative Frobenius commutes with base change along it, meaning the commutative square formed by $U \to W$, $U^{(p)} \to W^{(p)}$, $F_{U/V}$, and $F_{W/V}$ is Cartesian. Since $F_{U/V}$ is an isomorphism and $U^{(p)} \to W^{(p)}$ is an étale cover, descent implies $F_{W/V}$ is an isomorphism. As $F_{W/V}$ is the base change of $F_{\mathcal{X}/\mathcal{Y}}$ to V, it follows that $F_{\mathcal{X}/\mathcal{Y}}$ is an isomorphism, making f relatively perfect. □

prop:abstract-serre-tate

Proposition 3.2.4. *Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a formally étale morphism of stacks such that $\mathcal{Y} \simeq \lim \mathcal{Y}_n$ is the inverse limit of Artin stacks $\{\mathcal{Y}_n\}$. Assume further that $\mathcal{X}$ and $\mathcal{Y}_n$ are Noetherian Artin stacks with quasi-affine diagonal or they are quasicompact, separated, and tame Deligne–Mumford stacks. Then f satisfies the extension property with respect to arbitrary locally nilpotent thickenings*[1]*. That is, for every affine scheme S and every closed*

[1] Following [Stacks, Definition 00IL], an ideal I is called **locally nilpotent** if every element $x \in I$ is nilpotent. Equivalently, every finitely generated subideal of I is nilpotent, making I the filtered colimit of its nilpotent subideals.

subscheme $S_0 \subset S$ defined by a locally nilpotent ideal, the canonical functor

$$\mathcal{X}(S) \to \mathcal{X}(S_0) \times_{\mathcal{Y}(S_0)} \mathcal{Y}(S)$$

is an equivalence of categories.

Proof. Let R be a ring and $I \subset R$ a locally nilpotent ideal. We set $R_0 := R/I$. By [BH17, Corollary 1.5], the canonical map

$$\mathcal{X}(R) \xrightarrow{\sim} \lim \mathcal{X}(R/I^m)$$

is an equivalence. Similarly, for each n, $\mathcal{Y}_n(R) \xrightarrow{\sim} \lim_m \mathcal{Y}_n(R/I^m)$. Taking the inverse limit over n, we deduce that $\mathcal{Y}$ also satisfies this property:

$$\mathcal{Y}(R) \simeq \lim_n \mathcal{Y}_n(R) \simeq \lim_n \lim_m \mathcal{Y}_n(R/I^m) \simeq \lim_m \lim_n \mathcal{Y}_n(R/I^m) \simeq \lim_m \mathcal{Y}(R/I^m).$$

For each $m \geqslant 1$, the ideal $I/I^m \subset R/I^m$ is nilpotent. By formal étaleness, the functor

$$\mathcal{X}(R/I^m) \to \mathcal{X}(R_0) \times_{\mathcal{Y}(R_0)} \mathcal{Y}(R/I^m)$$

is an equivalence of categories. Taking the inverse limit over all $m \geqslant 1$ yields the desired equivalence

$$\mathcal{X}(R) \xrightarrow{\sim} \mathcal{X}(R_0) \times_{\mathcal{Y}(R_0)} \mathcal{Y}(R).$$ □

Corollary 3.2.5 (Serre–Tate for locally nilpotent thickenings)**.** *The syntomic realization $\varpi \colon \widehat{\mathscr{S}_K} \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v}$ satisfies the Serre–Tate property with respect to arbitrary locally nilpotent thickenings.*

`cor:mod-p-syntomic-realization-is-relatively-perfect`

Corollary 3.2.6 (Mod-p syntomic realization is relatively perfect)**.** *The map*

$$\varpi_k \colon \widehat{\mathscr{S}_K} \otimes k \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$$

of stacks over k is relatively perfect.

Proof. Let $\operatorname{Spec} A \hookrightarrow \widehat{\mathscr{S}_K} \otimes k$ be an affine open. Then the composite map of stacks $\operatorname{Spec} A \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$ is representable by affine schemes. Indeed, $\operatorname{Spec} A \to \mathrm{BT}_n^{\mathcal{G}^c, -\mu_v} \otimes k$ is an affine morphism since $\mathrm{BT}_n^{\mathcal{G}^c, -\mu_v}$ has affine diagonal, and the inverse limit of affine morphisms is representable by affine morphisms. Now the desired result is immediate by applying Proposition 3.2.3 to $\operatorname{Spec} A \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \otimes k$. □

3.3. The μ-ordinary locus.

We briefly recall the μ-ordinary locus, following [MY26, §3.6, §7.4].

`set:mu-ordinary-locus`

Setup 3.3.1. Maintain Setups 2.2.1 and 2.3.1. Suppose μ is chosen within its conjugacy class to factor through a maximal torus $T \subset \mathcal{G}$. Let $d = [\mathcal{O}[1/p] : \mathbf{Q}_p]$, and let $\nu = \sum_{i=0}^{d-1} \varphi^i(\mu)$, which defines a cocharacter of $\mathcal{G}$ defined over $\mathbf{Z}_p$. We let $\mathcal{P}_\nu^+ \subset \mathcal{G}$ denote the parabolic subgroup whose Lie algebra consists of the non-negative weight spaces for the adjoint action of ν, and let $\mathcal{M}_\nu$ be its Levi quotient (the centralizer of ν). By construction, μ is central in $\mathcal{M}_\nu$.

`def:mu-ordinary-stratum`

Definition 3.3.2 (The μ-ordinary locus)**.** Recall from Remark 2.3.8 that the stack $\mathcal{G}\text{-zip}_\mu$ admits a unique open dense stratum $\mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}$. For any $n \in \mathbf{N} \cup \{\infty\}$, the **$\mu$-ordinary locus** of the stack of apertures is the unique open formal substack $\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}} \subset \mathrm{BT}_n^{\mathcal{G},\mu}$ characterized by the property:

$$\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}} \otimes k = (\mathrm{BT}_n^{\mathcal{G},\mu} \otimes k) \times_{\mathcal{G}\text{-zip}_\mu} \mathcal{G}\text{-zip}_\mu^{\mathrm{ord}}.$$

`prop:mu-ordinary-apertures-parabolic`

Proposition 3.3.3. *The natural map $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{G},\mu}$ is an open embedding onto $\mathrm{BT}_n^{\mathcal{G},\mu,\mathrm{ord}}$.*

Proof. See [MY26, Proposition 3.6.12]. □

There is a canonical projection $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ given by quotienting by the unipotent radical of $\mathcal{P}_\nu^+$ at the level of syntomic torsors.

`thm:relative-classifying-stack`

Theorem 3.3.4. *The formal stack $\mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ is étale over $\mathcal{O}$ and is non-canonically isomorphic to the formal classifying stack of the finite group $\mathcal{M}_\nu(\mathbf{Z}/p^n\mathbf{Z})$ and $\mathrm{BT}_n^{\mathcal{P}_\nu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$ presents the source as the relative classifying stack of a certain (noncommutative) finite locally free group scheme over $\mathrm{BT}_n^{\mathcal{M}_\nu,\mu}$.*

Proof. See [MY26, Proposition 3.6.8]. □

Remark 3.3.5 (The case when μ is defined over $\mathbf{Z}_p$)**.** For our applications, we are primarily concerned with the case where μ is defined over $\mathbf{Z}_p$, which implies $\mu = \varphi(\mu)$. Let us explicate Theorem 3.3.4 in this setting by detailing the structure of the fiber of the map $\mathrm{BT}_n^{\mathcal{P}_\mu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$. Recall that $\mathcal{U}_\mu^+$ denotes the unipotent radical of $\mathcal{P}_\mu^+$. The Levi quotient $\mathcal{M}_\mu$ acts on $\mathcal{U}_\mu^+$ by conjugation. A point in $\mathrm{BT}_n^{\mathcal{M}_\mu,\mu}(R)$ corresponds to an $\mathcal{M}_\mu$-torsor $\mathcal{Q}$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$. Twisting the unipotent radical by this torsor yields a vector group $\mathcal{U} := \mathcal{Q} \times^{\mathcal{M}_\mu} \mathcal{U}_\mu^+$ over $R^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$. Moreover, since μ is minuscule, the Hodge–Tate weights of $\mathcal{U}$ are concentrated in $\{0,1\}$. The fiber of $\mathrm{BT}_n^{\mathcal{P}_\mu^+,\mu} \to \mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$ over $\mathcal{Q}$ is canonically identified with the assignment $C \mapsto \mathrm{Map}_{R^{\mathrm{syn}}\otimes\mathbf{Z}/p^n\mathbf{Z}}(C^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}, B\mathcal{U})$ which is precisely the classifying stack of the n-truncated Barsotti–Tate group over $\mathrm{BT}_n^{\mathcal{M}_\mu,\mu}$ corresponding to the vector bundle F-gauge $\mathcal{U}$ via [GM24, Theorem A].

`def:global-mu-ordinary-locus`

Definition 3.3.6 (The global μ-ordinary locus)**.** The **μ-ordinary locus** of the integral canonical model is the unique open formal subscheme $\widehat{\mathscr{S}}_K^{\mathrm{ord}} \subset \widehat{\mathscr{S}}_K$ whose special fiber is the preimage of the μ-ordinary stratum under the $\mathcal{G}^c$-zip realization map ζ (cf. Proposition 3.4.5):

$$\widehat{\mathscr{S}}_K^{\mathrm{ord}} \otimes k := \zeta^{-1}(\mathcal{G}^c\text{-zip}^{\mathrm{ord}}_{-\mu_v}).$$

By Corollary 3.4.9, this locus canonically extends to a formal affine open subscheme $\widehat{\mathscr{S}}_K^{*\mathrm{ord}}$ of the (formal) minimal compactification $\widehat{\mathscr{S}}_K^{*}$ as it coincides with the non-vanishing locus of the extended Hasse invariant.

Remark 3.3.7. To summarize, we have a diagram with Cartesian squares

$$\begin{tikzcd}
\widehat{\mathscr{S}}_K^{*\mathrm{ord}} \arrow[d, hook] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[l, hook'] \arrow[r] \arrow[d, hook] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \arrow[d, hook] \\
\widehat{\mathscr{S}}_K^{*} & \widehat{\mathscr{S}}_K \arrow[l, hook'] \arrow[r, "\varpi"'] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}
\end{tikzcd}$$

where the horizontal arrows on the right are the syntomic realization maps, the horizontal arrows on the left are open embeddings into the minimal compactifications, and the vertical maps are open embeddings.

`prop:canonical-frobenius-lifting-apertures`

Proposition 3.3.8 (Canonical Frobenius lift on $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}$)**.** *Let $q = |k|$ be the cardinality of the residue field of $\mathcal{O}$. There is a canonical endomorphism $\widetilde{\Phi}_{\mathrm{ap}} \colon \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}$ of p-adic formal stacks over* $\mathrm{Spf}\,\mathcal{O}$ *lifting the relative q-Frobenius endomorphism.*

Proof. See [MY26, Proposition 3.6.11]. □

We use the relative perfectness result from the previous section to give a formal explanation of existence of canonical Frobenius lifts on the μ-ordinary loci of Shimura varieties (cf. [MY26, Theorem 7.4.21] and [BST25b, Proposition 6.6]):

`thm:canonical-frobenius-lift-mu-ordinary`

Theorem 3.3.9 (Canonical Frobenius lift on the μ-ordinary locus)**.** *Let $q = |k|$ be the cardinality of the residue field k of $\mathcal{O}$. There exists a unique Cartesian square of p-adic formal stacks over* $\mathrm{Spf}\,\mathcal{O}$*:*

$$\begin{tikzcd}
\widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[r, "\widetilde{\Phi}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[d, "\varpi"] \\
\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}} \arrow[r, "\widetilde{\Phi}_{\mathrm{ap}}"] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v,\mathrm{ord}}
\end{tikzcd}$$

such that the vertical arrows are induced by syntomic realizations and the bottom horizontal arrow $\widetilde{\Phi}_{\mathrm{ap}}$ is the canonical Frobenius lifting of Proposition 3.3.8. Consequently, $\widetilde{\Phi}$ is a lift of the relative q-Frobenius endomorphism.

Proof. Following the argument of Theorem 5.1.2 (and [MY26, Lemma 7.4.10]), set $h = \nu(p)$. In what follows, $(-)^{\mathrm{an}}$ denotes the analytification functor over E_v. Recall the Hecke correspondence diagram associated to

$K_h := K \cap hKh^{-1} = K \cap \nu(p)K\nu(p)^{-1}$:

$$\begin{tikzcd} & \mathrm{Sh}_{K_h} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\ \mathrm{Sh}_K & & \mathrm{Sh}_K . \end{tikzcd}$$

Over $\mathcal{S}_K^{\mathrm{ord}}$, the p-adic étale realization map factors through $\mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)}$. The description in Remark 3.1.4 therefore shows that $t_h^{\mathrm{an}}|_{\mathcal{S}_K^{\mathrm{ord}}}$ is canonically isomorphic to the base-change of

$$[\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}} \times_{\mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)} \simeq [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}}$$

along $\mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)}$. Since $\mathcal{P}_\nu^+(\mathbf{Z}_p)\nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p) = \nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p)$, we have a canonical section:

$$\mathrm{Loc}^{\mathrm{an}}_{\mathcal{P}_\nu^+(\mathbf{Z}_p)} \xrightarrow{\sim} [\underline{\nu(p)\mathcal{P}_\nu^+(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}} \to [\underline{\mathcal{G}(\mathbf{Z}_p)}h\underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{P}_\nu^+(\mathbf{Z}_p)}]^{\mathrm{an}}.$$

This implies that we have a section

$$\mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Sh}^{\mathrm{an}}_{K_h} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}_K^{\mathrm{an}}} \mathcal{S}_K^{\mathrm{ord}}.$$

Composing this section with the map $s_h^{\mathrm{an}} \colon \mathrm{Sh}^{\mathrm{an}}_{K_h} \to \mathrm{Sh}^{\mathrm{an}}_K$ gives a map $\Phi' \colon \mathcal{S}_K^{\mathrm{ord}} \to \mathrm{Sh}_K^{\mathrm{an}}$ such that the diagram

$$\begin{tikzcd} \mathcal{S}_K^{\mathrm{ord}} \arrow[r, "\Phi'"] \arrow[d, "\varpi^{\mathrm{ad}}"'] & \mathrm{Sh}_K^{\mathrm{an}} \arrow[d] \\ \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}, \mathrm{ad}} \arrow[r, "\widetilde{\Phi}_{\mathrm{ap}}^{\mathrm{ad}}"] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \end{tikzcd}$$

commutes. By [MY26, Proposition 4.3.1], this identifies Φ' as the adic generic fiber of a morphism $\widetilde{\Phi} \colon \widehat{\mathscr{S}}_K^{\mathrm{ord}} \to \widehat{\mathscr{S}}_K$ which factors through $\widehat{\mathscr{S}}_K^{\mathrm{ord}}$.

It remains to check that the square

$$\begin{tikzcd} \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[r, "\widetilde{\Phi}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_K^{\mathrm{ord}} \arrow[d, "\varpi"] \\ \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}} \arrow[r, "\widetilde{\Phi}_{\mathrm{ap}}"] & \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v, \mathrm{ord}} \end{tikzcd}$$

is Cartesian. Since all the members of the square are smooth over $\mathrm{Spf}\,\mathcal{O}$, by [MY26, Proposition 4.3.1], it suffices to check it at adic generic fibers, which is definitional. Because this square is Cartesian and the mod-p syntomic realization ϖ_k is relatively perfect (Corollary 3.2.6), $\widetilde{\Phi}$ modulo p must be the q-Frobenius, as $\widetilde{\Phi}_{\mathrm{ap}}$ modulo p is the q-Frobenius. □

3.4. **Minimal compactifications.** To extend our perfectoidness results from the open Shimura variety to its minimal compactification, we will use Hartog's extension principle to uniquely extend functions and morphisms across the boundary.

Notation 3.4.1. For any open compact subgroup $K_p \subset \mathcal{G}(\mathbf{Z}_p)$, we denote the minimal (Baily–Borel–Satake) compactification of the generic fiber by $\mathrm{Sh}^*_{K_p}$. This is a normal projective variety over the reflex field.

`rem:automorphic-vector-bundles`

Remark 3.4.2 (Automorphic vector bundles)**.** By composing the syntomic realization $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v}$ with the filtered de Rham realization $\mathrm{BT}_\infty^{\mathcal{G}^c, -\mu_v} \to B\mathcal{P}^-_{-\mu_v}$ (Remark 2.3.14), we obtain a canonical classifying map $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to B\mathcal{P}^-_{-\mu_v}$. Any representation of $\mathcal{P}^-_{-\mu_v}$ thus induces an automorphic vector bundle over $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ (which algebraizes to $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$). In particular, an ample character $\chi \in \mathbf{X}^\bullet(\mathcal{M}_{\mu_v})$, which inflates to a character of $\mathcal{P}^-_{-\mu_v}$, induces an automorphic line bundle, denoted ω^χ. These automorphic bundles are canonically defined over $\overline{\mathbf{Q}}$ and by the classical Baily-Borel theorem, because the character χ is ample, the graded algebra of automorphic forms on the generic fiber is finitely generated, and the sections of $(\omega^\chi)^{\otimes n}$ for $n \gg 0$ define an open immersion of the Shimura variety into the resulting projective variety. Thus, for all sufficiently large p, this projective embedding actually extends integrally and in particular ω^χ is ample.

rem:levi-representations

Remark 3.4.3 (Automorphic vector bundles coming from the Levi)**.** In the context of Remark 3.4.2, the cocharacter μ_v associated to the Shimura datum is only well-defined up to conjugacy. As a result, the precise parabolic subgroup $\mathcal{P}^-_{-\mu_v}$ depends on the specific choice of the representative μ_v. If we were to construct an automorphic vector bundle from an arbitrary representation of $\mathcal{P}^-_{-\mu_v}$, the resulting bundle would depend on this non-canonical choice. However, the Levi quotient $\mathcal{M}_{\mu_v}$ of $\mathcal{P}^-_{-\mu_v}$ is canonically independent of the choice of representative (up to canonical isomorphism). Therefore, by restricting our attention to representations of $\mathcal{P}^-_{-\mu_v}$ that arise as inflations of representations of the Levi quotient $\mathcal{M}_{\mu_v}$ (such as the character χ), the induced automorphic vector bundles are canonical and genuinely independent of the choice of μ_v.

cons:minimal-compactification

Construction 3.4.4 (Minimal compactification)**.** As discussed in Remark 3.4.2, for sufficiently large p, we can construct the integral model of the minimal compactification as

$$\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} := \operatorname{Proj}\left(\bigoplus_{n\geqslant 0} \mathrm{H}^0(\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}, (\omega^\chi)^{\otimes n})\right).$$

This construction yields a normal projective scheme over $\mathcal{O}_{E,(v)}$ into which $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$ naturally embeds as an open dense subscheme. By standard properties of the minimal compactification, this resulting scheme $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ is canonically independent of the choice of ample character χ. By construction, ω^χ extends to $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ and we denote it by the same notation.

For any deeper level structure $K_p \subset \mathcal{G}(\mathbf{Z}_p)$ at p—such as the parahoric level $\Gamma_0(p^n)$, the pro-p Iwahori level $\Gamma_1(p^n)$, or the principal congruence subgroup $\Gamma(p^n)$—we define the integral model $\mathscr{S}^*_{K_p}$ of the minimal compactification simply as the normalization of $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ in the function field of $\mathrm{Sh}^*_{K_p}$. This normalization procedure yields a tower of normal projective integral models extending the tower of (open) integral canonical models $\mathscr{S}_{K_p}$.

The following is true essentially by construction:

prop:comparison-line-bundles

Proposition 3.4.5 (Comparison of automorphic line bundles)**.** *Let $\zeta\colon \mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k \to \mathcal{G}^c\text{-zip}_{-\mu_v}$ be the map classifying the universal $\mathcal{G}^c$-zip. Then the restriction of the automorphic line bundle ω^χ to the special fiber $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ is canonically isomorphic to $\zeta^*\mathcal{V}(\chi)$ (cf. Remark 2.4.2).*

Proof. Indeed, $(\mathcal{Q}, \mathcal{Q}^-, \mathcal{Q}^+, \alpha) \mapsto \mathcal{Q}^-$ gives a classifying map $\mathcal{G}^c\text{-zip}_{-\mu_v} \to B\mathcal{P}^-_{-\mu_v}$ whose composition with ζ is the mod p reduction of the filtered de Rham realization. ☐

Let $Z_{K_p} := \mathscr{S}^*_{K_p} \setminus \mathscr{S}_{K_p}$ denote the boundary of the compactification. Since the transition morphisms in the tower $\mathscr{S}^*_{K_p} \to \mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ are finite, the codimension of the boundary is independent of the choice of K_p.

assump:boundary-codimension

Assumption 3.4.6. We assume that the boundary $Z := \mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \setminus \mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)}$ has codimension $\geqslant 2$ in $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$. This is satisfied precisely when no $\mathbf{Q}$-simple factor of G^{ad} giving rise to a non-compact factor of $G(\mathbf{R})$ has $\mathbf{Q}$-rank 1. For instance, in the Siegel case, this is equivalent to excluding the modular curves. For Shimura varieties of exceptional type, the $\mathbf{Q}$-rank of any non-compact $\mathbf{Q}$-simple factor is bounded above by its $\mathbf{R}$-rank, which is at most 2 (for type E_6) or 3 (for type E_7). Our assumption therefore safely includes Shimura varieties of exceptional type, provided that their $\mathbf{Q}$-rank is not 1.

Because $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ is a normal scheme, the codimension $\geqslant 2$ assumption allows us to use classical extension principles, such as the algebraic Hartog's lemma:

prop:classical-hartog

Proposition 3.4.7 (Algebraic Hartog's lemma)**.** *Let R be a normal Noetherian ring, and let $Z \subset \operatorname{Spec} R$ be a closed subset everywhere of codimension $\geqslant 2$. Then the restriction map*

$$R \to \mathrm{H}^0(\operatorname{Spec} R \setminus Z, \mathcal{O}_{\operatorname{Spec} R})$$

is an isomorphism.

`prop:hasse-invariant-extension`

Proposition 3.4.8 (Extension of Hasse invariant)**.** *For ample characters χ, the Hasse invariant* Ha *uniquely extends to a global section of the automorphic line bundle ω^χ over the special fiber of the minimal compactification $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$.*

Proof. By Proposition 3.4.5, the restriction of ω^χ to the open Shimura variety $\mathscr{S}_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ canonically identifies with $\zeta^*\mathcal{V}(\chi)$, meaning Ha defines a regular section of ω^χ over this open locus. Since $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$ is normal and the boundary has codimension $\geqslant 2$ (Assumption 3.4.6), Proposition 3.4.7 implies that Ha extends uniquely to a global section over $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$. □

The following is a generalization of [GK19, Corollary I.2.6] for the μ-ordinary stratum and in particular also recovers [BST25b, Lemma 6.4]:

`cor:affine-extended-mu-ordinary-locus`

Corollary 3.4.9 (Affineness of extended μ-ordinary locus)**.** *The μ-ordinary locus of $\mathscr{S}^*_{\mathcal{G}(\mathbf{Z}_p)} \otimes k$, defined as the non-vanishing locus of the extended Hasse invariant* Ha*, is an affine open subscheme.*

`sec:exceptional-embedding`

3.5. **Embeddings of Shimura varieties of exceptional type.** The exceptional E_6 case presents two related difficulties. First, the Hodge cocharacter of the adjoint E_6-group does not lift to its simply connected cover. Second, the standard inclusion of E_6 in E_7 involves the 27-dimensional minuscule representation, which need not be defined over the field of definition of the group. We deal with the first problem by adjoining a norm-one torus, and with the second by passing to a suitable totally real extension that splits the relevant Tits algebra. Brown algebras then give the required form of E_7 and, more importantly, the inclusion itself. The construction is arranged so that every real factor of the target is of type E VII; this will make its reflex field equal to $\mathbf{Q}$.

`thm:exceptional-e6-embedding`

Theorem 3.5.1 (Exceptional embedding)**.** *Let (G, X) be a $\mathbf{Q}$-simple adjoint Shimura datum with $G \simeq \operatorname{Res}_{F_0/\mathbf{Q}} H_0$, where F_0 is totally real and H_0 is an absolutely simple adjoint group of outer type E_6. Let K_0/F_0 be its quadratic diagram field, namely the least extension over which H_0 becomes inner. Then K_0/F_0 is CM. Let $\pi\colon \widetilde{H}_0 \to H_0$ be the simply connected cover, and set $T := \operatorname{Res}^1_{K_0/F_0} \mathbf{G}_m$. The two finite group schemes $Z(\widetilde{H}_0)$ and $T[3]$ become μ_3 over K_0. The nontrivial element of $\operatorname{Gal}(K_0/F_0)$ interchanges ω_1 and ω_6, and hence acts by inversion on $Z(\widetilde{H}_0)$; it acts by inversion on $T[3]$ as well. The two group schemes therefore have the same descent datum. Fix an identification $Z(\widetilde{H}_0) \simeq T[3]$, and set*

$$L_0 := (\widetilde{H}_0 \times T)/Z(\widetilde{H}_0),$$

where $z \in Z(\widetilde{H}_0)$ is sent to (z, z^{-1}), using the fixed identification $Z(\widetilde{H}_0) \simeq T[3]$ in the second coordinate. Let $\mathbf{L} := \operatorname{Res}_{F_0/\mathbf{Q}} L_0$. Then the following hold:

`thm-item:lift`

(1) We have $\mathbf{L}^{\mathrm{ad}} \simeq G$, and the Shimura datum (G, X) lifts to a Shimura datum $(\mathbf{L}, X_L)$.

`thm-item:e7-embed`

(2) There exist a finite totally real extension F/F_0, an adjoint group G_F over F of type E_7, and a closed embedding $(L_0)_F \hookrightarrow G_F$ which, over $\overline{F}$, is conjugate to the standard inclusion $(E_6^{\mathrm{sc}} \times \mathbf{G}_m)/\mu_3 \hookrightarrow E_7^{\mathrm{ad}}$ described in Remark 3.5.3.

`thm-item:global-datum`

(3) For $\mathbf{G} := \operatorname{Res}_{F/\mathbf{Q}} G_F$, there is a Shimura datum $(\mathbf{G}, Y)$ with reflex field $\mathbf{Q}$ and a closed embedding

$$(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y).$$

`rem:failure-of-naive-weil-restriction-trick`

Remark 3.5.2 (Failure of naive Weil restriction trick)**.** On generic fibers, one can often change the local reflex field by a restriction-of-scalars argument. For example, one may choose a totally real field with a prescribed completion at p and then place the original datum in a larger restriction of scalars. This is not enough for the present application: knowing an integral canonical model for the source does not produce one for the auxiliary target, nor does it produce an integral extension of the generic-fiber map.

The point of Theorem 3.5.1 is that the group and its embedding are constructed over number fields from the outset. For compatible sufficiently small levels, the resulting map of Shimura varieties is a closed immersion [Del71, Proposition 1.15]. By the extension property of the chosen integral canonical models, this map extends uniquely after the bad primes are removed. Ordinary spreading out then shows that the extension remains a closed immersion after removing finitely many further primes. This integral extension is what allows us to pass perfectoidness from the ambient space to the source.

rem:standard-e6-e7-inclusion

Remark 3.5.3 (The standard inclusion of E_6 in E_7)**.** Let G_{ad} be the split adjoint algebraic group of type E_7 over a field k of characteristic zero. The Dynkin diagram of type E_7 contains a unique node (node 7 under the Bourbaki numbering [Bou68, Plate VI]) whose removal yields the Dynkin diagram of type E_6. The corresponding maximal parabolic has a Levi factor

$$M \simeq (E_6^{\mathrm{sc}} \times \mathbf{G}_m)/\mu_3,$$

whose derived group is the simply connected group E_6^{sc}. The quotient by μ_3 is important here. If $z \in Z(E_6^{\mathrm{sc}})$ acts on the 27-dimensional minuscule module by the scalar ζ, then it is paired with ζ^{-1} in the $\mathbf{G}_m$-factor. In other words, the kernel in $E_6^{\mathrm{sc}} \times \mathbf{G}_m$ is the anti-diagonal copy of μ_3. Thus $M \hookrightarrow G_{\mathrm{ad}}$ is well defined up to conjugacy.

Restriction of the adjoint representation gives

$$\mathfrak{e}_7 \simeq \mathfrak{e}_6 \oplus \mathfrak{gl}_1 \oplus V_{27} \oplus V_{27}^{\vee},$$

with torus weights $0, 0, 1, -1$, respectively, where V_{27} is the 27-dimensional minuscule representation and $V_{27}^{\vee}$ is its dual. The Levi and this decomposition are also described explicitly in [Spr06, the discussion preceding Lemma 2.5 and Corollary 2.7]. The decomposition explains the form of the descent used below: the outer involution of E_6 exchanges V_{27} and $V_{27}^{\vee}$, and therefore has to act on the one-dimensional torus by inversion.

lem:e6-archimedean-forms

Lemma 3.5.4. *At every real place v of F_0, the group $H_{0,v}$ is either compact of type $E_{6(-78)}$ or of type $E_{6(-14)}$. In the first case the v-component of a Deligne homomorphism in X is trivial; in the second it defines the exceptional Hermitian symmetric domain* E III*. Moreover, $K_0 \otimes_{F_0,v} \mathbf{R} \simeq \mathbf{C}$ in both cases.*

Proof. Deligne's axioms imply that every noncompact simple real factor of an adjoint Shimura datum is of Hermitian type. Among the real forms of E_6, the only one of Hermitian type is $E_{6(-14)}$, whose symmetric domain is E III; see, for example, [Hel01, Chapter X, Section 6]. If $H_{0,v}$ is compact, the Cartan involution is the identity. Since $H_{0,v}$ is adjoint and the Hodge types of its Lie algebra are confined to $(-1, 1)$, $(0, 0)$, and $(1, -1)$, the corresponding component h_v must be trivial.

Both $E_{6(-14)}$ and the compact form $E_{6(-78)}$ have nontrivial diagram action; see [GP07, Remark 2.7 and Example 4.6(b)]. Thus the quadratic diagram field does not split at any real place: $K_0 \otimes_{F_0,v} \mathbf{R} \simeq \mathbf{C}$ for every v. Since F_0 is totally real, this says exactly that K_0 is CM. □

rem:tits-algebras-general

Remark 3.5.5 (The Tits algebra obstruction)**.** Over an algebraic closure, irreducible representations of a semisimple group are classified by their dominant highest weights. Over the ground field, a Galois-stable highest weight λ need not be represented on an ordinary vector space. Instead, one obtains a module over a central simple algebra A_λ, called the Tits algebra of λ. The representation descends to a vector space over the ground field exactly when A_λ is split; see [Tit71, Théorème 3] or [KMRT98, Chapter VI, Section 27].

After base change to K_0, the group $\widetilde{H}_0$ is of inner type, so the minuscule weight ω_1 is Galois-stable. Its Tits algebra gives a class

$$[A_{\omega_1}] \in \mathrm{Br}(K_0)[3].$$

The exponent divides 3 because $P(E_6)/Q(E_6) \simeq \mathbf{Z}/3\mathbf{Z}$. The algebra associated with the dual minuscule weight ω_6 is the opposite algebra, and all the remaining Tits algebras are determined by these two classes. Splitting A_{ω_1} therefore removes the entire Tits obstruction. This is precisely the hypothesis in Garibaldi's Brown-algebra classification used below.

lem:annihilation-tits

Lemma 3.5.6 (Annihilation of the Tits algebra obstruction)**.** *There exists a finite totally real extension F/F_0 such that $A_{\omega_1} \otimes_{K_0} K_0F$ is split.*

Proof. Let Σ be the finite set of finite places of K_0 at which the local invariant $\mathrm{inv}_w(A_{\omega_1}) \in \mathbf{Q}/\mathbf{Z}$ is non-zero. Since K_0 is a CM field, it has no real completions, and all archimedean completions are complex, where the local Brauer group is trivial.

Let V be the set of places of F_0 below Σ. If V is empty, the class is already split everywhere, and we take $F = F_0$. Suppose that V is nonempty. For each $v \in V$, choose a separable irreducible cubic polynomial over $F_{0,v}$. We also choose, at every real place of F_0, a cubic polynomial with three distinct real roots. Irreducibility of a separable polynomial over a nonarchimedean local field is stable under a sufficiently small perturbation of its coefficients, by Krasner's lemma; the condition of having three distinct real roots is likewise open. Weak approximation on the coefficients therefore gives a single monic cubic $g \in F_0[x]$ satisfying all of these local conditions. Then

$$F := F_0[x]/(g)$$

is a field because g is irreducible at every $v \in V$, and it is totally real because g splits with three distinct roots at every real place. Notice that a single cubic extension handles all places in Σ simultaneously.

Put $K = K_0F$. Since F is totally real and K_0 is CM, the two fields are linearly disjoint over F_0. Let $w \in \Sigma$ lie above $v \in V$. Since g is irreducible over $F_{0,v}$, there is a unique place v' of F above v, and $[F_{v'} : F_{0,v}] = 3$. On the other hand, $[(K_0)_w : F_{0,v}]$ is either 1 or 2. These degrees are coprime, so the two local extensions are linearly disjoint and $(K_0)_w \otimes_{F_{0,v}} F_{v'}$ is a field. Consequently there is a unique place u of K above w, and

$$[K_u : (K_0)_w] = 3.$$

Under a finite extension of local fields, the Brauer invariant is multiplied by the local degree. Hence

$$\mathrm{inv}_u(A_{\omega_1} \otimes_{K_0} K) = 3\,\mathrm{inv}_w(A_{\omega_1}) = 0.$$

The last equality holds because $[A_{\omega_1}]$ has exponent dividing 3. At places outside Σ the invariant was already zero and remains so after base change. The Albert–Brauer–Hasse–Noether theorem [GS17, Theorem 6.5.1] now gives $[A_{\omega_1} \otimes_{K_0} K] = 0$.

We finish by spelling out why this also kills the Tits class over F, rather than only after passage to the inner field K. The character ω_1 generates the character group of the center of the inner E_6-group, so the preceding splitting says that the restriction to K of the Tits class of $\widetilde{H}_{0,F}$ is zero. This Tits class is 3-torsion. Since K/F is quadratic and cor $\circ$ res is multiplication by 2, restriction to K is injective on the 3-primary part. The Tits class over F is therefore already zero. Equivalently, all Tits algebras of $\widetilde{H}_{0,F}$ are trivial; in particular, the algebra for ω_6, which is opposite to that for ω_1, is split as well. □

rem:albert-freudenthal

Remark 3.5.7 (Brown algebras and Freudenthal systems)**.** A Brown algebra is a 56-dimensional central simple structurable algebra. Its type records whether the corresponding E_6-group is inner or outer; in type 2, the associated quadratic field is the diagram field. Garibaldi proves that the connected automorphism group $\mathrm{Aut}^+(\mathcal{B})$ is simply connected of type E_6 with trivial Tits algebras, and, conversely, that every simply connected E_6-group with trivial Tits algebras arises in this way [Gar01, Theorem 2.9(1)]. The converse is essential here: it produces a Brown algebra whose automorphism group is the given group $\widetilde{H}_{0,F}$, rather than merely another outer form with the same diagram field.

The same underlying 56-dimensional vector space carries a nondegenerate Freudenthal triple system. Its structure is given by an alternating bilinear form and a compatible quartic form, both obtained from the Brown algebra after choosing a generator of its one-dimensional space of skew elements. The connected semisimple group $\mathrm{Aut}^+(\mathcal{B})$ fixes this generator: its action on the skew line is a character, and it has no nontrivial characters. It consequently preserves both forms and acts on the triple system. The full isometry group $\mathrm{Inv}(\mathcal{B})$ is simply connected of type E_7 and again has trivial Tits algebras [Gar01, Theorem 4.15(1)]. Thus the Brown algebra supplies the E_6-group, the E_7-group, and the homomorphism between them in one construction. In particular, we do not need to assert that an arbitrary outer E_6-form descends from an Albert algebra of a prescribed kind.

cons:e7-embedding

Construction 3.5.8 (The E_7 embedding over F)**.** Put $K = K_0F$. This is the diagram field of $\widetilde{H}_{0,F}$, and Lemma 3.5.6 shows that all Tits algebras of $\widetilde{H}_{0,F}$ are trivial. Garibaldi's converse theorem therefore gives a Brown algebra $\mathcal{B}/F$ of type 2 and an isomorphism

$$\mathrm{Aut}^+(\mathcal{B}) \simeq \widetilde{H}_{0,F}.$$

Under this isomorphism, the quadratic field attached to $\mathcal{B}$ is necessarily K, since it is recovered from the diagram action on $\widetilde{H}_{0,F}$.

Let $\mathfrak{M}(\mathcal{B})$ denote the Freudenthal triple system associated with $\mathcal{B}$. The action of $\mathrm{Aut}^+(\mathcal{B})$ on $\mathcal{B}$ preserves the bilinear and quartic forms defining $\mathfrak{M}(\mathcal{B})$; see the formulas in [Gar01, Equations (4.3)–(4.4)]. It therefore gives a closed embedding

$$\widetilde{H}_{0,F} \hookrightarrow \widetilde{G}_F := \mathrm{Inv}(\mathcal{B}).$$

By [Gar01, Theorem 4.15(1)], $\widetilde{G}_F$ is simply connected of type E_7 and has trivial Tits algebras. We need an adjoint target, so we put $G_F = \widetilde{G}_F^{\mathrm{ad}}$. The restriction of the central quotient $\widetilde{G}_F \to G_F$ to $\widetilde{H}_{0,F}$ has trivial kernel:

that kernel is contained both in the order-3 center of $\widetilde{H}_{0,F}$ and in the order-2 center of $\widetilde{G}_F$. Hence the composite $\widetilde{H}_{0,F} \hookrightarrow G_F$ is still a closed embedding.

lem:e6-centralizer-torus

Lemma 3.5.9. *Let $S = Z_{G_F}(\widetilde{H}_{0,F})^0$. Then $S \simeq \mathrm{Res}^1_{K/F}\,\mathbf{G}_m$, and multiplication induces an isomorphism*

$$(\widetilde{H}_{0,F} \times S)/Z(\widetilde{H}_{0,F}) \xrightarrow{\sim} Z_{G_F}(S).$$

Under the fixed identification $Z(\widetilde{H}_0) \simeq T[3]$, this identifies $Z_{G_F}(S)$ with $(L_0)_F$.

Proof. We first work over $\overline{F}$. The Brown algebra is then split. After choosing an identification with the split Brown algebra, Garibaldi's formulas for the associated Freudenthal system [Gar01, Equations (4.3)–(4.4)] identify the natural map $\mathrm{Aut}^+(\mathcal{B}) \to \mathrm{Inv}(\mathcal{B})$ with Springer's explicit inclusion $H \subset H_1 = i(\mathbf{G}_m)H$ preceding [Spr06, Lemma 2.5]. That lemma identifies H_1, up to the central quotient already taken in the definition of G_F, with the node-7 Levi described in Remark 3.5.3. Thus the inclusion $\widetilde{H}_{0,\overline{F}} \hookrightarrow (G_F)_{\overline{F}}$ is the standard one. In particular, the connected centralizer $S_{\overline{F}}$ is the one-dimensional central torus of that Levi, and

$$\mathrm{Lie}(G_F)_{\overline{F}} \simeq \mathfrak{e}_6 \oplus \mathfrak{gl}_1 \oplus V_{27} \oplus V_{27}^\vee,$$

where S acts on V_{27} and $V_{27}^\vee$ through mutually inverse characters. This decomposition also determines the F-form of S. In the normalizer of the standard E_6-subgroup, inversion on the central torus exchanges the two 27-dimensional root spaces and induces the unique outer automorphism of E_6; E_7 itself has no diagram automorphism. Thus an element of $\mathrm{Gal}(\overline{F}/F)$ exchanges the two minuscule representations exactly when it acts nontrivially on the E_6-diagram; compare the discussion after [Spr06, Corollary 1.7]. To preserve the displayed decomposition, the same element must exchange the two characters of S, or equivalently act by -1 on $X^*(S_{\overline{F}}) \simeq \mathbf{Z}$. The Galois action on this character group is therefore the sign character of the quadratic extension K/F. A one-dimensional torus with this character module is the norm-one torus, so

$$S \simeq \mathrm{Res}^1_{K/F}\,\mathbf{G}_m.$$

It remains to identify the whole centralizer. Over $\overline{F}$, the standard root datum gives

$$\widetilde{H}_{0,F} \cap S = Z(\widetilde{H}_{0,F}) = S[3]$$

and identifies $Z_{G_F}(S)$ with the node-7 Levi. Since centralizers of tori in connected reductive groups are connected, these identities descend to F. There are two isomorphisms between S and $\mathrm{Res}^1_{K/F}\,\mathbf{G}_m$, differing by inversion. Exactly one of them makes the identification $S[3] \simeq Z(\widetilde{H}_{0,F})$ agree with the fixed identification $T_F[3] \simeq Z(\widetilde{H}_{0,F})$. With that choice, the kernel of multiplication

$$\widetilde{H}_{0,F} \times S \longrightarrow Z_{G_F}(S)$$

is $z \mapsto (z, z^{-1})$. Its quotient is therefore $(L_0)_F$. This proves both the lemma and part 2 of the theorem. ∎

lem:real-e6-e7-pairs

Lemma 3.5.10 (Real standard pairs)**.** *At every real place τ of F, the group $(G_F)_\tau$ is the real form $E_{7(-25)}$. If $v = \tau|_{F_0}$ is noncompact, the resulting real pair is*

$$E_{6(-14)} \cdot U(1) \subset E_{7(-25)},$$

and it gives the holomorphic embedding $\mathrm{E\,III} \hookrightarrow \mathrm{E\,VII}$. *If $H_{0,v}$ is compact, the pair is*

$$E_{6(-78)} \cdot U(1) \subset E_{7(-25)},$$

with the left-hand side a maximal compact subgroup.

Proof. The torus S_τ is the norm-one torus for $\mathbf{C}/\mathbf{R}$, and is therefore compact. We already know from Lemma 3.5.4 that the derived E_6-factor is either $E_{6(-14)}$ or $E_{6(-78)}$. The classification of the real Lie-algebra forms of the standard $E_6 \cdot U(1)$ centralizer pair identifies the corresponding pairs as

$$(\mathfrak{e}_{7(-25)}, \mathfrak{e}_{6(-14)} \oplus \mathfrak{u}(1)) \quad \text{and} \quad (\mathfrak{e}_{7(-25)}, \mathfrak{e}_{6(-78)} \oplus \mathfrak{u}(1));$$

see [Kan91, Section 5.2]. The global central quotients in the statement come from Lemma 3.5.9; the classification is used here only to identify the real forms. This proves the assertion about $(G_F)_\tau$ and also handles the real places lying over compact factors of H_0.

In the compact case, $E_{6(-78)} \cdot U(1)$ is the maximal compact subgroup of $E_{7(-25)}$. In the noncompact case, Ihara's root-theoretic criterion shows that the inclusion $E_{6(-14)} \hookrightarrow E_{7(-25)}$ satisfies the H1 condition and hence induces a holomorphic map of Hermitian symmetric domains [Iha67, Proposition 3 and Section 4.6]. This is the standard embedding $\mathrm{E\,III} \hookrightarrow \mathrm{E\,VII}$. □

cons:lifted-shimura-datum

Construction 3.5.11 (The lifted Shimura datum $(\mathbf{L}, X_L)$)**.** Under the identification $G_{\mathbf{R}} \simeq \prod_{v|\infty} H_{0,v}$, write $h = (h_v)_v \in X$, and let μ_v be the Hodge cocharacter of h_v. By Lemma 3.5.4, $\mu_v = 0$ when $H_{0,v}$ is compact. At a noncompact place, h_v defines one of the two complex orientations on $\mathrm{E\,III}$, and μ_v belongs accordingly to the Weyl orbit of $\omega_1^\vee$ or $\omega_6^\vee$.

For each v, choose a real maximal torus $D_v \subset H_{0,v}$ containing the image of h_v, and let $R_v \subset L_{0,v}$ be its inverse image. Then R_v is a real maximal torus of $L_{0,v}$ and contains the central torus T_v. We will choose the lifted cocharacter inside $X_*(R_{v,\mathbf{C}})$. This ensures that it commutes with its complex conjugate, which is needed when we reconstruct a real Deligne homomorphism below.

Let us first explain the obstruction to lifting μ_v. Fix a real place v and suppress it from the notation. If $P^\vee$ and $Q^\vee$ denote the coweight and coroot lattices of E_6, then

$$P^\vee/Q^\vee \simeq \mathbf{Z}/3\mathbf{Z}.$$

The cocharacter lattice of the simply connected group $\widetilde{H}_{0,\mathbf{C}}$ is $Q^\vee$, whereas the minuscule coweights $\omega_1^\vee$ and $\omega_6^\vee$ have nonzero image in $P^\vee/Q^\vee$. They therefore do not lift to $\widetilde{H}_{0,\mathbf{C}}$.

Passing from $\widetilde{H}_0$ to L_0 removes this obstruction. Quotienting $\widetilde{H}_0 \times T$ by the anti-diagonal 3-torsion allows the nonzero class in $P^\vee/Q^\vee$ to be cancelled by one third of a toric cocharacter. More precisely, the fixed identification of the centers determines an isomorphism

$$\delta\colon P^\vee/Q^\vee \xrightarrow{\sim} \left(\tfrac{1}{3}X_*(T_{\mathbf{C}})\right)/X_*(T_{\mathbf{C}}),$$

and the cocharacter lattice of R_v is

$$X_*(R_{v,\mathbf{C}}) = \left\{(\lambda, a) \in P^\vee \oplus \tfrac{1}{3}X_*(T_{\mathbf{C}}) \;\middle|\; a \bmod X_*(T_{\mathbf{C}}) = \delta(\lambda \bmod Q^\vee)\right\}.$$

Not every lift is compatible with the desired E_7-Shimura datum. After identifying $R_{v,\mathbf{C}}$ with a maximal torus in the standard Levi of Remark 3.5.3, let

$$\mathrm{pr}\colon X_*(R_{v,\mathbf{C}}) \longrightarrow X_*(D_{v,\mathbf{C}}) = P^\vee$$

be the map induced by $L_0 \to H_0$. The Hodge cocharacter for $\mathrm{E\,VII}$ is the special coweight attached to node 7. Let $\mathcal{C}_7$ be the Weyl orbit of $\omega_7^\vee$. This conjugacy class does not depend on the chosen identification with the standard Levi. Moreover, the embeddings into the factors $(G_F)_\tau$ for the different places $\tau \mid v$ are Galois-conjugate, and E_7 has no outer automorphisms, so the same class $\mathcal{C}_7$ is obtained in every factor. Projection to the adjoint E_6-factor gives the following root-data calculation:

$$\mathrm{pr}(\mathcal{C}_7) = \{0\} \cup W(E_6)\omega_1^\vee \cup W(E_6)\omega_6^\vee.$$

The fiber over 0 consists of two primitive, purely central cocharacters. The fibers over the other two orbits contain the corresponding lifts, whose toric components have absolute value one third. To see the calculation from the familiar branching rule, note that E_7 is simply laced and its 56-dimensional representation is minuscule. Its weights may therefore be identified with the coweights in $\mathcal{C}_7$. After choosing the sign of the torus generator, restriction to $E_6^{\mathrm{sc}} \cdot \mathbf{G}_m$ gives

$$56 = 1_3 \oplus 27_1 \oplus 27^\vee_{-1} \oplus 1_{-3}.$$

The two one-dimensional summands project to 0 and give primitive central cocharacters after dividing the displayed charges by 3; the other summands have highest weights ω_1 and ω_6 and give toric components $\pm\frac{1}{3}$. Compare the explicit action in [Spr06, Section 2.1 and Lemma 2.5].

We now choose $\mu_{L,v} \in X_*(R_{v,\mathbf{C}})$ above μ_v whose image in E_7 lies in $\mathcal{C}_7$. At a noncompact place, its toric sign is determined by whether μ_v lies in the $\omega_1^\vee$- or $\omega_6^\vee$-orbit. At a compact place, although $\mu_v = 0$, we take one of the two nonzero primitive central lifts. This last point is essential: a zero lift would give a trivial homomorphism in the noncompact group $E_{7(-25)}$ and would not define an $\mathrm{E\,VII}$ Shimura datum. The two possible nonzero lifts are conjugate in E_7.

Let $\overline{\mu}_{L,v}$ denote the complex-conjugate cocharacter. Since $\mathbf{S}_{\mathbf{C}} \simeq \mathbf{G}_m \times \mathbf{G}_m$, the formula

$$h_{L,v,\mathbf{C}}(z_1, z_2) = \mu_{L,v}(z_1)\,\overline{\mu}_{L,v}(z_2)$$

is a homomorphism because both cocharacters lie in the torus $R_{v,\mathbf{C}}$. Complex conjugation exchanges the two factors of $\mathbf{S}_{\mathbf{C}}$ and exchanges $\mu_{L,v}$ with $\overline{\mu}_{L,v}$, so the formula descends to a homomorphism $h_{L,v}\colon \mathbf{S} \to L_{0,v,\mathbf{R}}$. Its projection to $H_{0,v}$ is h_v. Taking the product over all real places gives

$$h_L = (h_{L,v})_v \colon \mathbf{S} \longrightarrow \mathbf{L}_{\mathbf{R}}.$$

Let X_L be its $\mathbf{L}(\mathbf{R})$-conjugacy class. We finally check compatibility with the target Hodge circle. At a noncompact place, Ihara's H1 condition says that the element defining the complex structure on E VII differs from the image of the corresponding element for E III by an element centralizing E_6. It therefore differs by a cocharacter of the central $U(1)$, and the $\pm\frac{1}{3}$ toric correction selected by $\mathcal{C}_7$ is exactly this difference. At a compact place, the primitive central cocharacter is the Hodge circle in the center of the maximal compact subgroup $E_{6(-78)} \cdot U(1)$. It follows that, for every $\tau \mid v$, the composite of $h_{L,v}$ with $(L_0)_v \to (G_F)_\tau$ is an E VII Deligne homomorphism.

lem:validity-of-lifted-datum

Lemma 3.5.12. *The pair $(\mathbf{L}, X_L)$ is a Shimura datum, and its projection to the adjoint quotient is (G, X).*

Proof. The projection $L_0 \to H_0$ has central kernel T and induces an isomorphism on adjoint groups. After restricting scalars, it gives $\mathbf{L}^{\mathrm{ad}} \simeq G$ and carries h_L to the original homomorphism h.

We check Deligne's axioms through this projection. On the derived Lie algebra, the Hodge decomposition defined by h_L is the same as the one defined by h, so its types are $(-1,1)$, $(0,0)$, and $(1,-1)$. The adjoint action of the central torus is trivial and contributes only type $(0,0)$. Similarly, conjugation by $h_L(i)$ induces on $\mathbf{L}^{\mathrm{ad}}_{\mathbf{R}}$ the same Cartan involution as conjugation by $h(i)$ on $G_{\mathbf{R}}$. The condition that no $\mathbf{Q}$-simple adjoint factor be killed is also inherited from (G, X).

Finally, $Z(\mathbf{L})^0 = \operatorname{Res}_{F_0/\mathbf{Q}} T$ is anisotropic over $\mathbf{R}$. Complex conjugation acts by -1 on its cocharacter group, so the two toric components in the formula for $h_{L,v,\mathbf{C}}$ cancel on the diagonal $\mathbf{G}_m \subset \mathbf{S}$. Thus the weight homomorphism is trivial, and in particular is defined over $\mathbf{Q}$. This proves that $(\mathbf{L}, X_L)$ is a Shimura datum and establishes part 1. □

lem:global-e7-datum

Lemma 3.5.13. *The group $\mathbf{G} := \operatorname{Res}_{F/\mathbf{Q}} G_F$ admits a Shimura datum $(\mathbf{G}, Y)$ with reflex field $\mathbf{Q}$, and there is a closed embedding $(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y)$.*

Proof. Base change from F_0 to F gives a diagonal map from L_0 to its copies over the embeddings of F above a fixed embedding of F_0. After restriction of scalars, and then composition with the embedding from Lemma 3.5.9, we obtain a closed embedding

$$j\colon \mathbf{L} = \operatorname{Res}_{F_0/\mathbf{Q}} L_0 \longrightarrow \operatorname{Res}_{F/\mathbf{Q}}(L_0)_F \longrightarrow \operatorname{Res}_{F/\mathbf{Q}} G_F = \mathbf{G}.$$

On real points, $\mathbf{G}_{\mathbf{R}}$ is the product of the groups $(G_F)_\tau$ over the real embeddings $\tau\colon F \hookrightarrow \mathbf{R}$. Let $v = \tau|_{F_0}$, let $j_\tau\colon (L_0)_v \to (G_F)_\tau$ be the induced real homomorphism, and set

$$y_\tau = j_\tau \circ h_{L,v}.$$

By Construction 3.5.11, each y_τ is an E VII Deligne homomorphism. Their product

$$y = (y_\tau)_\tau \colon \mathbf{S} \longrightarrow \mathbf{G}_{\mathbf{R}}$$

therefore satisfies the first two of Deligne's axioms, since these may be checked on the real simple factors. For the third, note that G_F is absolutely simple, so $\operatorname{Res}_{F/\mathbf{Q}} G_F$ is $\mathbf{Q}$-simple, and every component y_τ is nontrivial. Thus y is nontrivial on the unique $\mathbf{Q}$-simple adjoint factor of $\mathbf{G}$. It follows that $(\mathbf{G}, Y)$ is a Shimura datum, where Y is the $\mathbf{G}(\mathbf{R})$-conjugacy class of y. By construction $j_{\mathbf{R}} \circ h_L = y$, so j is an embedding of Shimura data, not merely an embedding of the underlying algebraic groups.

It remains to compute the reflex field. For an adjoint Shimura datum, Deligne describes it as the fixed field of the Galois subgroup preserving the marked special nodes on the geometric Dynkin diagram [Del71, Section 3.7]. Over $\overline{\mathbf{Q}}$, the Dynkin diagram of $\mathbf{G}$ is the disjoint union of E_7-diagrams indexed by the embeddings $F \hookrightarrow \overline{\mathbf{Q}}$. The cocharacter of y marks node 7 on every one of these components. This includes the components lying above compact factors of H_0, precisely because we used a nonzero central lift there.

The absolute Galois group permutes the components, but it does not change the marked node: the E_7-diagram has no nontrivial automorphism. Complex conjugation also preserves the cocharacter class, since the longest Weyl element of E_7 acts by -1, so $-\omega_7^\vee$ is Weyl-conjugate to $\omega_7^\vee$. Hence the entire tuple of cocharacter classes is fixed by $\operatorname{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. It follows that $E(\mathbf{G}, Y) = \mathbf{Q}$, proving part 3. □

4. Canonical subgroups

sec:canonical-subgroups

In this section, we prove the overconvergence of canonical subgroups integrally and produce lifts of Frobenii. A standing assumption is that μ is defined over $\mathbf{Z}_p$.

Setup 4.1. Maintain Setup 2.3.1. In what follows, $\mathcal{G}$ is *reductive*, μ is defined over $\mathbf{Z}_p$ so that $\varphi(\mu) = \mu$, and R is always a p-torsion free perfectoid ring unless otherwise stated. In particular, we say 'ordinary' instead of 'μ-ordinary'. We denote the tilt of p as $t \in R^\flat$, and the Frobenius on $R^\flat$ by σ. Let $(\mathcal{E}_1, m_1)$ be a 1-truncated $(\mathcal{G}, \mu)$-window over R. Because μ is minuscule, the maximal parabolic $\mathcal{P}_\mu^+$ corresponds to a fundamental weight χ_{fun} of $\mathcal{G}$. We fix a Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi))$ of fundamental weight $\chi = \chi_{\mathrm{fun}}$, supplied by Remark 2.4.6. This defines a Hasse invariant on $\mathrm{BT}_1^{\mathcal{G},\mu} \otimes k$ by Theorem 2.3.5 which we also denote by Ha. For instance, $\nu_t(\mathrm{Ha}(\mathcal{E}_1))$ denotes the t-adic valuation of the Hasse invariant of the $\mathcal{G}$-zip corresponding to $(\mathcal{E}_1, m_1)$ as in Remark 2.3.12 (we set $\nu_t(0) = \infty$).

Example 4.2 (Hasse invariants in the GL_n case)**.** Let us make this explicit in the case of $(\mathcal{G}, \mu) = (\mathrm{GL}_h, \mu_d)$, where $\mu_d(x) = \mathrm{diag}(x, \dots, x, 1, \dots, 1)$ with d copies of x. The associated parabolic subgroup $\mathcal{P}_{\mu_d}^+$ consists of block upper-triangular matrices with block sizes d and $h-d$, and its Levi quotient is $\mathcal{M}_{\mu_d} \cong \mathrm{GL}_d \times \mathrm{GL}_{h-d}$. The fundamental weight χ_{fun} corresponding to the unique simple root of GL_h not in $\mathcal{M}_{\mu_d}$ is the character $\mathcal{M}_{\mu_d} \to \mathbf{G}_{\mathrm{m}}$ given by the determinant of the top-left GL_d-block. In the context of a p-divisible group of height h and dimension d over a characteristic p ring, the reduction of structure of its de Rham cohomology to $\mathcal{P}_{\mu_d}^+$ corresponds to the Hodge filtration, and the line bundle $\mathcal{V}(\chi_{\mathrm{fun}})$ evaluates to the top exterior power of the Lie algebra, $\det(\mathrm{Lie}\, G) = \bigwedge^d \mathrm{Lie}\, G$. Under this identification, the Hasse invariant $\mathrm{Ha} \in \mathrm{H}^0(\mathcal{G}\text{-zip}_\mu, \mathcal{V}(\chi_{\mathrm{fun}}))$ corresponds exactly to the map $\det(\mathrm{Lie}\, G^{(p)}) \to \det(\mathrm{Lie}\, G)$ induced by the Verschiebung homomorphism $V\colon G^{(p)} \to G$. Thus, the chosen Hasse invariant is precisely the classical Hasse invariant, and $\nu_p(\mathrm{Ha}(G))$ recovers what is often called the *Hodge height*.

4.1. Canonical subgroups for BT_1.

Recall that we have canonical reductions $T_{\mathrm{dR}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}_\mu^-$ and $T_{\mathrm{conj}}(\mathcal{E}_1)\colon \operatorname{Spec} R^\flat/tR^\flat \to \mathcal{E}_1/\mathcal{P}_\mu^+$ as in Remark 2.3.12. Denote by $\mathcal{E}_{1,\mathrm{dR}} \subset \mathcal{E}_1|_{R^\flat/tR^\flat}$ the $\mathcal{P}_\mu^-$-torsor induced by $T_{\mathrm{dR}}(\mathcal{E}_1)$.

Notation 4.1.1. Let $\mathbf{Z}_p^{\mathrm{cycl}}$ be the ring of integers of the (p-adic completion of) maximal cyclotomic extension of $\mathbf{Q}_p$. Note that $\mathbf{Z}_p^{\mathrm{cycl}}$ contains elements of p-adic valuation $\frac{a}{(p-1)p^n}$ for any integers $a, n \geqslant 0$.

rem:replace

Remark 4.1.2. In what follows, $\mathbf{Z}_p^{\mathrm{cycl}}$ can be replaced by any (possibly finite) totally ramified extension of an unramified ring of integers as long as all the p-adic valuations make sense.

The goal of this section is to prove

thm:canonical-subgroup-for-1-truncated-apertures

Theorem 4.1.3 (Canonical subgroup for 1-truncated apertures)**.** *Let $\mathcal{E}_1 \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{F}_p$ be a $\mathcal{G}$-torsor representing a 1-truncated $(\mathcal{G}, \mu)$-aperture $\mathcal{A}\colon \mathfrak{X} \to \mathrm{BT}_1^{\mathcal{G},\mu}$ for a flat p-adic formal scheme $\mathfrak{X}$ over $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$. Let $x_{\mathrm{dR}}\colon \mathfrak{X}_{\mathbf{F}_p} \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{F}_p$ be the mod p version of the de Rham section from Recollection 2.1.12. Let $(\mathcal{Q} := (x_{\mathrm{dR}})^*\mathcal{E}_1, \mathcal{Q}^+, \mathcal{Q}^-, \alpha) \in \mathcal{G}\text{-zip}_\mu(\mathfrak{X}_{\mathbf{F}_p})$ be the $\mathcal{G}$-zip corresponding to $\mathcal{A}|_{\mathfrak{X}_{\mathbf{F}_p}}$. If $\frac{p}{p+1} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{\mathfrak{X}_{\mathbf{F}_p}}))$, then there exists a unique reduction $\mathcal{E}_1^+ \subset \mathcal{E}_1$ to $\mathcal{P}_\mu^+$ such that $(x_{\mathrm{dR}})^*\mathcal{E}_1^+$ agrees with $\mathcal{Q}^+$ modulo $p^{1-\varepsilon}$.*

See Section 4.3 for a discussion on how the above relates to classical results on canonical subgroups of Barsotti–Tate groups. We will first prove it for perfectoid rings and then use a moduli argument combined with quasisyntomic descent to deduce the general case.

rem:canonical-big-cell

Remark 4.1.4 (Big cell in relative Grassmannian)**.** Set $\mathrm{Gr}^+_{\mathcal{E}_1,\mu} := \mathcal{E}_1/\mathcal{P}_\mu^+$. There is a canonical open dense 'big cell'

$$\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat} \subset \mathrm{Gr}^+_{\mathcal{E}_1,\mu}|_{R^\flat/tR^\flat}$$

given by the image of the open embedding

$$\mathcal{E}_{1,\mathrm{dR}} \times^{\mathcal{P}_\mu^-} (\mathcal{P}_\mu^- \mathcal{P}_\mu^+/\mathcal{P}_\mu^+) \hookrightarrow \mathcal{E}_{1,\mathrm{dR}} \times^{\mathcal{P}_\mu^-} (\mathcal{G}/\mathcal{P}_\mu^+) \xrightarrow{\sim} \mathrm{Gr}^+_{\mathcal{E}_1,\mu}|_{R^\flat/tR^\flat}.$$

This open subscheme $\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat}$ is canonically a torsor for the unipotent group scheme $\mathcal{E}_{1,\mathrm{dR}} \times^{\mathcal{P}_\mu^-} \mathcal{U}_\mu^-$. Observe that $(\mathcal{E}_1, m_1)$ is ordinary if and only if $T_{\mathrm{conj}}(\mathcal{E}_1)$ factors through $\mathcal{C}_{\mathrm{big}}$.

Definition 4.1.5 (m_1-stable reduction of structure)**.** A reduction of structure $\mathcal{E}_1^+ \subset \mathcal{E}_1$ to a smooth subgroup scheme $\mathcal{H} \subset \mathcal{G}$ is said to be m_1**-stable** if the restriction of the modification m_1 to $\varphi^*\mathcal{E}_1^+[1/t]$ factors through $\mathcal{E}_1^+[1/t]$, thereby inducing an isomorphism $m_1^+ \colon \varphi^*\mathcal{E}_1^+[1/t] \xrightarrow{\sim} \mathcal{E}_1^+[1/t]$ of $\mathcal{H}$-torsors over $R^\flat[1/t]$.

Proposition 4.1.6. *$(\mathcal{E}_1, m_1)$ is ordinary if and only if there exists a m_1-stable reduction $\mathcal{E}_1^+ \subset \mathcal{E}_1$ to $\mathcal{P}_\mu^+$ lifting $T_{\mathrm{conj}}(\mathcal{E}_1)$.*

Proof. If $(\mathcal{E}_1, m_1)$ is ordinary then the desired result is clear because such a window reduces structure to a 1-truncated $(\mathcal{P}_\mu^+, \mu)$-window. Conversely, suppose $\mathcal{E}_1^+ \subset \mathcal{E}_1$ is a m_1-stable reduction of structure as in the proposition inducing a modification $m_1^+ \colon \varphi^*\mathcal{E}_1^+[1/t] \xrightarrow{\sim} \mathcal{E}_1^+[1/t]$. By assumption, the reduction of structure as in Remark 2.2.10 of $\mathcal{E}_1^+|_{R^\flat/tR^\flat}$ induced by m_1^+ agrees with $T_{\mathrm{conj}}(\mathcal{E}_1)$. This implies that m_1^+ has type μ as a modification of $\mathcal{P}_\mu^+$-torsors which is to say that $(\mathcal{E}_1^+, m_1^+)$ is a 1-truncated $(\mathcal{P}_\mu^+, \mu)$-window. □

We now prove the key overconvergence result: `prop:fixed-point`

Proposition 4.1.7 (Overconvergence for 1-truncated windows)**.** *If $\frac{p}{p+1} > \varepsilon \geqslant \nu_t(\mathrm{Ha}\,\mathcal{E}_1)$, then there exists a unique m_1-stable reduction $\mathcal{E}_1^+ \subset \mathcal{E}_1$ to $\mathcal{P}_\mu^+$ such that $\mathcal{E}_1^+$ agrees with $T_{\mathrm{conj}}(\mathcal{E}_1)$ modulo $t^{1-\varepsilon}$.*

Proof. The modification m_1 induces an isomorphism

$$\mathrm{Gr}^+_{\varphi^*\mathcal{E}_1,\mu}[1/t] \xrightarrow{\sim} \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t],$$

which when composed with the relative Frobenius yields a σ-linear endomorphism

$$\Psi \colon \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t] \to \mathrm{Gr}^+_{\mathcal{E}_1,\mu}[1/t].$$

Choose an $R^\flat$-point of $\mathcal{G}\text{-zip}_\mu$ lifting the canonical map $\operatorname{Spec} R^\flat/tR^\flat \to \mathcal{G}\text{-zip}_\mu$, which is possible, for instance, by the smoothness of $\mathcal{G}\text{-zip}_\mu$. This gives an $R^\flat$-point $y_0 \in \mathrm{Gr}^+_{\mathcal{E}_1,\mu}(R^\flat)$ lifting $T_{\mathrm{conj}}(\mathcal{E}_1)$ and defines an open dense "big cell" $\mathcal{C}_{\mathrm{big}} \subset \mathrm{Gr}^+_{\mathcal{E}_1,\mu}$ as in Remark 4.1.4 lifting $\mathcal{C}_{\mathrm{big}}|_{R^\flat/tR^\flat}$. Since $\mathrm{Ha}(\mathcal{E}_1)$ has positive t-adic valuation, it becomes invertible in $R^\flat[1/t]$, which implies that $y_0[1/t] \in \mathcal{C}_{\mathrm{big}}[1/t]$. We wish to show that there exists $x \in \mathrm{Gr}^+_{\mathcal{E}_1,\mu}(R^\flat[1/t])$ satisfying

$$\Psi(x) = x \quad \text{and} \quad x \equiv y_0 \pmod{t^{1-\varepsilon}}.$$

After p-completely étale localization on R, we may assume that $\mathcal{E}_1$ is a trivial torsor. Let the modification m_1 be represented by $\mu(t)M \in \mathcal{G}(R^\flat[1/t])$ where $M \in \mathcal{G}(R^\flat)$. By Remarks 2.3.13 and 2.4.6, we have $M \in \mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$ since the Hasse invariant has positive t-adic valuation. Denote by $\mathfrak{u}_\mu^-$ the Lie algebra of the unipotent group $\mathcal{U}_\mu^-$. By centering coordinates of the big cell $\mathcal{C}_{\mathrm{big}}$ around y_0, we identify $\mathcal{C}_{\mathrm{big}}(R^\flat[1/t])$ with $\mathfrak{u}_\mu^-(R^\flat[1/t])$ so that $y_0[1/t]$ corresponds to 0. Under this coordinate chart, any point $x \in \mathcal{C}_{\mathrm{big}}(R^\flat[1/t])$ is represented by the coset $\exp(Z)\mathcal{P}_\mu^+$ for a unique Lie algebra element $Z \in \mathfrak{u}_\mu^-(R^\flat[1/t])$. Since the action of Ψ sends $\exp(Z)\mathcal{P}_\mu^+$ to $\mu(t)M\exp(\sigma(Z))\mathcal{P}_\mu^+$, the fixed-point condition $\Psi(x) = x$ is equivalent to the containment

$$\exp(Z)^{-1}\mu(t)M\exp(\sigma(Z)) \in \mathcal{P}_\mu^+(R^\flat[1/t]).$$

Using the conjugation identity $\mu(t)\exp(Z)\mu(t)^{-1} = \exp(t^{-1}Z)$ together with $\mu(t) \in \mathcal{P}_\mu^+(R^\flat[1/t])$, this containment simplifies to

$$M\exp(\sigma(Z)) \in \exp(t^{-1}Z)\mathcal{P}_\mu^+(R^\flat[1/t]).$$

Let $\pi_- \colon \mathcal{P}_\mu^-\mathcal{P}_\mu^+ \twoheadrightarrow \mathcal{U}_\mu^-$ denote the projection along $\mathcal{P}_\mu^+$. Assuming $M\exp(\sigma(Z))$ lies in the big cell $\mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$, taking its unipotent projection yields $\exp(t^{-1}Z) = \pi_-(M\exp(\sigma(Z)))$. Applying the logarithm $\log \colon \mathcal{U}_\mu^- \xrightarrow{\sim} \mathfrak{u}_\mu^-$ and multiplying by t, we define the operator

$$\mathbf{F}(Z) := t\log\pi_-\big(M\exp(\sigma(Z))\big),$$

so that finding a fixed point of Ψ reduces to solving the equation $Z = \mathbf{F}(Z)$.

A priori, $\mathbf{F}$ is only well-defined on points Z where $M\exp(\sigma(Z))$ lies in the big cell $\mathcal{P}_\mu^-\mathcal{P}_\mu^+(R^\flat[1/t])$. We show that this holds on the closed t-adic ball of radius $1-\varepsilon$, defined by

$$\mathcal{B}_{1-\varepsilon} := \{Z \in \mathfrak{u}_\mu^-(R^\flat[1/t]) \mid \nu_t(Z) \geqslant 1-\varepsilon\}.$$

Let $\Delta \in \mathcal{O}(\mathcal{G}_{R^\flat})$ be the regular function descending to the Hasse invariant Ha that cuts out the big cell $\mathcal{P}_\mu^-\mathcal{P}_\mu^+$. By assumption, $\nu_t(\Delta(M)) \leqslant \varepsilon$. The map $X \mapsto \Delta(M\exp(X))$ is a polynomial function on the Lie algebra $\mathfrak{u}_\mu^-(R^\flat)$.

For any $Z \in \mathcal{B}_{1-\varepsilon}$, the Frobenius action multiplies valuations by p, so $\nu_t(\sigma(Z)) \geqslant p(1-\varepsilon)$. Taylor expanding this polynomial around $X = 0$ yields

$$\Delta(M \exp(\sigma(Z))) = \Delta(M) + P(\sigma(Z)),$$

where P is a polynomial without a constant term whose coefficients lie in $R^\flat$. Because $\sigma(Z)$ has valuation at least $p(1-\varepsilon)$, we obtain the valuation bound

$$\nu_t\big(\Delta(M \exp(\sigma(Z))) - \Delta(M)\big) \geqslant \nu_t(\sigma(Z)) \geqslant p(1-\varepsilon).$$

Since $\varepsilon < \frac{p}{p+1} < 1$, we have $p(1-\varepsilon) > \varepsilon \geqslant \nu_t(\Delta(M))$. By the ultrametric inequality,

$$\nu_t\big(\Delta(M \exp(\sigma(Z)))\big) = \min\big(\nu_t(\Delta(M)),\, \nu_t(P(\sigma(Z)))\big) = \nu_t(\Delta(M)) \leqslant \varepsilon.$$

In particular, $\Delta(M \exp(\sigma(Z)))$ is invertible in $R^\flat[1/t]$, which proves that $M \exp(\sigma(Z)) \in \mathcal{P}_\mu^- \mathcal{P}_\mu^+(R^\flat[1/t])$ and hence that $\mathbf{F}$ is well-defined on $\mathcal{B}_{1-\varepsilon}$.

To compute $\mathbf{F}(Z)$ explicitly, let $V = V_0 \oplus V_1 \oplus \cdots$ be the minuscule fundamental representation of $\mathcal{G}$ corresponding to $\mathcal{P}_\mu^+$, graded relative to the Levi subgroup with highest weight vector $v_0 \in V_0$. For any $y = U_y P_y \in \mathcal{P}_\mu^- \mathcal{P}_\mu^+(R^\flat[1/t])$, the canonical projection $\Pi\colon V \to V_0 \oplus V_1$ satisfies

$$\Pi(y v_0) = \Delta(y)\big(v_0 + (\log U_y) v_0\big).$$

Applying this to M shows that $\Delta(M) \log U$ is integral, where $\log U := \log \pi_-(M)$, so the t-adic poles of $\log U$ have order at most $\nu_t(\Delta(M)) \leqslant \varepsilon$. For $y := M \exp(\sigma Z)$, expanding $\Pi(y v_0) = \Pi(M v_0) + \Pi(M \sigma Z v_0)$ in components along $V_0 \oplus V_1 \simeq R^\flat \oplus \mathfrak{u}_\mu^-$ yields integral linear maps $a\colon \mathfrak{u}_\mu^- \to R^\flat$ and $X\colon \mathfrak{u}_\mu^- \to \mathfrak{u}_\mu^-$ such that

$$\Delta(y) = \Delta(M) + a(\sigma Z) \quad \text{and} \quad \Delta(y) \log U_y = \Delta(M) \log U + X(\sigma Z).$$

Defining the linear operator $B(W) := X(W) - a(W) \log U$, dividing by $\Delta(y)$, and multiplying by t, we obtain

$$\mathbf{F}(Z) = t \log U + t \frac{B(\sigma Z)}{\Delta(M) + a(\sigma Z)}.$$

Since B introduces t-adic poles of order at most $\nu_t(\Delta(M)) \leqslant \varepsilon$, the initial point $Z_1 := \mathbf{F}(0) = t \log U$ satisfies $\nu_t(Z_1) \geqslant 1 - \varepsilon$, proving $\mathbf{F}(0) \in \mathcal{B}_{1-\varepsilon}$.

Finally, for any $Z_1, Z_2 \in \mathcal{B}_{1-\varepsilon}$ with $\Delta_i := \Delta(M) + a(\sigma Z_i)$, taking the difference gives

$$\mathbf{F}(Z_1) - \mathbf{F}(Z_2) = t \frac{\Delta(M) B(\sigma(Z_1 - Z_2)) + B(\sigma Z_1) a(\sigma(Z_2 - Z_1)) + a(\sigma Z_1) B(\sigma(Z_1 - Z_2))}{\Delta_1 \Delta_2}.$$

The denominator has valuation $2\nu_t(\Delta(M)) \leqslant 2\varepsilon$. In the numerator, since B introduces poles of order at most $\nu_t(\Delta(M))$, the first term has valuation $\geqslant p\nu_t(Z_1 - Z_2)$. Because $\nu_t(\sigma Z_i) \geqslant p(1-\varepsilon)$ and $\varepsilon < \frac{p}{p+1}$, we have $p(1-\varepsilon) - \nu_t(\Delta(M)) \geqslant p(1-\varepsilon) - \varepsilon > 0$, so the cross terms also have valuation strictly greater than $p\nu_t(Z_1 - Z_2)$. Combining these bounds yields

$$\nu_t(\mathbf{F}(Z_1) - \mathbf{F}(Z_2)) \geqslant 1 - 2\varepsilon + p\nu_t(Z_1 - Z_2).$$

Strict contraction requires $(p-1)\nu_t(Z_1 - Z_2) > 2\varepsilon - 1$, which holds for all $Z_i \in \mathcal{B}_{1-\varepsilon}$ because $(p-1)(1-\varepsilon) > 2\varepsilon - 1 \iff \varepsilon < \frac{p}{p+1}$. Since $\mathbf{F}(0) \in \mathcal{B}_{1-\varepsilon}$ and $\mathbf{F}$ is strictly contractive, $\mathbf{F}$ maps $\mathcal{B}_{1-\varepsilon}$ into itself and has a unique fixed point, proving the proposition. □

Remark 4.1.8 (Sharpness of the bound)**.** The bound $t^{1-\varepsilon}$ is sharp in the generic case where $\nu_t(\Delta(M)) = \varepsilon$. In the proof, the canonical subgroup corresponds to the limit Z of the sequence Z_n obtained by iterating the operator $\mathbf{F}$, starting at $Z_0 = 0$. The first step of the iteration is $Z_1 = \mathbf{F}(0) = t \log U$. As shown, $\Delta(M) \log U$ is an integral matrix, and $\nu_t(\Delta(M)) = \varepsilon$, so the t-adic poles of $\log U$ are bounded by ε. However, for a generic matrix M on the locus where $\nu_t(\Delta(M)) = \varepsilon$, there is no reason for $\Delta(M) \log U$ to be divisible by any positive powers of t. Thus, generically, the exact valuation is $\nu_t(Z_1) = 1 - \varepsilon$. Because $\mathbf{F}$ is a strict contraction, the distance between subsequent iterates is strictly greater than the initial step size. Indeed, using $\nu_t(Z_2 - Z_1) \geqslant 1 - 2\varepsilon + p(1-\varepsilon)$ and the assumption $\varepsilon < \frac{p}{p+1}$, a quick check shows $\nu_t(Z_2 - Z_1) > 1 - \varepsilon$. By the non-Archimedean triangle inequality, terms of strictly higher valuation do not alter the exact valuation of the sum $Z = Z_1 + (Z_2 - Z_1) + \cdots$. Therefore, generically, $\nu_t(Z) = \nu_t(Z_1) = 1 - \varepsilon$. This implies that the fixed point lies at a distance of exactly $1 - \varepsilon$ from 0 (which corresponds to $T_{\mathrm{conj}}(\mathcal{E}_1)$).

Remark 4.1.9 (Dependence on the choice of Hasse invariant)**.** If one instead works with a Hasse invariant of a different weight χ, the overconvergence bound becomes strictly worse. Because μ is minuscule, the associated parabolic $\mathcal{P}_\mu^+$ is maximal, meaning the Picard group of the flag variety $\mathcal{G}/\mathcal{P}_\mu^+$ has rank 1 and its ample cone is generated by the line bundle associated to χ_{fun}. Consequently, any ample character χ on the Levi subgroup $\mathcal{M}_\mu$ must restrict to a strictly positive integer multiple of χ_{fun} on the derived group. Thus, for χ to be ample, it must be of the form $k\chi_{\mathrm{fun}} + \eta$ for some integer $k \geqslant 1$ and a character η of the whole group $\mathcal{G}$. The generalized minor Δ cutting out the big cell would then be exactly Δ_{fun}^k (up to a unit), so its valuation would be $\varepsilon' := k\nu_t(\Delta_{\mathrm{fun}}(M))$. In this case, the bound on the poles of the iteration operator is still determined by $\Delta_{\mathrm{fun}}(M)^2$, which translates to $\frac{2}{k}\varepsilon'$. The strict contraction inequality would then require $(p-1)(1-\varepsilon') > \frac{2}{k}\varepsilon' - 1$, leading to the condition $\varepsilon' < \frac{kp}{kp+2-k}$. However, translating this bound back to the valuation of the fundamental Hasse invariant $\varepsilon = \varepsilon'/k$, the condition restricts the locus to $\varepsilon < \frac{p}{kp+2-k}$. Since the function $k \mapsto \frac{p}{kp+2-k}$ is strictly decreasing for $p \geqslant 2$, choosing $k = 1$ (the fundamental weight) yields $\frac{p}{p+1}$, which provides the largest and therefore optimal radius of overconvergence.

`cons:epsilon-neighborhood-of-ordinary-locus`

Construction 4.1.10 (ε-neighbourhood of ordinary locus)**.** Fix an element $p^\varepsilon \in \mathbf{Z}_p^{\mathrm{cycl}}$ with p-adic valuation $0 \leqslant \varepsilon < 1$. We define the p-adic formal stack $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ over $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$. For any p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra S, the groupoid $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}(S)$ consists of tuples $(f, g, \alpha, \widetilde{\mathrm{Ha}}, s)$ where:

- $f \in \mathrm{BT}_1^{\mathcal{G},\mu}(S)$ is a 1-truncated $(\mathcal{G},\mu)$-aperture over S, yielding a map $f_{\mathrm{zip}} \colon \operatorname{Spec} S/pS \to \mathcal{G}\text{-zip}_\mu$,
- $g \colon \operatorname{Spf} S' \to \operatorname{Spf} S$ is a p-completely étale cover,
- $\alpha \colon \mathcal{O}_{S'/pS'} \xrightarrow{\sim} (g \otimes \mathbf{F}_p)^*(f_{\mathrm{zip}})^*\mathcal{V}(\chi)$ is a line bundle trivialization over $\operatorname{Spec} S'/pS'$,
- $\widetilde{\mathrm{Ha}} \in S'$ is a lift of $\alpha^{-1}(g \otimes \mathbf{F}_p)^* \mathrm{Ha} \in S'/pS'$ to characteristic zero,
- $s \in S'$ is an element witnessing $s\widetilde{\mathrm{Ha}} = p^\varepsilon$.

A morphism $(f_1, g_1, \alpha_1, \widetilde{\mathrm{Ha}}_1, s_1) \to (f_2, g_2, \alpha_2, \widetilde{\mathrm{Ha}}_2, s_2)$ is the data of an equivalence class of tuples $(\gamma, g_{12}, h_1, h_2)$ where:

- $\gamma \colon f_1 \xrightarrow{\sim} f_2$ is an isomorphism in $\mathrm{BT}_1^{\mathcal{G},\mu}(S)$,
- $g_{12} \colon \operatorname{Spf} S'' \to \operatorname{Spf} S$ is a p-completely étale cover,
- $h_i \colon \operatorname{Spf} S'' \to \operatorname{Spf} S_i'$ are p-completely étale morphisms factoring the covers (so $g_1 \circ h_1 = g_{12} = g_2 \circ h_2$),

such that the local witnesses are strictly compatible under pullback. Explicitly, γ induces an isomorphism of line bundles $\gamma_{\mathrm{zip}}^* \colon (f_{2,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{\sim} (f_{1,\mathrm{zip}})^*\mathcal{V}(\chi)$ over $\operatorname{Spec} S/pS$. Pulling back to $\operatorname{Spec} S''/pS''$, the trivializations are related by a unit $u \in (S''/pS'')^\times$ uniquely determined by the composition

$$u \colon \mathcal{O}_{S''/pS''} \xrightarrow{(h_2 \otimes \mathbf{F}_p)^*\alpha_2} (g_{12} \otimes \mathbf{F}_p)^*(f_{2,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{(g_{12} \otimes \mathbf{F}_p)^*\gamma_{\mathrm{zip}}^*} (g_{12} \otimes \mathbf{F}_p)^*(f_{1,\mathrm{zip}})^*\mathcal{V}(\chi) \xrightarrow{((h_1 \otimes \mathbf{F}_p)^*\alpha_1)^{-1}} \mathcal{O}_{S''/pS''}.$$

If $\widetilde{u} \in (S'')^\times$ is any lift of u to characteristic zero, the pulled-back local lifts of the Hasse invariant satisfy $h_2^*\widetilde{\mathrm{Ha}}_2 \equiv \widetilde{u}h_1^*\widetilde{\mathrm{Ha}}_1 \pmod{p}$, meaning there exists some $\delta \in S''$ such that $h_2^*\widetilde{\mathrm{Ha}}_2 = \widetilde{u}h_1^*\widetilde{\mathrm{Ha}}_1 + p\delta$. The compatibility of witnesses requires the strict equality

$$h_2^* s_2 = h_1^* s_1 \widetilde{u}^{-1}(1 + h_1^* s_1 \widetilde{u}^{-1} p^{1-\varepsilon}\delta)^{-1}$$

in S''. Because $1 - \varepsilon > 0$ and S'' is p-complete, this geometric series converges perfectly. Two such tuples $(\gamma, g_{12}, h_1, h_2)$ and $(\gamma', g_{12}', h_1', h_2')$ are equivalent if $\gamma = \gamma'$ and there exists a further common p-completely étale refinement dominating both g_{12} and g_{12}' where the connecting maps commute.

`prop:lci-stack`

Proposition 4.1.11. *There exists a smooth atlas* $\operatorname{Spf} A_0 \to \mathrm{BT}_1^{\mathcal{G},\mu}$ *and a nonzerodivisor* $f \in A_0$ *such that for* $A := A_0 \otimes_{\mathbf{Z}_p} \mathbf{Z}_p^{\mathrm{cycl}}$*, there is a Cartesian diagram*

$$\begin{tikzcd}
\operatorname{Spf} A\langle s\rangle/(sf - p^\varepsilon) \arrow[r] \arrow[d] & \mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu} \arrow[d] \\
\operatorname{Spf} A \arrow[r] & \mathrm{BT}_1^{\mathcal{G},\mu} \otimes \mathbf{Z}_p^{\mathrm{cycl}}
\end{tikzcd}$$

In particular, $\mathrm{BT}_{1,|\mathrm{Ha}|\geqslant\varepsilon}^{\mathcal{G},\mu}$ *is a syntomic (i.e., flat and locally complete intersection) p-adic formal Artin stack over* $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$ *with affine diagonal.*

Proof. We show that the natural forgetful morphism $\pi\colon \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon} \to \mathrm{BT}^{\mathcal{G},\mu}_{1} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ is representable by affine formal schemes. Given a morphism from a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R to $\mathrm{BT}^{\mathcal{G},\mu}_{1}$, we must show that the fiber product $\mathcal{X} := \mathrm{Spf}\, R \times_{\mathrm{BT}^{\mathcal{G},\mu}_{1}} \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ is an affine formal scheme. Because relative affine formal schemes satisfy effective descent in the p-completely étale topology, we may pass to a p-completely étale cover of $\mathrm{Spf}\, R$ to assume the pullback of $\mathcal{V}(\chi)$ is trivialized, identifying the Hasse invariant with an element $\mathrm{Ha}_R \in R/pR$. Then $\mathcal{X}$ is represented by the p-adic completion of $\mathrm{Spec}\, R[s]/(s\widetilde{\mathrm{Ha}}_R - p^{\varepsilon})$, where $\widetilde{\mathrm{Ha}}_R \in R$ is any lift of Ha_R.

By Theorem 2.2.3, $\mathrm{BT}^{\mathcal{G},\mu}_{1}$ is smooth over $\mathbf{Z}_p$, so we can choose a $\mathbf{Z}_p$-smooth affine atlas $\mathrm{Spf}\, A_0 \to \mathrm{BT}^{\mathcal{G},\mu}_{1}$ over which the pullback of $\mathcal{V}(\chi)$ is trivialized. We can choose a lift $f \in A_0$ of the Hasse invariant Ha_{A_0}. Because the ordinary locus is dense, Ha_{A_0} is a nonzerodivisor in A_0/pA_0. This implies that f is a nonzerodivisor in A_0. Let $A := A_0 \otimes_{\mathbf{Z}_p} \mathbf{Z}_p^{\mathrm{cycl}}$. By the representability shown above, the fiber product of $\mathrm{Spf}\, A \to \mathrm{BT}^{\mathcal{G},\mu}_{1} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ along π is exactly $\mathrm{Spf}\, A\langle s\rangle/(sf - p^{\varepsilon})$, which gives the desired Cartesian diagram. Finally, Ha_{A_0} being a nonzerodivisor in A_0/pA_0 implies that $sf - p^{\varepsilon}$ and p form a regular sequence in $A_0[s]$. Since A_0 is a regular ring (being smooth over $\mathbf{Z}_p$), we get the asserted syntomicity. $\square$

Remark 4.1.12 (Relation to admissible formal blowups)**.** The construction of $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ can be viewed as taking an admissible formal blowup in the sense of Raynaud (specifically blowing up the ideal generated by (f, p^{ε})), but extended from formal schemes to formal stacks. The condition $sf - p^{\varepsilon} = 0$ in the local charts imposes the locus where the valuation of the Hasse invariant is bounded by ε.

Remark 4.1.13 (Descent to smaller base rings)**.** Although we defined the formal stack $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ over $\mathrm{Spf}\, \mathbf{Z}_p^{\mathrm{cycl}}$, it naturally descends to a formal stack over any p-adic ring of integers where the valuation ε makes sense (i.e., any p-adic ring of integers containing an element of p-adic valuation exactly ε).

The following result can be thought of as an analogue of the classical (and easy) fact that if G is an n-truncated Barsotti-Tate group over a p-torsion free $\mathbf{Z}_p$-algebra R then a finite locally free subgroup scheme $H \subset G$ is determined by $H[1/p]$ as a subgroup of $G[1/p]$.

`lem:generic-fiber-functor`

Lemma 4.1.14. *Let $\mathfrak{X}$ be a p-quasisyntomic flat p-adic formal scheme and $\mathcal{P} \subset \mathcal{G}$ a parabolic subgroup scheme. Let $\mathcal{E} \to \mathfrak{X}^{\mathrm{syn}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ be a $\mathcal{G}$-torsor and $T_{\mathrm{\acute{e}t}}(\mathcal{E})$ its étale realization, a pro-étale $\mathcal{G}(\mathbf{Z}/p^n\mathbf{Z})$-local system on $\mathfrak{X}^{\mathrm{ad}}$. The canonical functor from the groupoid of reductions of structure of $\mathcal{E}$ to $\mathcal{P}$ to the groupoid of reductions of structure of $T_{\mathrm{\acute{e}t}}(\mathcal{E})$ to $\mathcal{P}(\mathbf{Z}/p^n\mathbf{Z})$ is fully faithful.*

Proof. By quasisyntomic descent, we may therefore assume that $\mathfrak{X} = \mathrm{Spf}\, R$, where R is a p-torsion free qrsp ring. Its absolute prismatic cohomology is a p-torsion free δ-ring $\mathbb{\Delta}_R$ with an invertible ideal I such that (p, I) is a regular sequence in $\mathbb{\Delta}_R$. Two reductions s_1, s_2 of $\mathcal{E}$ give sections

$$s_i\colon \mathrm{Spec}\, \mathbb{\Delta}_R/p^n \longrightarrow \mathcal{E}_{\mathbb{\Delta}_R}/\mathcal{P},$$

where $\mathcal{E}_{\mathbb{\Delta}_R}$ is the evaluation of $\mathcal{E}$ at $(\mathbb{\Delta}_R, I)$. Equality of the reductions after étale realization implies

$$s_1|_{\mathrm{Spec}\, \mathbb{\Delta}_R[1/I]/p^n} = s_2|_{\mathrm{Spec}\, \mathbb{\Delta}_R[1/I]/p^n}.$$

The flag space $\mathcal{E}_{\mathbb{\Delta}_R}/\mathcal{P}$ is separated, hence $s_1 = s_2$. In other words, s_1 and s_2 agree over the de Rham open $\mathfrak{X}^{\mathbb{\Delta}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ of the filtered prismatization. The de Rham open is schematically dense in $\mathfrak{X}^{\mathcal{N}} \otimes \mathbf{Z}/p^n\mathbf{Z}$. The pullbacks of s_1 and s_2 to $\mathfrak{X}^{\mathcal{N}} \otimes \mathbf{Z}/p^n\mathbf{Z}$ therefore agree. Finally, the desired result follows because $\mathfrak{X}^{\mathcal{N}} \to \mathfrak{X}^{\mathrm{syn}}$ is an effective epimorphism. $\square$

Proof of Theorem 4.1.3. If the base ring is perfectoid, existence and uniqueness follow directly from Proposition 4.1.7 and Theorem 2.2.15. For a general p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, the condition on the Hasse invariant defines a map to the stack $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$. Because reduction of structure satisfies effective descent in the quasisyntomic topology, it suffices to construct a unique such reduction in the universal case.

By the density of the value group of $\mathbf{Z}_p^{\mathrm{cycl}}$, we may approximate ε from above and assume without loss of generality that there is an element $\pi \in \mathbf{Z}_p^{\mathrm{cycl}}$ with p-adic valuation exactly ε. This element π generates a finite, totally ramified extension $K = \mathbf{Q}_p(\pi)$.

By Proposition 4.1.11, we can choose an affine atlas $\operatorname{Spf} A$ for $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ given by

$$A = A_0\langle s\rangle/(s\widetilde{\mathrm{Ha}}_A - \pi),$$

where A_0 is a p-completely smooth $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, and $\widetilde{\mathrm{Ha}}_A \in A_0$ is a characteristic zero lift of the Hasse invariant. In particular, $(p, \widetilde{\mathrm{Ha}}_A)$ is a regular sequence. The Čech nerve of $\operatorname{Spf} A \to \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ is the base change of that of $\operatorname{Spf} A_0 \to \mathrm{BT}^{\mathcal{G},\mu}_1 \otimes \mathbf{Z}_p^{\mathrm{cycl}}$ along $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon} \to \mathrm{BT}^{\mathcal{G},\mu}_1 \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. This implies that the Čech nerve of $\operatorname{Spf} A \to \mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ consists of terms of the form $A_0^{(n)}\langle s\rangle/(s\,\mathrm{pr}_1^*\,\widetilde{\mathrm{Ha}}_A - \pi)$ where $A_0^{(n)}$ are Čech nerve terms of the latter, and in particular p-completely smooth. Since smooth morphisms are in particular quasisyntomic, it therefore suffices to solve the problem for rings of the form $B := B'\langle t\rangle/(tf - \pi)$ where $B' = B_0 \otimes_{\mathbf{Z}_p} \mathcal{O}_K$ with B_0 a p-completely smooth ring and $f \in B_0$ maps to a nonzerodivisor in B_0/pB_0. Note that even though B is not a $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, the desired statement still makes sense.

By choosing a monic Eisenstein polynomial $E(u) \in \mathbf{Z}_p[u]$ such that $E(\pi) = 0$, we can write $B \simeq B_0\langle t\rangle/(E(tf))$. By p-complete étale localization, we may choose a smooth framing $B_0 = \mathbf{Z}_p\langle T_1^{\pm1}, \ldots, T_n^{\pm1}\rangle$. This lifts to a δ-ring $P_0 = \mathbf{Z}_p\langle T_1^{\pm1}, \ldots, T_n^{\pm1}\rangle$ determined by $\delta(T_i) = 0$. Let $\widetilde{f} \in P_0$ be any lift of f, and extend P_0 to a δ-ring $P_1 = P_0\langle t\rangle$ by setting $\delta(t) = 0$. Since $E(u)$ is an Eisenstein polynomial, $d := E(t\widetilde{f})$ is a distinguished element of P_1, and its (p, d)-adic completion $\underline{P} = (P := (P_1)^{\wedge}_{(p,d)}, (d))$ forms a bounded prism with quotient $P/(d) \simeq B$. Let B_∞ be the Hodge–Tate quotient of the perfection of $\underline{P}$. Then B_∞ is a perfectoid ring and the natural map $B \to B_\infty$ is a quasisyntomic cover as it is a p-completed base-change of

$$\pi_\infty\colon \mathbf{Z}_p\langle T_1^{\pm1}, \ldots, T_n^{\pm1}, t\rangle \to \mathbf{Z}_p\langle T_1^{\pm1/p^\infty}, \ldots, T_n^{\pm1/p^\infty}, t^{1/p^\infty}\rangle.$$

The perfectoid cover $B \to B_\infty$ comes naturally equipped with a continuous action of the profinite group $\Gamma := \mathbf{Z}_p(1)^{n+1} \simeq \varprojlim \mu_{p^m}^{n+1}$. Explicitly, an element $(\zeta_{1,m}, \ldots, \zeta_{n,m}, \xi_m)_m \in \Gamma$ acts on B_∞ via continuous B-algebra automorphisms by scaling the chosen p^m-th roots: $T_i^{1/p^m} \mapsto \zeta_{i,m}T_i^{1/p^m}$ and $t^{1/p^m} \mapsto \xi_m t^{1/p^m}$. This yields a pro-étale Galois cover when p is inverted. Indeed, the variables T_i are units in B by definition and t becomes a unit when p is inverted because $tf = \pi$. Therefore, at each finite level m, the $B[1/p]$-subalgebra generated by T_i^{1/p^m} and t^{1/p^m} is a finite étale Kummer extension of $B[1/p]$ with Galois group $\mu_{p^m}^{n+1}$. Passing to the inverse limit over m, $B[1/p] \to B_\infty[1/p]$ becomes a Γ-torsor in pro-étale topology.

Let $\mathcal{A}$ be a 1-truncated $(\mathcal{G}, \mu)$-aperture over B. We know how to solve the problem for the base change to the perfectoid cover $\mathcal{A}_{B_\infty}$ and obtain a (unique) reduction of structure to $\mathcal{P}_\mu^+$. To show that this descends to B, we have to show the induced reductions on the intersection $B_\infty \otimes_B B_\infty$ coincide. By Lemma 4.1.14, it suffices to show that the reductions induced on the associated $\mathcal{G}(\mathbf{Z}/p\mathbf{Z})$-local systems agree. But this is true by pro-étale descent because the Γ-action on $\mathcal{A}_{B_\infty}$ must preserve the reduction due to the uniqueness assertion. □

4.2. **Overconvergent canonical quasi-isogeny.** For abelian schemes and p-divisible groups, quotienting by the (overconvergent) canonical subgroup yields an isogeny which we call the (overconvergent) canonical isogeny. This isogeny has the property that it lifts the relative Frobenius isogeny modulo a suitable power of p depending on the valuation of the Hasse invariant. In this section, we construct the overconvergent canonical (quasi-)isogeny for $(\mathcal{G}, \mu)$-apertures. As before, we assume that μ is defined over $\mathbf{Z}_p$.

Setup 4.2.1. Let R be a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra, and let $\mathcal{A} \in \mathrm{BT}^{\mathcal{G},\mu}_\infty(R)$ be a $(\mathcal{G}, \mu)$-aperture over R. This corresponds to a $\mathcal{G}$-torsor $\mathcal{E}$ over R^{syn}. Let $\mathcal{A}_1 \in \mathrm{BT}^{\mathcal{G},\mu}_1(R)$ denote the 1-truncated aperture over R obtained by reducing $\mathcal{A}$ modulo p, corresponding to the $\mathcal{G}$-torsor $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ over $R^{\mathrm{syn}} \otimes \mathbf{F}_p$. We assume that the Hasse invariant of $\mathcal{A}_1$ satisfies $\frac{p}{p+1} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{R/pR}))$. Under this assumption, Theorem 4.1.3 provides a unique reduction of structure $\mathcal{E}_1^+ \subset \mathcal{E}_1$ to $\mathcal{P}_\mu^+$ that agrees with $\mathcal{Q}^+$ modulo $p^{1-\varepsilon}$.

Construction 4.2.2 (Parahoric subgroup)**.** Let $\mathcal{H}_\mu$ be the parahoric group scheme over $\mathbf{Z}_p$ defined as the dilatation of $\mathcal{G}$ along the closed subgroup scheme $\mathcal{P}^+_{\mu,\mathbf{F}_p} \subset \mathcal{G}_{\mathbf{F}_p}$. For any p-torsion free $\mathbf{Z}_p$-algebra B, its B-points are given by the Cartesian square:

$$\begin{array}{ccc}
\mathcal{H}_\mu(B) & \longrightarrow & \mathcal{G}(B) \\
\big\downarrow & & \big\downarrow \\
\mathcal{P}_\mu^+(B/pB) & \longrightarrow & \mathcal{G}(B/pB)
\end{array}$$

Conjugation by $\mu(p)$ (defined as $x \mapsto \mu(p)x\mu(p)^{-1}$) on the generic fiber extends to a homomorphism of group schemes

$$\mathrm{inn}_{\mu(p)} \colon \mathcal{H}_\mu \to \mathcal{G}.$$

cons:modified-aperture

Construction 4.2.3 (The modified aperture)**.** The $\mathcal{P}_\mu^+$-reduction $\mathcal{E}_1^+$ over the special fiber tautologically lifts to a reduction of structure $\mathcal{E}^+ \subset \mathcal{E}$ to $\mathcal{H}_\mu$. We define a new $\mathcal{G}$-torsor over R^{syn} by pushing forward $\mathcal{E}^+$ along $\mathrm{inn}_{\mu(p)}$:

$$\mathcal{E}' := \mathcal{E}^+ \times^{\mathcal{H}_\mu, \mathrm{inn}_{\mu(p)}} \mathcal{G}.$$

As in Remark 2.1.23, we get a canonical modification $\mathcal{E} \dashrightarrow \mathcal{E}'$ along $p = 0$ bounded by μ.

Proposition 4.2.4. *The $\mathcal{G}$-torsor $\mathcal{E}'$ defines a $(\mathcal{G}, \mu)$-aperture over R, which we denote by $\mathcal{A}'$.*

Proof. We verify the condition on a geometric point $R \to \kappa$ of characteristic p by evaluating on the prism $(W(\kappa), (p))$. The $\mathcal{G}$-torsor $\mathcal{E}$ evaluates to a $\mathcal{G}$-torsor $\mathcal{E}_{W(\kappa)}$ over $W(\kappa)$ equipped with a Frobenius modification $m \colon \varphi^*\mathcal{E}_{W(\kappa)}[1/p] \xrightarrow{\sim} \mathcal{E}_{W(\kappa)}[1/p]$ of type μ. Locally trivializing the $\mathcal{H}_\mu$-reduction $\mathcal{E}^-_{W(\kappa)}$, the parahoric pushforward via $\mathrm{inn}_{\mu(p)}$ yields the Frobenius modification $m' = \mu(p)m\varphi(\mu(p))^{-1} = \mu(p)m\mu(p)^{-1}$ for $\mathcal{E}'_{W(\kappa)}$ over $W(\kappa)[1/p]$ since μ is defined over $\mathbf{Z}_p$. Since $\mathcal{E}^-_{W(\kappa)}$ lifts the conjugate reduction to $\mathcal{P}_\mu^+$ modulo p (which is preserved by m), m acts on $\mathcal{E}^-_{W(\kappa)}$, which means $m \in \mathcal{H}_\mu(W(\kappa)[1/p])$. Because $\mathcal{H}_\mu$ is the dilatation along $\mathcal{P}_\mu^+$ and μ is minuscule, conjugation by $\mu(p)$ maps $\mathcal{H}_\mu$ into $\mathcal{G}$, so $m' \in \mathcal{G}(W(\kappa))$. As m' is conjugate to m over $W(\kappa)[1/p]$, their invariant factors agree. Since μ is minuscule, any integral matrix bounded by μ has type exactly μ. Thus m' has type μ, making $\mathcal{E}'$ an aperture. □

Definition 4.2.5 (Overconvergent canonical quasi-isogeny)**.** The quasi-isogeny $\mathrm{can}_{\mathcal{A}} \colon \mathcal{A} \dashrightarrow \mathcal{A}'$ bounded by μ corresponding to the modification $\mathcal{E} \dashrightarrow \mathcal{E}'$ defined in Construction 4.2.3 is called the **overconvergent canonical quasi-isogeny of level** 1.

thm:frobenius-lifts

Theorem 4.2.6 (Frobenius lifts)**.** *The overconvergent canonical quasi-isogeny* $\mathrm{can}_{\mathcal{A}}$ *reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$.*

Proof. Recall from the setup that $\mathcal{E}_1 = \mathcal{E} \otimes \mathbf{F}_p$ is the $\mathcal{G}$-torsor over $R^{\mathrm{syn}} \otimes \mathbf{F}_p$ corresponding to the mod p fiber $\mathcal{A}_1$. The conjugate realization of $\mathcal{E}_1$, denoted $T_{\mathrm{conj}}(\mathcal{E}_1)$, naturally provides a canonical reduction of structure of $\mathcal{E}_1$ to $\mathcal{P}_\mu^+$. By Theorem 2.6.6, the relative Frobenius quasi-isogeny on $\mathcal{A}_1$ corresponds precisely to the parahoric modification induced by this conjugate reduction $T_{\mathrm{conj}}(\mathcal{E}_1)$. On the other hand, the overconvergent canonical quasi-isogeny $\mathrm{can}_{\mathcal{A}}$ is defined as the parahoric modification associated to the reduction $\mathcal{E}^+$, which modulo p yields $\mathcal{E}_1^+ \subset \mathcal{E}_1$. By Theorem 4.1.3, the reduction $\mathcal{E}_1^+$ agrees with the conjugate realization $T_{\mathrm{conj}}(\mathcal{E}_1)$ modulo $p^{1-\varepsilon}$. This congruence implies that the induced modifications agree modulo $p^{1-\varepsilon}$. Thus, $\mathrm{can}_{\mathcal{A}}$ reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$. □

rem:analytic-fiber-of-overconvergent-canonical-quasi-isogeny

Remark 4.2.7 (Analytic fiber of the overconvergent canonical quasi-isogeny)**.** Denote by $\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\leqslant\epsilon}$ the pullback of $\mathrm{BT}^{\mathcal{G},\mu}_{1,|\mathrm{Ha}|\leqslant\epsilon}$ under the 1-truncation map $\mathrm{BT}^{\mathcal{G},\mu}_{\infty} \otimes \mathbf{Z}_p^{\mathrm{cycl}} \to \mathrm{BT}^{\mathcal{G},\mu}_{1} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. For any p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, we have the following commuting diagram:

$$\begin{tikzcd}
\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\geqslant p^{-1}\epsilon}(R) \arrow[r, "T_{\text{ét}}"] \arrow[d] & \mathrm{Loc}_{\mathcal{H}_\mu(\mathbf{Z}_p)}(R[1/p]) \arrow[d, "\mathrm{inn}_{\mu(p)}"] \\
\mathrm{BT}^{\mathcal{G},\mu}_{\infty,|\mathrm{Ha}|\geqslant\epsilon}(R) \arrow[r, "T_{\text{ét}}"'] & \mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}(R[1/p])
\end{tikzcd}$$

We emphasize that our work on the overconvergent canonical quasi-isogeny implies that even though the top horizontal arrow *a priori* maps to $\mathrm{Loc}_{\mathcal{G}(\mathbf{Z}_p)}$, it actually admits a canonical factorization via $\mathrm{Loc}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$ by virtue of the functor in Lemma 4.1.14. Here, the right vertical arrow is induced by the homomorphism $\mathcal{H}_\mu(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}_p)$ given by the adjoint action of $\mu(p)$. Indeed, this commutativity is essentially immediate from the construction of $\mathrm{can}_{\mathcal{A}}$. The left vertical arrow has the property that it reduces to the relative Frobenius modulo $p^{1-\varepsilon}$.

thm:level-n-frobenius-lifts

Theorem 4.2.8 (Level-n Frobenius lifts)**.** *Let $n \geqslant 1$ be an integer. Suppose that the Hasse invariant of $\mathcal{A}_1$ satisfies the stricter bound*

$$\frac{p}{p^{n-1}(p+1)} > \varepsilon \geqslant \nu_p(\mathrm{Ha}(\mathcal{A}|_{R/pR})).$$

Then one can iteratively apply the overconvergent canonical quasi-isogeny n times to obtain a composition

$$\mathcal{A} = \mathcal{A}^{(0)} \dashrightarrow \mathcal{A}^{(1)} \dashrightarrow \cdots \dashrightarrow \mathcal{A}^{(n)}$$

denoted $\mathrm{can}^{(n)}_{\mathcal{A}} \colon \mathcal{A} \dashrightarrow \mathcal{A}^{(n)}$ and called the ***overconvergent canonical quasi-isogeny of level*** *n. Furthermore, $\mathrm{can}^{(n)}_{\mathcal{A}}$ reduces to the n-th relative Frobenius quasi-isogeny modulo $p^{1-p^{n-1}\varepsilon}$.*

Proof. We proceed by induction on n, the base case $n = 1$ being Theorem 4.2.6. For $n \geqslant 2$, the level-1 modification $\mathrm{can}^{(1)}_{\mathcal{A}} \colon \mathcal{A} \dashrightarrow \mathcal{A}^{(1)}$ reduces to the relative Frobenius quasi-isogeny modulo $p^{1-\varepsilon}$. Consequently, the Hasse invariant of the modified aperture satisfies $\mathrm{Ha}(\mathcal{A}^{(1)}|_{R/pR}) \equiv \mathrm{Ha}(\mathcal{A}|_{R/pR})^p \pmod{p^{1-\varepsilon}}$. Since $p\varepsilon \leqslant p^{n-1}\varepsilon < \frac{p}{p+1} < 1 - \varepsilon$, this congruence forces the valuation to be exactly $\nu_p(\mathrm{Ha}(\mathcal{A}^{(1)}|_{R/pR})) = p\varepsilon$. Applying the inductive hypothesis to $\mathcal{A}^{(1)}$ at level $n-1$ produces a sequence of modifications up to $\mathcal{A}^{(n)}$ which reduces to the $(n-1)$-th relative Frobenius modulo $p^{1-p^{n-2}(p\varepsilon)} = p^{1-p^{n-1}\varepsilon}$. Composing this with $\mathrm{can}^{(1)}_{\mathcal{A}}$ yields the result, noting that the initial congruence modulo $p^{1-\varepsilon}$ is strictly finer since $1 - \varepsilon > 1 - p^{n-1}\varepsilon$. □

sec:comparison

4.3. Comparison with existing results.

Canonical subgroups for abelian varieties and Barsotti–Tate groups have been extensively investigated, notably by Abbes–Mokrane [AM04], Andreatta–Gasbarri [AG06], Tian [Tia10], Rabinoff [Rab12], Fargues [Far11], and Hattori [Hat13]. Scholze in [Sch15, §3.2.1] also develops a theory of canonical subgroups *integrally* by using Illusie's work on the cotangent complex. Rabinoff's result is close enough to the present theorem to deserve a separate comparison, but his notion of canonical subgroup is defined by a different criterion on the rigid generic fiber. Because the work of Fargues encompasses and refines the earlier results of Abbes–Mokrane, Andreatta–Gasbarri, and Tian, we shall restrict the remaining detailed comparison to [Far11] and [Hat13]. Furthermore, the overconvergence bounds established here improve upon those obtained by Scholze in [Sch15] and Fargues [Far11]. A unifying characteristic of these classical approaches is their restriction to complete rank-1 valuation rings of mixed characteristic. In contrast, the linear algebraic approach taken using prismatic Dieudonné theory in this paper yields an integral, group-theoretic construction that works uniformly over arbitrary flat p-adic formal bases while recovering and strengthening existing results.

Setting $\mathcal{G} = \mathrm{GL}_h$ and $\mu(z) = \mathrm{diag}(z, \dots, z, 1, \dots, 1)$ with d occurrences of z and $h-d$ occurrences of 1, we obtain:

thm:canonical-subgroup-for-1-truncated-barsotti-tate-groups

Theorem 4.3.1 (Canonical subgroup for 1-truncated Barsotti–Tate groups)**.** *Let G be a 1-truncated Barsotti–Tate group of height h and dimension d over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, with $G_0 := G|_{R/pR}$ its reduction modulo p. Assume that the Hasse invariant of G_0 has p-adic valuation $\varepsilon := \nu_p(\mathrm{Ha}\, G_0) < \frac{p}{p+1}$. Then G admits a unique finite locally free subgroup scheme $C_1 \subset G$ of rank p^d which agrees with the kernel of the relative Frobenius $\mathrm{Ker}(G_0 \to G_0^{(p)})$ modulo $p^{1-\varepsilon}$.*

Proof. The existence, uniqueness, and congruence condition specialize directly from Theorem 4.1.3 applied to $\mathcal{G} = \mathrm{GL}_h$ and $\mu(z) = \mathrm{diag}(z, \dots, z, 1, \dots, 1)$. To determine the rank of the finite locally free subgroup scheme $C_1 \subset G$ over R, let $I = (p^{1-\varepsilon}) \subset R$. Since R is p-complete and $1 - \varepsilon > 0$, the ideal I is contained in the Jacobson radical of R; hence the rank of any finite locally free R-scheme is determined by its base change to R/I. Modulo I, the congruence condition identifies $C_1 \times_R R/I$ with $\mathrm{Ker}(G_0 \to G_0^{(p)}) \times_{R/pR} R/I$. Because G_0 is a 1-truncated Barsotti–Tate group of dimension d over R/pR, its relative Frobenius kernel is finite locally free of rank p^d, whence C_1 has rank p^d over R. □

rem:general-base-rings

Remark 4.3.2 (General bases)**.** The assumption requiring the base ring R to be a $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra is not essential for the validity of Theorems 4.3.1 and 4.3.3. The results carry over to an arbitrary p-complete flat $\mathbf{Z}_p$-algebra R, provided that references to the congruence condition "modulo $p^{1-\varepsilon}$" are systematically replaced by congruence modulo the ideal generated by all elements of R having p-adic valuation at least $1 - \varepsilon$.

We also have a higher-level analogue where we recover the Frobenius-kernel lift and the expected structure of geometric fibers in the full overconvergent range: thm:level-n-canonical-subgroup-for-p-divisible-groups

Theorem 4.3.3 (Level-n canonical subgroup for p-divisible groups)**.** *Let $n \geqslant 1$ be an integer, and let G be a p-divisible group of height h and dimension d over a p-complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R, with reduction $G_0 := G|_{R/pR}$ modulo p. Assume that the Hasse invariant of G_0 satisfies the valuation bound*

$$\varepsilon := \nu_p(\mathrm{Ha}\, G_0) < \frac{1}{p^{n-2}(p+1)}.$$

Then the p^n-torsion subgroup $G[p^n]$ admits a unique finite locally free subgroup scheme $C_n \subset G[p^n]$ of rank p^{nd}, called the ***canonical subgroup of level*** *n, satisfying the following properties:*

(1) ***Congruence with Frobenius kernel:*** *The subgroup scheme C_n agrees with the kernel of the n-th iterated relative Frobenius isogeny $\mathrm{Ker}(G_0 \to G_0^{(p^n)})$ modulo $p^{1-p^{n-1}\varepsilon}$.*
(2) ***Quotient compatibility and Hasse invariant scaling:*** *For any integer $1 \leqslant k < n$, let $G^{(k)} := G/C_k$ denote the quotient by the level-k canonical subgroup. Then the Hasse invariant of $G^{(k)}$ has valuation $\nu_p(\mathrm{Ha}(G^{(k)}|_{R/pR})) = p^k\varepsilon$, and under the canonical identification $(G/C_k)[p^{n-k}] \simeq G[p^n]/C_k$, the level-$(n-k)$ canonical subgroup of $G^{(k)}$ coincides with the quotient C_n/C_k.*
(3) ***Cartier duality:*** *The Cartier dual p-divisible group G^D of height h and dimension $h-d$ has Hasse invariant valuation $\varepsilon < \frac{1}{p^{n-2}(p+1)}$, and its canonical subgroup of level n, denoted $C_n^D \subset G^D[p^n]$, has rank $p^{n(h-d)}$ and coincides with the orthogonal complement of C_n under the perfect Weil pairing:*

$$C_n^D = C_n^\perp \subset G^D[p^n].$$

(4) ***Geometric points and truncation:*** *For any characteristic 0 geometric point $R \to \overline{\kappa}$ and any integer $1 \leqslant k < n$, one has*

$$C_n(\overline{\kappa})[p^k] = C_k(\overline{\kappa}).$$

In particular,

$$C_n(\overline{\kappa}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^d.$$

Proof. The existence, rank assertion, Frobenius congruence, and quotient compatibility are the specialization of Theorem 4.2.8 to $\mathcal{G} = \mathrm{GL}_h$. Under Cartier duality, the general linear group GL_h and cocharacter μ are replaced by the dual representation, which transposes dimensions: the dual p-divisible group G^D has height h and dimension $h-d$. Because the Hasse invariant is invariant under Cartier duality ($\mathrm{Ha}(G^D) = \mathrm{Ha}(G)$), G^D satisfies the same overconvergence bound $\varepsilon < \frac{1}{p^{n-2}(p+1)}$ and therefore admits a level-n canonical subgroup $C_n^D \subset G^D[p^n]$ of rank $p^{n(h-d)}$. In our group-theoretic framework, Cartier duality corresponds to applying a Chevalley involution $C\colon \mathcal{G} \to \mathcal{G}$ (cf. Remark 2.6.8), under which the overconvergent canonical quasi-isogeny and the properties established in Theorem 4.2.8 are canonically preserved. Consequently, the dual canonical subgroup C_n^D coincides precisely with the orthogonal complement of C_n under the perfect Weil pairing: $C_n^D = C_n^\perp$.

Finally, for any characteristic-0 geometric point $R \to \overline{\kappa}$, the assertions $C_n(\overline{\kappa})[p^k] = C_k(\overline{\kappa})$ and $C_n(\overline{\kappa}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^d$ follow by induction on n from quotient compatibility (2). Indeed, for $n = 1$, $C_1(\overline{\kappa})$ is a subgroup of $G[p](\overline{\kappa}) \simeq (\mathbf{Z}/p\mathbf{Z})^h$ of order p^d, hence isomorphic to $(\mathbf{Z}/p\mathbf{Z})^d$. For $n > 1$ and $1 \leqslant k < n$, quotient compatibility yields a short exact sequence

$$0 \to C_k(\overline{\kappa}) \to C_n(\overline{\kappa}) \to C_{n-k}^{(k)}(\overline{\kappa}) \to 0$$

of finite abelian groups inside $G[p^n](\overline{\kappa}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^h$, which identifies $C_n(\overline{\kappa})[p^k]$ with $C_k(\overline{\kappa})$ and forces $C_n(\overline{\kappa}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^d$. □

rem:comparison-fargues

Remark 4.3.4 (Comparison with Fargues [Far11])**.** Although the overconvergence bound on the Hasse invariant established above is strictly stronger than that of [Far11, Théorème 4], a direct comparison warrants caution. Fargues characterizes the canonical subgroup via the ramification filtrations of Abbes–Saito, whereas our construction relies on lifting the kernel of the relative Frobenius morphism. As we do not establish a comparison with the Abbes–Saito filtration, it is not right to directly compare our results with that of Fargues.

rem:comparison-hattori

Remark 4.3.5 (Comparison with Hattori [Hat13])**.** Over complete discrete valuation rings $\mathcal{O}_K$, Hattori [Hat13, Theorem 1.1] constructs level-n canonical subgroups of truncated Barsotti–Tate groups via Breuil–Kisin modules under the same bound $\varepsilon < \frac{1}{p^{n-2}(p+1)}$, proving the same congruence modulo $p^{1-p^{n-1}\varepsilon}$ with the iterated relative

Frobenius kernel. In *loc. cit.*, the geometric-fiber structure $C_n(\overline{K}) \simeq (\mathbf{Z}/p^n\mathbf{Z})^d$ and compatibility across levels are established under the more restrictive assumption $\varepsilon < \frac{p-1}{p^n-1}$ for general truncated Barsotti–Tate groups. For p-divisible groups, as in Theorem 4.3.3, these properties hold automatically throughout the full range $\varepsilon < \frac{1}{p^{n-2}(p+1)}$ by iterating the level-1 construction.

Furthermore, to construct canonical subgroups in rigid-analytic families, Hattori imposes an even stronger valuation bound so as to identify the canonical subgroup with an upper ramification subgroup and glue over admissible covers [Hat13, Corollary 1.2]. Working directly with prismatic Dieudonné modules over general flat p-adic formal schemes avoids this rigid-analytic gluing step entirely, so our existence bound $\varepsilon < \frac{1}{p^{n-2}(p+1)}$ remains uniform in families.

rem:when-rabinoff-lifts-frobenius

Remark 4.3.6 (Comparison with Rabinoff [Rab12])**.** Let $R = \mathcal{O}_K$ be a complete discrete valuation ring with perfect residue field, and let G be a p-divisible group over R with Hasse invariant valuation $\varepsilon < \frac{p-1}{p^n}$. In this range, Rabinoff [Rab12, §1.6] defines the level-n canonical subgroup $C_n^{\mathrm{Rab}} = G^{\circ}_{\leqslant\rho} \subset G[p^n]_K$ as the lower ramification subgroup at break $r = \frac{1}{p^{n-1}(p-1)} - \frac{\varepsilon}{p-1}$. The comparison between Rabinoff's construction and our canonical subgroup is essentially due to Hattori. By [Hat14, Theorem 1.2], its schematic closure $\overline{C}_n^{\mathrm{Rab}} \subset G[p^n]$ coincides with the level-n canonical subgroup constructed from Breuil–Kisin modules. Since the latter satisfies the same Frobenius-kernel congruence modulo $p^{1-p^{n-1}\varepsilon}$ by [Hat13, Theorem 1.1], the uniqueness in Theorem 4.3.3 identifies $\overline{C}_n^{\mathrm{Rab}} = C_n$.

5. Perfectoid Shimura varieties

sec:perfectoid-shimura-varieties

We prove Theorem A by following Scholze's method [Sch15].

Setup 5.1. Maintain the notations of Section 3.1. For brevity, we rename $-\mu_v$ to $-\mu$.

assump:main

Assumption 5.2. The Shimura datum has reflex field $\mathbf{Q}_p$, i.e., $-\mu$ is defined over $\mathbf{Z}_p$. We also assume that the boundary of minimal compactification has codimension at least 2 (cf. Assumption 3.4.6).

5.1. The overconvergent anticanonical tower.

We use the description of p-Hecke correspondences in Remark 3.1.4, together with the analytic description of the overconvergent canonical quasi-isogeny in Remark 4.2.7, to define the anticanonical tower. Let us first make some preliminary definitions:

cons:eps-neighbourhoods

Construction 5.1.1. Recall that we have a formally étale syntomic realization $\varpi\colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)} \to \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$ by Theorem 3.1.5.

- (Ordinary loci) Let $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ be the adic generic fiber of $\widehat{\mathscr{S}}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ from Definition 3.3.6, an open adic subspace of $\S_{\mathcal{G}(\mathbf{Z}_p)}$. In the notations of Remark 3.1.4, let $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)_h}$ be the preimage of $\mathcal{S}^{\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ along the analytification of the level map s_h. Let $\mathcal{S}^{*,\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)}$ be the adic generic fiber of the unique open formal subscheme of $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$ whose special fiber is the extended ordinary locus supplied by Corollary 3.4.9. Define $\mathcal{S}^{*,\mathrm{ord}}_{\mathcal{G}(\mathbf{Z}_p)_h}$ analogously as the preimage along analytification of level maps.
- (ε-neighbourhoods) For $0 \leqslant \varepsilon < 1$, we define $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ and $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ as the formal schemes over $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$ representing the functors which assign to any p-adically complete flat $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra R the set of pairs (f, u) where $f\colon \operatorname{Spf} R \to \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ (resp. $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$) and $u \in \mathrm{H}^0(\operatorname{Spf} R, f^*\omega^{-\chi})$ is a section such that $u\,\mathrm{Ha}(f) = p^{\varepsilon}$ modulo p (up to natural equivalence). By Proposition 3.4.8, these are admissible blow-ups and their adic generic fibers, denoted $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$ and $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$, are exactly the open adic subspaces of $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}$ and $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}$ where $|\,\mathrm{Ha}\,| \geqslant |p|^{\varepsilon}$. For any deeper level K_p, we define $\mathcal{S}_{K_p}(\varepsilon)$ and $\mathcal{S}^*_{K_p}(\varepsilon)$ as their preimages under the level maps. The adic spaces involved in this item are all analytic adic space over $\operatorname{Spa}(\mathbf{Q}_p^{\mathrm{cycl}}, \mathbf{Z}_p^{\mathrm{cycl}})$.
- (ε-neighbourhoods in $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}$) We define the formal stack $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(\varepsilon)$ as the base change of the open formal substack $\mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ along the natural 1-truncation map $\mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v} \otimes \mathbf{Z}_p^{\mathrm{cycl}} \to \mathrm{BT}_1^{\mathcal{G}^c,-\mu_v} \otimes \mathbf{Z}_p^{\mathrm{cycl}}$. Here $\mathrm{BT}^{\mathcal{G}^c,-\mu_v}_{1,|\mathrm{Ha}|\geqslant\varepsilon}$ is the locus where the Hasse invariant satisfies $|\,\mathrm{Ha}\,| \geqslant |p|^{\varepsilon}$ (see Construction 4.1.10).

lem:canonical-frobenius-lifts

Theorem 5.1.2 (Canonical Frobenius lifts modulo $p^{1-\varepsilon}$)**.** *Let $0 \leqslant \varepsilon < \frac{p}{p+1}$. There exists a unique Cartesian square of p-adic formal stacks over* $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$

$$\begin{tikzcd} \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] \arrow[d, "\varpi"'] & \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[d, "\varpi"] \\ \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(p^{-1}\varepsilon) \arrow[r] & \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(\varepsilon) \end{tikzcd}$$

such that the vertical arrows are induced by syntomic realizations, the horizontal arrows are unique lifts of the relative Frobenii modulo $p^{1-\varepsilon}$, and the bottom horizontal arrow is induced by the overconvergent canonical quasi-isogeny of level 1 *(cf. Theorem 4.2.6).*

Proof. Set $h = \mu(p)^{-1}$. In what follows, $(-)^{\mathrm{an}}$ denotes the analytification functor. Recall that we have a Hecke correspondence diagram:

$$\begin{tikzcd} & \mathrm{Sh}_{\Gamma_0(p)} \arrow[dl, "s_h"'] \arrow[dr, "t_h"] & \\ \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} & & \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)} . \end{tikzcd}$$

Over $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon)$, the p-adic étale realization map factors through $\mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$ (cf. Remark 4.2.7). The description in Remark 3.1.4 therefore shows that $t_h^{\mathrm{an}}|_{\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon)}$ is canonically isomorphic to the base-change of

$$[\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}} \times_{\mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \simeq [\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}$$

along $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$. In other words, we have the following Cartesian squares:

$$\begin{tikzcd} \mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] \arrow[d, "t_h^{\mathrm{an}}"'] & {[\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}} \arrow[r] \arrow[d] & {[\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{G}(\mathbf{Z}_p)}]^{\mathrm{an}}} \arrow[d, "\alpha_h^{\mathrm{an}}"] \\ \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \end{tikzcd}$$

We have a conjugation map $\mathrm{inn}_{\mu(p)} \colon \mathcal{H}_\mu(\mathbf{Z}_p) \to \mathcal{G}(\mathbf{Z}_p)$. This gives a section:

$$\mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \xrightarrow{\simeq} [h\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}} \to [\underline{\mathcal{G}(\mathbf{Z}_p)} h \underline{\mathcal{G}(\mathbf{Z}_p)}/\underline{\mathcal{H}_\mu(\mathbf{Z}_p)}]^{\mathrm{an}}.$$

This implies that we have a section

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \times_{t_h^{\mathrm{an}}, \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon).$$

Composing this section with the map $s_h^{\mathrm{an}} \colon \mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p)} \to \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}$ gives a map $\Phi' \colon \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}$ such that the diagram

$$\begin{tikzcd} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r] \arrow[d, "\Phi'"'] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)} \arrow[d, "\mathrm{inn}_{\mu(p)}"] \\ \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \arrow[r] & \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} \end{tikzcd}$$

commutes. By Remark 4.2.7, the composition of the top horizontal arrow with the right vertical arrow factors as

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \xrightarrow{\varpi^{\mathrm{ad}}} \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(p^{-1}\varepsilon)^{\mathrm{ad}} \xrightarrow{\text{Theorem 4.2.6}} \mathrm{BT}_\infty^{\mathcal{G}^c,-\mu_v}(\varepsilon)^{\mathrm{ad}} \to \mathrm{Loc}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)} .$$

The argument from [MY26, Proposition 4.3.1] now shows that this is the adic generic fiber of a morphism $\widetilde{F} \colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \to \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}$ and considerations with the Hasse invariant tells us that it actually factors through $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$. That the commuting square in the theorem statement is Cartesian also follows from [MY26, Proposition 4.3.1] since all of its members are normal (see Proposition 4.1.11). The reduction modulo $p^{1-\varepsilon}$ property follows from Corollary 3.2.6. □

The following is the analogue of [Sch15, Theorem 3.2.15]:

thm:overconvergent-anticanonical-tower

Theorem 5.1.3. *Let $0 \leqslant \varepsilon < \frac{p}{p+1}$.*

(a) There is a unique commutative diagram of formal schemes over $\operatorname{Spf} \mathbf{Z}_p^{\mathrm{cycl}}$*:*

$$\begin{tikzcd} \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] \arrow[d, hook] & \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[d, hook] \\ \widehat{\mathscr{S}}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \arrow[r, "\widetilde{F}"] & \widehat{\mathscr{S}}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \end{tikzcd}$$

where the horizontal arrows modulo $p^{1-\varepsilon}$ agree with the relative Frobenii.

(b) For any integer $m \geqslant 0$, the inverse of the overconvergent canonical quasi-isogeny of level m (cf. Theorem 4.2.8) induces a morphism on the adic generic fibers

$$\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \to \mathcal{S}_{\Gamma_0(p^m)}$$

which extends uniquely to a morphism of adic spaces $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \to \mathcal{S}^*_{\Gamma_0(p^m)}$. These morphisms are open embeddings. Denote their images by $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ inside $\mathcal{S}_{\Gamma_0(p^m)}$ and $\mathcal{S}^*_{\Gamma_0(p^m)}$, respectively. Moreover, for $m \geqslant 1$, the diagram

$$\begin{tikzcd} \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m-1}\varepsilon) \arrow[r, hook] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^*_{\Gamma_0(p^{m+1})} \arrow[d] \\ \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \arrow[r, hook] & \mathcal{S}^*_{\Gamma_0(p^m)} \end{tikzcd}$$

is commutative and Cartesian.

(c) The images $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ are open and closed adic subspaces of the ε-neighbourhoods $\mathcal{S}_{\Gamma_0(p^m)}(\varepsilon)$ and $\mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon)$, respectively.

Proof. We first work over the open Shimura varieties. For part (a), Theorem 5.1.2 gives the unique lift

$$\widetilde{F}\colon \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon) \longrightarrow \widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$$

of relative Frobenius modulo $p^{1-\varepsilon}$. We extend this map to the minimal compactification below.

We next construct the morphism in part (b). Set $h = \mu(p)^{-1}$. Suppose $m \geqslant 1$. With the convention of Notation 3.1.3 and under Assumption 5.2, we have

$$\mathcal{G}(\mathbf{Z}_p)_{h^m} = \mathcal{G}(\mathbf{Z}_p) \cap h^m \mathcal{G}(\mathbf{Z}_p) h^{-m} = \Gamma_0(p^m).$$

Let $\mathcal{A}$ be the pullback of the universal aperture to $\widehat{\mathscr{S}}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$. By Remark 4.2.7, at every step defining the level-m canonical quasi-isogeny the étale realization factors canonically through $\mathrm{Loc}^{\mathrm{an}}_{\mathcal{H}_\mu(\mathbf{Z}_p)}$, and changing structure group by $\mathrm{inn}_{\mu(p)}$ gives the étale realization after applying the canonical quasi-isogeny. In the double-coset description of Remark 3.1.4, this is precisely the Hecke modification represented by h.

The level-m canonical quasi-isogeny has constant type $m\mu$. By Remark 2.2.10, this quasi-isogeny consequently gives a canonical reduction to $\mathcal{H}_{m\mu}$. Since μ is minuscule, the root-subgroup description in Remark 2.2.11 gives

$$\mathcal{H}_{m\mu}(\mathbf{Z}_p) = \mathcal{G}(\mathbf{Z}_p) \cap \mu(p)^{-m} \mathcal{G}(\mathbf{Z}_p) \mu(p)^m = \mathcal{G}(\mathbf{Z}_p)_{h^m} = \Gamma_0(p^m).$$

Extension of structure group along $\Gamma_0(p^m) \hookrightarrow \mathcal{G}(\mathbf{Z}_p)$ recovers $T_{\text{ét}}(\mathcal{A})$, while extension along $\mathrm{inn}_{\mu(p)^m}$ gives $T_{\text{ét}}(\mathcal{A}^{(m)})$. Equivalently, the étale realization of $(\mathrm{can}^{(m)}_{\mathcal{A}})^{-1}$ has relative position $\mathcal{G}(\mathbf{Z}_p) h^m \mathcal{G}(\mathbf{Z}_p)$.

The resulting $\Gamma_0(p^m)$-reduction is a section of the finite étale cover obtained by pulling back $\alpha^{\mathrm{an}}_{h^m}$ along the classifying map for $T_{\text{ét}}(\mathcal{A})$. By Remark 3.1.4, this pullback is canonically the cover

$$\mathrm{Sh}^{\mathrm{an}}_{\Gamma_0(p^m)} \times_{t^{\mathrm{an}}_{h^m}, \mathrm{Sh}^{\mathrm{an}}_{\mathcal{G}(\mathbf{Z}_p)}} \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \longrightarrow \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon).$$

Consequently, the section gives the desired morphism

$$f\colon \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \longrightarrow \mathcal{S}_{\Gamma_0(p^m)}.$$

Its two projections satisfy

$$t^{\mathrm{an}}_{h^m} \circ f = \big(\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \hookrightarrow \mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}\big), \qquad s^{\mathrm{an}}_{h^m} \circ f = (\widetilde{F}^m)^{\mathrm{ad}}.$$

In particular, viewed as a map to the base change of $t^{\mathrm{an}}_{h^m}$, the map f is a section of a finite étale morphism and hence is an open immersion. Its composite with the open immersion from that base change to $\mathcal{S}_{\Gamma_0(p^m)}$ is therefore an open immersion as well.

Compatibility of the canonical reductions under iteration makes the square in part (b) commutative over the open Shimura varieties. The square is Cartesian there by a degree calculation. Observe that $\widetilde{F}$ must be finite locally free. Its special fiber is relative Frobenius, so its degree is $p^{\dim \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}}$; after passing to the adic generic fiber it is finite étale. For $m \geqslant 1$, the source map on the right is finite étale of the same degree: the number of $\mathbf{F}_p$-points of the relevant unipotent radical, namely $p^{\dim \mathrm{Sh}_{\mathcal{G}(\mathbf{Z}_p)}}$. The induced open immersion from the upper-left corner into the fiber product is therefore a map between finite étale covers of the same degree, and hence is an isomorphism.

We now pass to minimal compactifications. The boundary has codimension at least 2 by Assumption 5.2. The Hartogs argument of [Sch15, Lemma 3.2.10 and proof of Theorem 3.2.15] therefore extends $\widetilde{F}$ uniquely to $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-1}\varepsilon)$, proving part (a), and extends f uniquely to

$$f^*\colon \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \longrightarrow \mathcal{S}^*_{\Gamma_0(p^m)}.$$

More precisely, after restricting to an affinoid in $\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$, one applies Hartogs to the section of the finite normal pullback of the level map. The extension is finite and generically an open immersion, so normality makes it an open and closed immersion into that pullback; in particular, f^* is an open immersion. The same Hartogs argument preserves the commutativity and Cartesian property of the squares. This proves part (b).

Finally, the displayed identities for the two Hecke projections show that the images of f and f^* lie in the respective ε-neighbourhoods at level $\Gamma_0(p^m)$. They are open by part (b). They are also closed: the composite

$$\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon) \xrightarrow{\sim} \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \hookrightarrow \mathcal{S}^*_{\Gamma_0(p^m)}(\varepsilon) \longrightarrow \mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)$$

is $(\widetilde{F}^m)^{\mathrm{ad}}$ and is finite, while the last level map is finite. Hence the inclusion of the image is finite, and therefore closed. Taking the preimage of the open Shimura variety gives the corresponding assertion without compactifications. This proves part (c). □

rem:overconvergent-anticanonical-tower

Remark 5.1.4 (The overconvergent anticanonical tower)**.** The **overconvergent anticanonical tower** is the inverse system on the right side of the following commutative diagram of Cartesian squares:

$$\begin{tikzcd}
\vdots \arrow[d] & \vdots \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-(n+1)}\varepsilon) \arrow[r, "\sim"] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{n+1})}(\varepsilon) \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(p^{-n}\varepsilon) \arrow[r, "\sim"] \arrow[d, "(\widetilde{F})^{\mathrm{ad}}"'] & \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{n})}(\varepsilon) \arrow[d] \\
\vdots \arrow[d] & \vdots \arrow[d] \\
\mathcal{S}^*_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon) \arrow[r, equal] & \mathcal{S}^{*,\mathrm{anti}}_{\mathcal{G}(\mathbf{Z}_p)}(\varepsilon)
\end{tikzcd}$$

A completely analogous statement holds for the open Shimura variety $\mathcal{S}$. The upshot is that the tower on the right has an integral model given by the obvious integral model of the tower on the left and this integral model has the crucial property that it reduces to a tower of relative Frobenii modulo $p^{1-\varepsilon}$.

5.2. **Perfectoid anticanonical neighbourhood.** This subsection essentially mirrors [Sch15, §3.2.3]. We show that the inverse limit of the overconvergent anticanonical tower constructed in Remark 5.1.4 is a perfectoid space.

lem:affinoidness

Lemma 5.2.1 (Anticanonical neighbourhoods are affinoid). *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. For m sufficiently large, the adic space $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ is affinoid.*

Proof. By Theorem 5.1.3(b), $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \simeq \mathcal{S}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$. Since ω^{χ} is ample on $\mathscr{S}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$, for $m \gg 0$, the higher cohomology of $(\omega^{\chi})^{\otimes p^m}$ vanishes, allowing us to lift Ha^{p^m} to a global section $\widetilde{\mathrm{Ha}}_m$ over $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}$. The locus $\mathcal{S}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$ where $|\,\mathrm{Ha}\,| \geqslant |p|^{p^{-m}\varepsilon}$ is equivalently defined by $|\widetilde{\mathrm{Ha}}_m| \geqslant |p|^{\varepsilon}$. As $\widetilde{\mathrm{Ha}}_m$ is a global section of an ample line bundle, this defines an affinoid open subspace. $\square$

Fix $t \in (\mathbf{Z}_p^{\mathrm{cycl}})^{\flat}$ with $|t| = |p|$, chosen to admit compatible p-power roots, yielding $(\mathbf{Z}_p^{\mathrm{cycl}})^{\flat} \simeq \mathbf{F}_p[\![t^{1/p^{\infty}}]\!]$. Let $\mathcal{S}'^{*}$ be the t-adic completion of $\mathscr{S}^{*}_{\mathcal{G}(\mathbf{Z}_p)} \otimes_{\mathbf{Z}_{(p)}} \mathbf{F}_p[\![t^{1/p^{\infty}}]\!]$ over $\mathbf{F}_p[\![t^{1/p^{\infty}}]\!]$, and $\mathcal{S}'^{*}$ its generic fiber. As in characteristic 0, let $\mathcal{S}'^{*}(\varepsilon)$ be the locus where $|\,\mathrm{Ha}\,| \geqslant |t|^{\varepsilon}$. Taking the inverse limit along relative Frobenius yields a perfect formal scheme $\mathcal{S}'^{*}(\varepsilon)_{\mathrm{perf}} = \varprojlim_{\Phi} \mathcal{S}'^{*}(\varepsilon)$ with perfectoid generic fiber $\mathcal{S}'^{*}_{\mathrm{perf}}(\varepsilon)$.

thm:perfectoid-anticanonical-locus

Theorem 5.2.2 (Perfectoid anticanonical neighborhood). *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. There are unique perfectoid spaces $\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$ over $\mathbf{Q}_p^{\mathrm{cycl}}$ such that*

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon) \sim \varprojlim_{m} \mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^{m})}(\varepsilon) \quad \textit{and} \quad \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon) \sim \varprojlim_{m} \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{m})}(\varepsilon).$$

Moreover, the tilt $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)^{\flat}$ identifies naturally with the open subset $\mathcal{S}'^{*}_{\mathrm{perf}}(\varepsilon) \subset \mathcal{S}'^{*}_{\mathrm{perf}}$ where $|\,\mathrm{Ha}\,| \geqslant |t|^{\varepsilon}$. In particular, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$ is affinoid perfectoid, and the boundary

$$\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon) \subset \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$$

is strongly Zariski closed.

Proof. We focus on the minimal compactification; the open Shimura variety follows analogously. By Theorem 5.1.3(b), $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)$ admits the integral model $\widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$. Crucially, the transition maps in this tower reduce modulo $p^{1-\varepsilon}$ to the relative Frobenius map. Let $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon) := \varprojlim_m \widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$. Working affine locally with $\operatorname{Spf} R_m \subset \widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)$, the inverse limit yields a flat p-adically complete $\mathbf{Z}_p^{\mathrm{cycl}}$-algebra $R_{\infty} = (\varinjlim_m R_m)^{\wedge}_p$. Since the transition maps reduce to Frobenius modulo $p^{1-\varepsilon}$, absolute Frobenius induces an isomorphism

$$R_{\infty}/p^{(1-\varepsilon)/p} \xrightarrow{\sim} R_{\infty}/p^{1-\varepsilon}.$$

This implies that R_{∞} is a perfectoid algebra. Thus, its generic fiber is an affinoid perfectoid space over $\mathbf{Q}_p^{\mathrm{cycl}}$, which guarantees the existence and uniqueness of $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$ (cf. [SW13, Definition 2.4.1, Proposition 2.4.2, Proposition 2.4.5]).

The canonical identification $\mathcal{S}'^{*}(p^{-m}\varepsilon)/t^{1-\varepsilon} = \widehat{\mathscr{S}}^{*}_{\mathcal{G}(\mathbf{Z}_p)}(p^{-m}\varepsilon)/p^{1-\varepsilon}$ from Theorem 5.1.3(a) passes to the limit, yielding an isomorphism of coordinate rings modulo $t^{1-\varepsilon}$ and $p^{1-\varepsilon}$. This implies that the generic fibers are tilts of each other. Since the tilt $\mathcal{S}'^{*}_{\mathrm{perf}}(\varepsilon)$ is affinoid with a Zariski closed boundary, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^{\infty})}(\varepsilon)$ is affinoid perfectoid with a strongly Zariski closed boundary. $\square$

5.3. **Passage to full level.** We now pass to the full level tower $\Gamma(p^m)$ at p. For any $m \geqslant 0$, let

$$\pi_m \colon \mathcal{S}_{\Gamma(p^m)} \to \mathcal{S}_{\Gamma_0(p^m)} \quad \text{and} \quad \pi_m^{*} \colon \mathcal{S}^{*}_{\Gamma(p^m)} \to \mathcal{S}^{*}_{\Gamma_0(p^m)}$$

be the natural level projection maps. Since these projection maps are finite, we can define the anticanonical loci at full level by pulling back the corresponding loci of parahoric level.

def:anticanonical-loci-full-level

Definition 5.3.1 (Anticanonical loci at level $\Gamma(p^m)$). Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. For any integer $m \geqslant 0$, we define the **anticanonical locus of level** $\Gamma(p^m)$ **and radius** ε on the open Shimura variety and its minimal compactification to be the preimages:

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) := \pi_m^{-1}\Big(\mathcal{S}^{\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)\Big)$$

and

$$\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) := (\pi^*_m)^{-1}\Big(\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon)\Big)$$

respectively. Since the anticanonical loci of level $\Gamma_0(p^m)$ are open and closed in the respective ε-neighbourhoods, the same holds for $\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$.

To study the limit of these spaces as $m \to \infty$, we will use the following general result on inverse limits of adic spaces along finite transition maps:

lem:hansen-johansson-5-9

Lemma 5.3.2 ([HJ23, Lemma 5.9])**.** *Let $(X_m)_{m\in I} \to (Y_m)_{m\in I}$ be a morphism of cofiltered inverse systems of locally Noetherian adic spaces. Assume that the maps $f_m\colon X_m \to Y_m$ and the transition maps in both inverse systems are finite, and that $Y_\infty := \varprojlim_m Y_m^\diamondsuit$ is perfectoid. Then $X_\infty := \varprojlim_m X_m^\diamondsuit$ is perfectoid, and the morphism $f_\infty\colon X_\infty \to Y_\infty$ is quasicompact. Moreover, if $U \subset Y_\infty$ is an open affinoid perfectoid subset which arises as the preimage of an open affinoid $U_m \subset Y_m$ for some m, then $f_\infty^{-1}(U) \subset X_\infty$ is also affinoid perfectoid. In particular, f_∞ is affinoid in the sense of [HJ23, Theorem 5.8].*

Applying this lemma, we obtain the analogue of Theorem 5.2.2 at full level:

thm:perfectoid-anticanonical-locus-full-level

Theorem 5.3.3 (Perfectoid anticanonical neighborhood at full level)**.** *Fix $0 \leqslant \varepsilon < \frac{p}{p+1}$. There are unique perfectoid spaces $\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ and $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ over $\mathbf{Q}_p^{\mathrm{cycl}}$ such that*

$$\mathcal{S}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) \quad \textit{and} \quad \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \sim \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon).$$

Moreover, the natural projection

$$f_\infty\colon \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$$

is an affinoid morphism of perfectoid spaces (in the sense of [HJ23, Theorem 5.8]). In particular, $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is affinoid perfectoid, and the boundary

$$\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \subset \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$$

is strongly Zariski closed.

Proof. We focus on the minimal compactification; the statement for the open Shimura variety follows by taking the open part. For each $m \geqslant 0$, the level projection $\pi^*_m\colon \mathcal{S}^*_{\Gamma(p^m)} \to \mathcal{S}^*_{\Gamma_0(p^m)}$ is a finite morphism of locally Noetherian adic spaces. Restricting this map to the anticanonical loci, we obtain finite maps of locally Noetherian adic spaces:

$$f_m\colon \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon).$$

Furthermore, the transition maps in both towers are induced by the natural level projections, which are finite. Therefore, the collection of maps $(f_m)_m$ defines a morphism of cofiltered inverse systems of locally Noetherian adic spaces with finite maps and finite transition maps. By Theorem 5.2.2, the limit

$$Y_\infty := \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^m)}(\varepsilon) \simeq \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$$

is a perfectoid space, which is moreover affinoid perfectoid. Applying Lemma 5.3.2, we deduce that the limit

$$\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) := \varprojlim_m \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^m)}(\varepsilon)$$

is a perfectoid space, and the projection map $f_\infty\colon \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon) \to \mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is an affinoid morphism. Since $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is affinoid perfectoid and f_∞ is affinoid, we conclude that $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is affinoid perfectoid.

Finally, the boundary $\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is the preimage of the boundary $\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ under f_∞. Since $\mathcal{Z}^{\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ is strongly Zariski closed in $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma_0(p^\infty)}(\varepsilon)$ (by Theorem 5.2.2) and f_∞ is an affinoid morphism, it follows that $\mathcal{Z}^{\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$ is strongly Zariski closed in $\mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$. ∎

5.4. **The Hodge–Tate period map.** Let us explain how to construct the Hodge–Tate period map in diamonds for toroidal and minimal compactifications. The contents of this subsection are well known to experts.

Setup 5.4.1. In this subsection, we impose no restrictions on the Shimura datum: the reflex field is not assumed to be $\mathbf{Q}_p$, and no codimension hypothesis is imposed on the boundary of the minimal compactification. All adic spaces and diamonds below are viewed over a complete algebraically closed extension of E_v, where E_v is the completion of the reflex field at our chosen place $v \mid p$. Throughout, all diamonds are assumed spatial.

`rem:hodge-tate-filtration`

Remark 5.4.2 (Construction of the Hodge–Tate filtration)**.** Consider a choice of log-smooth toroidal compactification $\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}$ of the Shimura variety. For any algebraic representation W of $\mathcal{G}^c$, one associates a local system $\mathcal{W}_p$ on the pro-Kummer-étale site of $\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}$, alongside a filtered vector bundle $\mathcal{W}_{\mathrm{dR}}$ equipped with an integrable log-connection. The work of Diao–Lan–Liu–Zhu [DLLZ23, Theorem 5.3.1] establishes that these two objects are canonically associated via the relative period sheaves. Specifically, there is an isomorphism

$$\mathcal{W}_p \otimes_{\mathbf{Q}_p} \mathcal{O}\mathbf{B}_{\mathrm{dR,log}} \simeq \mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \mathcal{O}\mathbf{B}_{\mathrm{dR,log}}$$

which respects the filtrations, the connections, and the Hecke action.

Leveraging this canonical comparison, we construct the Hodge–Tate filtration on the pro-Kummer-étale vector bundle $\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}$. We introduce the two $\mathbf{B}^+_{\mathrm{dR}}$-local systems

$$\mathbf{M} := \mathcal{W}_p \otimes_{\mathbf{Q}_p} \mathbf{B}^+_{\mathrm{dR,log}} \quad \text{and} \quad \mathbf{M}_0 := (\mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \mathcal{O}\mathbf{B}^+_{\mathrm{dR,log}})^{\nabla=0}.$$

These serve as lattices inside the common $\mathbf{B}_{\mathrm{dR}}$-local system $\mathbf{M} \otimes_{\mathbf{B}^+_{\mathrm{dR}}} \mathbf{B}_{\mathrm{dR}} \simeq \mathbf{M}_0 \otimes_{\mathbf{B}^+_{\mathrm{dR}}} \mathbf{B}_{\mathrm{dR}}$. Because $\mathbf{M}$ and $\mathbf{M}_0$ both carry filtrations, they jointly define an ascending filtration via

$$\mathrm{Fil}_{-j}\big(\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}\big) := (\mathbf{M} \cap \mathrm{Fil}^j \mathbf{M}_0)\big/(\mathrm{Fil}^1 \mathbf{M} \cap \mathrm{Fil}^j \mathbf{M}_0).$$

This ascending filtration satisfies the grading relation $\mathrm{Gr}_j\big(\mathcal{W}_p \otimes_{\mathbf{Q}_p} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}\big)(j) \simeq \mathrm{Gr}^j \mathcal{W}_{\mathrm{dR}} \otimes_{\mathcal{O}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}} \widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor}}_{K,\Sigma}}$.

The preceding comparison is functorial in W and compatible with tensor products, duals, and morphisms of representations. Consequently, it admits a Tannakian interpretation in terms of principal bundles. Let $\mathcal{G}_{\text{p-ét},p}$ be the pro-Kummer-étale $\mathcal{G}^c(\mathbf{Q}_p)$-torsor and let $\mathcal{G}_{\mathrm{dR}}$ be the $\mathcal{G}^c$-torsor on the étale site. The de Rham filtration is decreasing and defines a $\mathcal{P}^{c,+}_{\mu}$-reduction $\mathcal{P}_{\mathrm{dR}}$ of $\mathcal{G}_{\mathrm{dR}}$, whereas the Hodge–Tate filtration is increasing and defines a $\mathcal{P}^{c,-}_{\mu}$-reduction $\mathcal{P}_{\mathrm{HT}}$ of $\mathcal{G}_{\text{p-ét},p} \times^{\mathcal{G}^c(\mathbf{Q}_p)} \mathcal{G}^c$. Passing to the common Levi quotient, we put

$$\mathcal{M}_{\mathrm{dR}} := \mathcal{P}_{\mathrm{dR}} \times^{\mathcal{P}^{c,+}_{\mu}} \mathcal{M}^c_{\mu}, \qquad \mathcal{M}_{\mathrm{HT}} := \mathcal{P}_{\mathrm{HT}} \times^{\mathcal{P}^{c,-}_{\mu}} \mathcal{M}^c_{\mu}.$$

If W_j denotes the μ-weight space corresponding to $\mathrm{Gr}^j\, \mathcal{W}_{\mathrm{dR}}$, then the graded comparison above can be written as

$$\Big(\mathcal{M}_{\mathrm{HT}} \times^{\mathcal{M}^c_{\mu}} W_j\Big)(j) \simeq \mathcal{M}_{\mathrm{dR}} \times^{\mathcal{M}^c_{\mu}} W_j.$$

These isomorphisms are compatible with all tensor operations and therefore identify the two $\mathcal{M}^c_{\mu}$-valued graded fiber functors, up to the cyclotomic twist induced by μ.

In particular, let $\chi\colon \mathcal{P}^{c,-}_{\mu} \to \mathbf{G}_{\mathrm{m}}$ be a character. Since χ is trivial on $\mathcal{U}^{c,-}_{\mu}$, it factors through $\mathcal{M}^c_{\mu}$. If $a_\chi \in \mathbf{Z}$ is determined by $\chi \circ \mu(z) = z^{a_\chi}$, then applying χ to the graded comparison above gives

$$\Big(\mathcal{P}_{\mathrm{HT}} \times^{\mathcal{P}^{c,-}_{\mu}} \mathbf{A}^1_{\chi}\Big)(a_\chi) \simeq \mathcal{P}_{\mathrm{dR}} \times^{\mathcal{P}^{c,+}_{\mu}} \mathbf{A}^1_{\chi}.$$

Thus the Hodge–Tate and de Rham automorphic character lines differ precisely by the Tate twist prescribed by the μ-weight of χ. In the notation used below, if $\mathcal{L}_{\chi,\mathrm{HT}} \simeq \mathcal{L}_{\chi,\mathrm{dR}}(n_\chi)$, then $n_\chi = -a_\chi$.

`conv:compatible-toroidal-tower`

Setup 5.4.3 (A compatible toroidal tower and Hecke equivariance)**.** Choose a neat bottom level $K_{p,0} \subset \mathcal{G}(\mathbf{Z}_p)$ and a cofinal decreasing sequence of open normal subgroups $K_{p,m} \lhd K_{p,0}$. Choose a smooth projective $K^pK_{p,0}$-admissible cone decomposition Σ_0. For any index $m \geqslant 0$, let $\mathcal{S}^{\mathrm{tor}}_{K^pK_{p,m},\Sigma_m}$ be the normalization of $\mathcal{S}^{\mathrm{tor}}_{K^pK_{p,0},\Sigma_0}$ inside $\mathcal{S}_{K^pK_{p,m}}$, equipped with its divisorial log structure. This yields the toroidal compactification associated with the pullback cone decomposition Σ_m, and the transition maps across the level tower are finite Kummer–étale and surjective. We form the infinite-level inverse limits of spatial diamonds

$$\mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} := \varprojlim_m \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^pK_{p,m},\Sigma_m} \qquad \text{and} \qquad \mathcal{S}^{*,\diamondsuit}_{K^p} := \varprojlim_m \mathcal{S}^{*,\diamondsuit}_{K^pK_{p,m}}.$$

Because our chosen sequence of levels is cofinal, these coincide with the inverse limits over all p-power levels $K_p \subset \mathcal{G}(\mathbf{Z}_p)$. The toroidal-to-minimal projections $q_m \colon \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^pK_{p,m},\Sigma_m} \to \mathcal{S}^{*,\diamondsuit}_{K^pK_{p,m}}$ are compatible and induce the infinite-level projection $q \colon \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} \to \mathcal{S}^{*,\diamondsuit}_{K^p}$. We write $\rho_m \colon \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} \to \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^pK_{p,m},\Sigma_m}$ and $\eta_m \colon \mathcal{S}^{*,\diamondsuit}_{K^p} \to \mathcal{S}^{*,\diamondsuit}_{K^pK_{p,m}}$ for the respective projection maps to finite levels.

Because a single cone decomposition is rarely preserved by all of $G(\mathbf{Q}_p)$, we use the phrase "$G(\mathbf{Q}_p)$-equivariant" on the toroidal tower in the standard Hecke-compatible sense: for any element $g \in G(\mathbf{Q}_p)$, after passing to a common refinement of the source and target cone decompositions, our period maps commute with the Hecke action induced by g. Over the minimal tower there is no choice of cone decomposition, and $G(\mathbf{Q}_p)$-equivariance holds literally. On the fixed tower chosen above, the toroidal maps are literally $K_{p,0}$-equivariant.

`lem:positive-character-for-minimal`

Lemma 5.4.4. *There exists a character $\chi \colon \mathcal{P}^{c,-}_{\mu} \to \mathbf{G}_{\mathrm{m}}$ and an index $m \geqslant 0$ such that the line bundle $\mathscr{A}_\chi := \mathcal{G}^c_{E_v} \times^{\mathcal{P}^{c,-}_\mu} \mathbf{A}^1_\chi$ is very ample on $\mathcal{F}l_{\mathcal{G},\mu}$, and the canonical extension $\omega^{\mathrm{tor}}_{\chi,m}$ of the automorphic line bundle at level $K^pK_{p,m}$ descends to an ample line bundle $\omega^*_{\chi,m}$ on the minimal compactification:*

$$\omega^{\mathrm{tor}}_{\chi,m} \simeq q_m^*\omega^*_{\chi,m}.$$

Moreover, this isomorphism is compatible with level transitions.

Proof. We start by choosing a regular dominant automorphic character χ_0 lying in the interior of the ample cone of the compact dual flag variety. On the homogeneous projective variety $\mathcal{F}l_{\mathcal{G},\mu} \simeq \mathcal{G}^c_{E_v}/\mathcal{P}^{c,-}_\mu$, every ample line bundle becomes very ample upon taking a sufficiently high positive power. Meanwhile, by the fundamental construction of the Baily–Borel minimal compactification and the Koecher principle, a sufficiently divisible positive power of the canonical extension of the automorphic line bundle at finite level descends to an ample line bundle on the normal minimal compactification $\mathcal{S}^*_{K^pK_{p,m}}$. Replacing χ_0 by a common high power $\chi := \chi_0^N$ (and by the norm over its Galois conjugates if defined over a splitting field) yields both claims. Compatibility across level transitions follows from the functoriality of canonical extensions. □

`lem:infinite-level-Stein`

Lemma 5.4.5. *The natural projection $q \colon \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} \to \mathcal{S}^{*,\diamondsuit}_{K^p}$ is quasi-compact and surjective, hence a v-cover of spatial diamonds. Furthermore, let $m \geqslant 0$, let $\mathcal{E}_m$ be a finite locally free sheaf on $\mathcal{S}^{*,\diamondsuit}_{K^pK_{p,m}}$, and let $\mathcal{L}_m$ be an invertible sheaf on $\mathcal{S}^{*,\diamondsuit}_{K^pK_{p,m}}$. For any integer $r \in \mathbf{Z}$, set*

$$\mathcal{E} := \eta_m^*\mathcal{E}_m \otimes_{\mathcal{O}_{\mathcal{S}^{*,\diamondsuit}_{K^p}}} \widehat{\mathcal{O}}_{\mathcal{S}^{*,\diamondsuit}_{K^p}} \qquad \text{and} \qquad \mathcal{L} := \eta_m^*\mathcal{L}_m \otimes_{\mathcal{O}_{\mathcal{S}^{*,\diamondsuit}_{K^p}}} \widehat{\mathcal{O}}_{\mathcal{S}^{*,\diamondsuit}_{K^p}}(r).$$

Then pullback along q induces a canonical identification

$$\mathrm{Hom}_{\widehat{\mathcal{O}}_{\mathcal{S}^{*,\diamondsuit}_{K^p}}}(\mathcal{E},\mathcal{L}) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma}}}(q^*\mathcal{E}, q^*\mathcal{L}).$$

Proof. Quasi-compactness and surjectivity follow from the description of inverse limits of spatial diamonds in [Sch17, Lemmas 2.11 and 11.22]. The argument proving [**CS24**], applied to the tower $(q_n)_n$, gives

$$\widehat{\mathcal{O}}_{\mathcal{S}^{*,\diamondsuit}_{K^p}} \xrightarrow{\sim} q_*\widehat{\mathcal{O}}_{\mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma}}.$$

The continuity input used there is stated for arbitrary locally spatial diamonds in [**CS24**]; in particular, this argument does not use perfectoid representability of either infinite-level diamond. The asserted identification is now the finite-locally-free projection formula applied to $\mathcal{E}^\vee \otimes \mathcal{L}$, including the Tate twist. □

`thm:hodge-tate-period-map`

Theorem 5.4.6 (The Hodge–Tate period map in diamonds)**.** *There exists a $G(\mathbf{Q}_p)$-equivariant map of spatial diamonds*

$$\pi^{\mathrm{tor}}_{\mathrm{HT}} \colon \mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} \longrightarrow \mathcal{F}l^{\diamondsuit}_{\mathcal{G},\mu}.$$

Furthermore, $\pi^{\mathrm{tor}}_{\mathrm{HT}}$ factors uniquely through a $G(\mathbf{Q}_p)$-equivariant map on the infinite-level minimal compactification:

$$\begin{tikzcd}
\mathcal{S}^{\mathrm{tor},\diamondsuit}_{K^p,\Sigma} \arrow[rr, "\pi^{\mathrm{tor}}_{\mathrm{HT}}"] \arrow[dr, "q"'] & & \mathcal{F}l^{\diamondsuit}_{\mathcal{G},\mu} \\
& \mathcal{S}^{*,\diamondsuit}_{K^p} \arrow[ur, "\pi^*_{\mathrm{HT}}"'] &
\end{tikzcd}$$

Proof. The construction of $\pi_{\mathrm{HT}}^{\mathrm{tor}}$, including its tensor and Hecke compatibilities, follows from Remark 5.4.2, based on [DLLZ23, Theorem 5.3.1]; the equivariance convention is that of Setup 5.4.3.

Choose (χ, m) as in Lemma 5.4.4. The character-line comparison of Remark 5.4.2 and the identity $\omega_{\chi,m}^{\mathrm{tor}} \simeq q_m^* \omega_{\chi,m}^*$ give

$$(\pi_{\mathrm{HT}}^{\mathrm{tor}})^* \mathscr{A}_\chi \simeq q^* \Big(\eta_m^* \omega_{\chi,m}^* \otimes \widehat{\mathcal{O}}_{\mathcal{S}_{K^p}^{*,\lozenge}}(n_\chi) \Big) .$$

Embed $\mathcal{F}l_{\mathcal{G},\mu}$ by the complete linear system of $\mathscr{A}_\chi$. By Lemma 5.4.5, the corresponding invertible quotient descends uniquely across q, and hence so does $\pi_{\mathrm{HT}}^{\mathrm{tor}}$. Since q is a v-cover, factorization through the closed subdiamond $\mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}$ is v-local; uniqueness and the common-refinement convention of Setup 5.4.3 give $G(\mathbf{Q}_p)$-equivariance on the minimal tower. □

5.5. **Global perfectoidness.** Using Hecke-equivariance of the Hodge–Tate period map, we spread perfectoidness of the anticanonical locus ε-neighbourhood to the whole infinite-level minimal compactification. Maintain Assumption 5.2.

Notation 5.5.1. Since the principal congruence subgroups are cofinal in the p-level tower, we also write

$$\mathcal{S}_{\Gamma(p^\infty)}^{*,\lozenge} = \mathcal{S}_{K^p}^{*,\lozenge}.$$

Define $\mathcal{S}_{\Gamma(p^\infty)}^*(0)$ to be the inverse image in this diamond of the ordinary locus $\mathcal{S}_{\mathcal{G}(\mathbf{Z}_p)}^*(0)$ at hyperspecial level.
lem:mu-ordinary-preimage

Lemma 5.5.2. *The preimage of $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p) \subset |\mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}|$ under $|\pi_{\mathrm{HT}}^*|$ is the closure of $\mathcal{S}_{\Gamma(p^\infty)}^*(0)$.*

Proof. The fact that the μ-ordinary locus can be seen from pointwise isocrystals with $\mathcal{G}$-structure is explained in Remark 2.6.5. The work of Caraiani and Scholze [CS17, §3] then proves the desired result on the open Shimura variety locus. For the boundary, an argument similar to [CS24, §4.2] works except that [CS24, Theorem 4.2.1] needs a different proof. Thankfully, this is provided by [MW26, Lemmas 5.11–5.12, Corollaries 5.13–5.14], together with the uniqueness in [MW26, Lemma 2.54]. □
lem:mu-ordinary-neighborhood

Lemma 5.5.3. *Fix $0 < \varepsilon < 1$. There is an open subset $U \subset \mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}$ containing $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ such that*

$$|\pi_{\mathrm{HT}}^*|^{-1}(U) \subset |\mathcal{S}_{\Gamma(p^\infty)}^*(\varepsilon)|.$$

Proof. The period map $\pi_{\mathrm{HT}}^* \colon \mathcal{S}_{K^p}^{*,\lozenge} \to \mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}$ is a map of spatial diamonds. By Lemma 5.5.2, the preimage $(\pi_{\mathrm{HT}}^*)^{-1}(\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p))$ is the closure of the extended μ-ordinary tube. For every $\varepsilon > 0$, this closure is contained in $|\mathcal{S}_{\Gamma(p^\infty)}^*(\varepsilon)|$. Since $|\mathcal{S}_{K^p}^{*,\lozenge}|$ is a spectral space, the closed subset $|\mathcal{S}_{K^p}^{*,\lozenge}| \setminus |\mathcal{S}_{\Gamma(p^\infty)}^*(\varepsilon)|$ is quasi-compact for the constructible topology. As the intersection of the closed subsets $(\pi_{\mathrm{HT}}^*)^{-1}(U)$ over all open neighborhoods $U \supset \mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ is exactly $(\pi_{\mathrm{HT}}^*)^{-1}(\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p))$, which is disjoint from this complement, compactness implies that there exists some open $U \supset \mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ satisfying $|\pi_{\mathrm{HT}}^*|^{-1}(U) \subset |\mathcal{S}_{\Gamma(p^\infty)}^*(\varepsilon)|$. □
lem:flag-variety-orbit

Lemma 5.5.4. *For any open subdiamond $U \subset \mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}$ containing a $\mathbf{Q}_p$-rational point, we have*

$$\mathcal{G}(\mathbf{Q}_p) \cdot U = \mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}.$$

Proof. Let $C/\mathbf{Q}_p$ be the complete algebraically closed field over which the diamonds are viewed. Let $\mathcal{P}_\mu^- \subset \mathcal{G}$ be the inverse image of $\mathcal{P}_\mu^{c,-} \subset \mathcal{G}^c$ under the cuspidal central quotient. This quotient induces an isomorphism

$$X_0 := \mathcal{G}_{\mathbf{Q}_p}/\mathcal{P}_\mu^- \xrightarrow{\sim} \mathcal{G}_{\mathbf{Q}_p}^c/\mathcal{P}_\mu^{c,-} = \mathcal{F}l_{\mathcal{G},\mu}.$$

Thus, under the standing base-change convention, $\mathcal{F}l_{\mathcal{G},\mu}^{\lozenge}$ is the diamond associated with $X = (X_{0,C})^{\mathrm{ad}}$. First note that $\mathcal{G}(\mathbf{Z}_p)$ acts transitively on $X_0(\mathbf{Q}_p)$. Indeed, properness of the integral flag scheme $\mathcal{G}/\mathcal{P}_\mu^-$ extends every point of $X_0(\mathbf{Q}_p)$ to a $\mathbf{Z}_p$-point. Its inverse image under $\mathcal{G} \to \mathcal{G}/\mathcal{P}_\mu^-$ is a $\mathcal{P}_\mu^-$-torsor. The special fiber of this torsor is trivial by Lang's theorem, since $\mathcal{P}_\mu^-$ is connected, and a point lifts to $\mathbf{Z}_p$ by smoothness. After translating U, we may therefore assume that it contains the base point $x = e\mathcal{P}_\mu^-$.

Let $\mathcal{U}_\mu^+$ be the unipotent radical of the opposite parabolic and put

$$\mathcal{C}_0 = \mathcal{U}_{\mu,\mathbf{Q}_p}^+ \cdot x \subset X_0.$$

This is the algebraic big cell. The positive μ-weights on $\mathfrak{u}_\mu^+ = \operatorname{Lie}\mathcal{U}^+_{\mu,\mathbf{Q}_p}$ are all equal to 1, because μ is minuscule. In particular, $[\mathfrak{u}_\mu^+, \mathfrak{u}_\mu^+]$ has weight 2 and hence vanishes. In characteristic zero the exponential map consequently gives a μ-equivariant isomorphism of algebraic groups

$$\exp\colon \mathfrak{u}_\mu^+ \xrightarrow{\sim} \mathcal{U}^+_{\mu,\mathbf{Q}_p}.$$

Choose a $\mathbf{Q}_p$-basis of $\mathfrak{u}_\mu^+$. After base change to C, it identifies the adic big cell $\mathcal{C} = (\mathcal{C}_{0,C})^{\mathrm{ad}}$ with $(\mathbf{A}^d_C)^{\mathrm{ad}}$, carrying x to the origin, and the element $\gamma = \mu(p) \in \mathcal{G}(\mathbf{Q}_p)$ acts in these coordinates by

$$(T_1, \ldots, T_d) \longmapsto (pT_1, \ldots, pT_d).$$

Since U is an open neighborhood of x, there is an integer $r \geqslant 0$ such that the affinoid subdiamond

$$D_r^\diamondsuit = \{|T_i| \leqslant |p|^r \text{ for all } i\}$$

is contained in $U \cap \mathcal{C}^\diamondsuit$. For $m \geqslant 0$, set

$$C_m = \{|T_i| \leqslant |p|^{-m} \text{ for all } i\}.$$

The C_m are affinoid domains and $\mathcal{C}^\diamondsuit = \bigcup_{m\geqslant 0} C_m^\diamondsuit$. If $n \geqslant m + r$, then $\gamma^n C_m^\diamondsuit \subset D_r^\diamondsuit$, and therefore

$$C_m^\diamondsuit \subset \gamma^{-n} U \subset \mathcal{G}(\mathbf{Q}_p) \cdot U.$$

It follows that $\mathcal{C}^\diamondsuit \subset \mathcal{G}(\mathbf{Q}_p) \cdot U$.

It remains to see that the rational translates of the big cell cover the flag variety. On the algebraic flag variety, put

$$\Omega = \bigcup_{g \in \mathcal{G}(\mathbf{Q}_p)} g\mathcal{C}_0, \qquad Z = X_0 \setminus \Omega.$$

The subset Z is Zariski closed and $\mathcal{G}(\mathbf{Q}_p)$-stable. For any $z \in Z(\overline{\mathbf{Q}_p})$, the inverse image of $Z_{\overline{\mathbf{Q}_p}}$ under the orbit map

$$\mathcal{G}_{\overline{\mathbf{Q}_p}} \longrightarrow X_{0,\overline{\mathbf{Q}_p}}, \qquad g \longmapsto gz,$$

is closed and contains $\mathcal{G}(\mathbf{Q}_p)$. Since the connected group $\mathcal{G}_{\mathbf{Q}_p}$ has Zariski-dense $\mathbf{Q}_p$-points, this inverse image is all of $\mathcal{G}_{\overline{\mathbf{Q}_p}}$. Thus $Z_{\overline{\mathbf{Q}_p}}$ is $\mathcal{G}_{\overline{\mathbf{Q}_p}}$-stable. The transitive action on $X_{0,\overline{\mathbf{Q}_p}}$ forces Z to be empty, since $\mathcal{C}_0 \subset \Omega$. Hence the translates of $\mathcal{C}_0$ cover X_0; after base change to C, analytification, and diamondification, the translates of $\mathcal{C}^\diamondsuit$ cover $\mathcal{F}l^\diamondsuit_{\mathcal{G},\mu}$. The preceding inclusion for $\mathcal{C}^\diamondsuit$ now proves the claim. $\square$

thm:global-perfectoidness

Theorem 5.5.5. *The infinite-level minimal compactification $\mathcal{S}^*_{\Gamma(p^\infty)}$ is a perfectoid space.*

Proof. Fix a sufficiently small radius $0 < \varepsilon < 1/2$. By Lemma 5.5.3, we may find a quasicompact open subset $U \subset \mathcal{F}l^\diamondsuit_{\mathcal{G},\mu}$ containing the $\mathbf{Q}_p$-rational points $\mathcal{F}l_{\mathcal{G},\mu}(\mathbf{Q}_p)$ such that its preimage under the Hodge–Tate period map satisfies

$$|\pi^*_{\mathrm{HT}}|^{-1}(U) \subset |\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon)|.$$

By Lemma 5.5.4, the $\mathcal{G}(\mathbf{Q}_p)$-translates of U cover the entirety of $\mathcal{F}l^\diamondsuit_{\mathcal{G},\mu}$. Because $\mathcal{F}l^\diamondsuit_{\mathcal{G},\mu}$ is quasicompact, we can extract a finite subcover. That is, there exist finitely many elements $\gamma_1, \ldots, \gamma_k \in \mathcal{G}(\mathbf{Q}_p)$ such that

$$\mathcal{F}l^\diamondsuit_{\mathcal{G},\mu} = \bigcup_{i=1}^k \gamma_i \cdot U.$$

Since the Hodge–Tate period map π^*_{HT} is equivariant for the action of $\mathcal{G}(\mathbf{Q}_p)$, we derive

$$|\mathcal{S}^*_{\Gamma(p^\infty)}| = \bigcup_{i=1}^k \gamma_i \cdot |\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon)|.$$

Since $\mathcal{S}^*_{\Gamma(p^\infty)}(\varepsilon) = \mathcal{G}(\mathbf{Z}_p) \cdot \mathcal{S}^{*,\mathrm{anti}}_{\Gamma(p^\infty)}(\varepsilon)$, the desired result follows from Theorem 5.2.2. $\square$

5.6. **Conclusion.** Finally, we conclude by reducing the problem of infinite level perfectoidness of general Shimura varieties to that of Shimura varieties with reflex field $\mathbf{Q}$.

Setup 5.6.1. Suppose (G, X) is a general Shimura data. We do not enforce Assumption 5.2.

We borrow the following piece of terminology from [HJ23]:

Definition 5.6.2 (Property P)**.** We say that a Shimura datum (G, X) satisfies Property P if for every open compact subgroup $K^p \subset G(\mathbf{A}_f^p)$, the diamond of the minimal compactification at infinite level $\mathrm{Sh}^*_{K^p}(G, X) := \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathrm{Sh}^*_{K^pK_p}(G, X)^\diamond$ is a perfectoid space. Similarly, a connected Shimura datum (G, X^+) satisfies Property P if for every arithmetic subgroup $\Gamma \subset G(\mathbf{Q})^+$, the diamond $\mathrm{Sh}^*_{\Gamma,\infty}(G, X^+) := \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathrm{Sh}^*_{\Gamma \cap K_p}(G, X^+)^\diamond$ is a perfectoid space.

prop:hj-adjoint

Proposition 5.6.3. *A Shimura datum (G, X) satisfies Property P if and only if its adjoint connected Shimura datum (G^{ad}, X^+) does.*

Proof. See [HJ23, Proposition 5.18]. □

Remark 5.6.4 (Reduction to absolutely simple Shimura data)**.** A standard dévissage allows one to reduce the question of Property P for Shimura varieties to the case of absolutely simple Shimura data. The adjoint group G^{ad} decomposes into a direct product $\prod_{i=1}^m G_i$ of $\mathbf{Q}$-simple adjoint groups. Correspondingly, the connected Shimura datum (G^{ad}, X^+) decomposes as a product $\prod_{i=1}^m (G_i, X_i^+)$, where each factor (G_i, X_i^+) is an absolutely simple connected Shimura datum. Because fiber products of perfectoid spaces are perfectoid, Property P is preserved under finite products. Consequently, by Proposition 5.6.3, to establish Property P for Shimura varieties of a certain type, it suffices to prove it for absolutely simple connected Shimura data.

Theorem 5.6.5. *(G, X) satisfies Property P for all sufficiently large primes p.*

Proof. By the preceding remark, we may assume (G, X) is a $\mathbf{Q}$-simple adjoint Shimura datum. If (G, X) is of pre-abelian type, we know that integral canonical models exist for all primes and hence a standard Weil restriction argument allows us to reduce to a Shimura datum with reflex field $\mathbf{Q}$ because a closed adic subspace of a perfectoid space is automatically perfectoid. So this is solved by Theorem 5.5.5. Therefore, we only consider $\mathbf{Q}$-simple adjoint Shimura data of exceptional type. If it is of type E_7, its reflex field is totally real by [Del71, Proposition 3.8]. The same restriction-of-scalars argument reduces to reflex field $\mathbf{Q}$; the resulting global map extends to the integral models after excluding finitely many primes.

It remains to consider outer type E_6. Let $(\mathbf{L}, X_L) \hookrightarrow (\mathbf{G}, Y)$ be the embedding of Theorem 3.5.1. The target has reflex field $\mathbf{Q}$, so it satisfies Property P by Theorem 5.5.5. The embedding extends to a closed immersion of integral models for all sufficiently large p, and hence $(\mathbf{L}, X_L)$ satisfies Property P. Since $\mathbf{L}^{\mathrm{ad}} \simeq G$, Proposition 5.6.3 then gives Property P for (G, X). □

The Hodge–Tate period map a priori exists as a map of diamonds. However, since the source $\mathcal{S}^*_{K^p}$ is now the diamond of a perfectoid space, this map upgrades to a morphism of adic spaces:

thm:perfectoid-shimura-varieties-body

Theorem 5.6.6. *Let (G, X) be a Shimura datum with reflex field E. For all sufficiently large primes p, the following holds. Fix a complete algebraically closed field $C/\mathbf{Q}_p$ and an embedding $E \hookrightarrow C$. Fix any open compact subgroup $K^p \subset G(\mathbf{A}_f^p)$. For any open compact subgroup $K_p \subset G(\mathbf{Q}_p)$, let $\mathcal{S}^*_{K^pK_p}$ denote the adic space over $\operatorname{Spa} C$ associated with the base change of the minimal compactification $\mathrm{Sh}^*_{K^pK_p}(G, X)$ along $E \hookrightarrow C$. Then there is a perfectoid space $\mathcal{S}^*_{K^p}$ such that*

$$\mathcal{S}^*_{K^p} \sim \varprojlim_{K_p \subset G(\mathbf{Q}_p)} \mathcal{S}^*_{K^pK_p}$$

as diamonds over $\operatorname{Spd} C$. Moreover, there is a canonical $G(\mathbf{Q}_p)$-equivariant Hodge–Tate period map

$$\pi_{\mathrm{HT}} \colon \mathcal{S}^*_{K^p} \to \mathscr{F}\ell_{G,\mu}$$

of adic spaces over $\operatorname{Spa} C$ which is functorial in the tame level. Finally, the boundary of $\mathcal{S}^*_{K^p}$ is Zariski-closed.

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